

“It Takes Two to Tango”: Joint Location Choices and Local Shocks*

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Abstract

Most adults in the US live in couples, so decisions about where to live and whether to work are rarely made alone. I examine how joint decision-making shapes adjustment to local shocks and who bears them. Combining geocoded data from the Panel Study of Income Dynamics with individual-level measures of exposure to the China Trade Shock, I find that workers in exposed industries are more likely to leave their commuting zones than their non-exposed neighbors. Couples, however, move only when the husband is exposed, not when the wife is. Wives, in turn, are more likely to participate in the labor force when their husbands are exposed and less likely when they are exposed themselves. Couples thus adjust through two margins: migration and wives' labor supply. I then extend a dynamic spatial and trade equilibrium model to include couples who jointly choose where to live, each spouse's sector, and whether the wife works. Accounting for couples changes who gains from the shock far more than how much the economy gains. Because the shock raises husbands' wages along with wives', households rely less on wives' earnings, so fewer women enter the labor market than in an economy without couples, and female wages bear more of the adjustment. Joint decisions therefore shape the gains of singles as well: without couples, single women would gain 17% less, and single men 6% more. For couples, the two margins are asymmetric substitutes: couples move more when wives cannot respond, but wives do not work more when couples cannot move. Wives' participation insures every couple, whereas migration insures only the exposed and spreads gains more evenly across states.

Keywords: Local labor market, Joint Decision-Making, Migration decisions

JEL Codes: F16, J12, J61, R23

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1 Introduction

For most adults, the response to a local shock is a joint decision. According to the 2020 Census, roughly two-thirds of US households consisted of couples (Anderson et al., 2024). Unlike single individuals, couples must weigh two labor-market exposures against each other: a move that pays off for one spouse may not pay off for the other, making couples less likely to relocate in pursuit of better opportunities. At the same time, couples have an insurance margin that singles lack: the added-worker effect (Lundberg, 1985; Attanasio et al., 2005). When one spouse loses a job or earnings, the other spouse, usually the wife, may increase her labor supply to offset the household’s lost income. In the presence of local shocks, therefore, couples face a trade-off between lower spatial mobility and an additional insurance margin. How couples resolve this trade-off shapes not only how the economy adjusts to a local shock, but also who bears its effects and where.

Yet spatial models of local shocks abstract from this trade-off, because they assume that each worker decides alone. A substantial body of theoretical literature suggests that local labor markets should adjust to local economic shocks through worker mobility (Allen and Arkolakis, 2025), but empirical evidence generally indicates that workers migrate less in response to such shocks than the theory predicts (Autor et al., 2013; Dix-Carneiro and Kovak, 2017; Borusyak et al., 2026). While migration costs are the leading explanation for this disconnect (Allen and Arkolakis, 2025; Desmet and Parro, 2025), another potential factor remains underexplored: most workers do not make these decisions alone. If couples move less but insure more within the household, a model without couples may misread both how mobile workers are and who bears the cost of the shock.

In this paper, I examine how joint decision-making shapes adjustment to local shocks and who bears them. I focus on two research questions. First, do couples respond to local shocks by moving, as singles would, or by reallocating labor within the household? Second, what does joint decision-making imply for the shock’s aggregate employment and welfare effects, and for how these effects are distributed across household types and states? To answer them, I combine new reduced-form evidence on how couples respond to local shocks in the US with a dynamic spatial and trade model in which couples decide together where to live and whether the wife works. My main finding is that accounting for couples changes who gains from local shocks far more than how much the economy gains.

To study how couples respond, I use geocoded data from the Panel Study of Income Dynamics (PSID) and compare exposed and non-exposed households living in the same commuting zone. I exploit the China Trade Shock, i.e., the increase in import competition faced by US manufacturing after China joined the WTO in 2001 (Autor et al., 2013, 2021),

as the local shock and estimate its effects using a difference-in-differences design. Unlike previous work, which measured the shock at the commuting zone level, I measure exposure at the individual level and classify an individual as exposed if they were employed in 1999 in industries adversely affected by Chinese import competition. Because the PSID follows households over time, this design allows me to track household composition, migration, and labor-market outcomes, and, in particular, to distinguish couples in which the husband is exposed from couples in which the wife is. Based on this design, I establish three empirical facts.

The first fact is that, within the same commuting zone, exposed individuals experienced worse labor-market outcomes and were more likely to leave their commuting zone than non-exposed individuals. On average, exposed individuals were 6.3 p.p. less likely to be employed and 3.9 p.p. less likely to participate in the labor force during the shock, worked 2 fewer hours per week, and earned 10% lower hourly wages. Their probability of out-migration rose by an average of 4.7 p.p., with the gap widening by approximately 2 p.p. every two years and reaching 6 p.p. by the end of the shock. Because I measure exposure at the individual level, I am the first to document these differences within commuting zones. Although my analysis uses a different dataset, sample definition, and empirical design, these results are consistent with previous evidence that increased import competition from China harmed more exposed workers and places, with persistent employment effects on manufacturing workers, even though it improved aggregate welfare in the US economy (Autor et al., 2013, 2021; Caliendo et al., 2019).

The second fact is that couples, who already migrate less than singles, move in response to the shock only when the husband is exposed. Relative to couples with non-exposed husbands, out-migration rises by 5 p.p. when the husband is exposed but does not change when the wife is exposed. Changes in household composition do not explain this asymmetry: the shock has no effect on marriage, divorce, or fertility. Nor is the asymmetry attributable to differences in migration responses among parents or women.

The third fact is that wives' labor force participation also responds asymmetrically: it rises when their husbands are exposed and falls when they themselves are exposed. When husbands are exposed, wives are 5.4% more likely to participate in the labor force, consistent with an added-worker effect. By contrast, exposed wives are 13% less likely to be employed and 14% less likely to participate in the labor force than non-exposed wives, and I find no offsetting response in husbands' extensive-margin labor supply, although husbands' hourly wages decline. Notably, exposure has no statistically significant effect on household income, regardless of which spouse is exposed. Together, the three facts suggest that couples mitigate the shock through a mix of spatial insurance (migration) and intra-household insurance (the

added-worker effect), and that which margin they use depends on which spouse is hit.

I next extend an existing quantitative spatial and trade equilibrium model (Caliendo et al., 2019) to rationalize the empirical patterns documented above and to quantify what joint decision-making implies for the economy as a whole. The model is dynamic, multi-region, and multi-sector. Households consist of single men, single women, or couples and choose among location-sector bundles each period, subject to migration frictions and idiosyncratic preference shocks. Because location is the key choice couples must make jointly, I follow Rodríguez-Clare et al. (2026) and nest the preference shocks, allowing households to substitute across sectors within a location at a different rate than across locations. On the production side, each region contains multiple sectors that trade with one another, and firms combine male and female labor using a CES technology with sector-specific gender intensities. Thus, shocks can affect men and women differently, even within the same household.

The model’s main distinctive feature is that a couple jointly chooses where to live, which sectors the spouses are attached to, and whether the wife works in the market or remains at home. This participation decision depends on the spouses’ relative wages and a home-production productivity draw, thus embedding an endogenous participation margin and the economies of scale from marriage within a dynamic spatial and trade model. This structure captures the interaction among migration, sectoral reallocation, and intra-household labor supply in response to local shocks. To account for the economies of scale of marriage and to compare welfare across household types, I develop a welfare measure in money-metric consumption-equivalent terms, following Chiappori and Meghir (2015). As in Caliendo et al. (2019) (hereafter CDP), I solve the model using Dynamic Hat Algebra, which expresses the equilibrium without requiring the recovery of the levels of fundamentals, such as mobility costs.

To bring the model to the data, I align the quantitative environment with CDP: 50 US states, 37 foreign trade partners including China, 22 sectors, and non-employment, at a quarterly frequency for 200 periods starting in 2000Q1. I take the dispersion of preferences across sectors from the literature and, following Rodríguez-Clare et al. (2026), calibrate the dispersion across locations by indirect inference, so that the model reproduces the population response to import competition estimated by Autor et al. (2013). Of the four parameters that govern the couple’s problem, I calibrate two using consumption expenditure data and discipline the other two with the reduced-form evidence. The calibration already shows that couples matter for mobility: when I split every couple into two singles and calibrate to the same target, the resulting economy implies a higher migration elasticity. Ignoring couples, therefore, makes individuals appear more mobile than they are.

In the quantitative exercise, the China Trade Shock reduces manufacturing employment

but is, in the aggregate, a boom for US workers. In the long run, the decline in manufacturing jobs per capita is close to CDP's estimate (-0.21 vs. -0.23 p.p.), so adding couples who choose where to live together and decide whether the wife works changes neither where the shock lands nor its magnitude. Cheaper Chinese goods lower the consumption price index and increase the real return to market work relative to non-employment, and services absorb more workers than manufacturing sheds; consequently, the employment rate rises for everyone. This aggregate picture, however, masks substantial heterogeneity across household types. The decline in manufacturing employment is about twice as large for men as for women, and married men experience the largest decline (-0.41 p.p., compared with -0.29 p.p. for single men) but the smallest employment gain: because they are already employed, they adjust primarily by reallocating across sectors rather than by entering or leaving the labor force.

For married women, by contrast, the employment gain reflects an added-worker response, reproducing the third empirical fact. By disproportionately affecting male-intensive manufacturing, the shock raises wives' wages relative to their husbands' and draws wives into the labor force. This response occurs where husbands lose their jobs: across states, increases in married women's labor force participation are positively correlated with declines in manufacturing employment.

The average individual gains 0.18% in money-metric consumption-equivalent terms, which is similar to previous estimates of the welfare consequences of the shock, but these gains differ sharply across household types. Single women gain the most (34% more than the average individual) because they are concentrated in expanding sectors, underrepresented in the manufacturing sectors most affected by the shock, and unconstrained by a joint location decision; single men gain the least (10% less than the average individual), and couples fall in between, closer to single men. Among couples, gains decline monotonically with exposure, and the husband's exposure determines their order: compared with couples in which neither spouse initially works in manufacturing, the ones in which only the wife does gain 10% less, those in which only the husband does gain 22% less, and those in which both do gain 30% less. When the husband is exposed, the shock hits the household's main source of income; when only the wife is exposed, the couple can cushion part of her loss because she can leave the market and work at home.

Across states, the gains are smallest where people leave. The states that gain the least are those that lose the most population, led by Louisiana, Kentucky, Indiana, and Mississippi, and this geography is nearly identical across household types. Couples' welfare gains are, however, the most geographically unequal, because spouses cannot take jobs in different states.

Comparing these results with those from a *single-agent economy* shows that accounting for couples changes who gains from the shock far more than how much the economy gains. To build this counterfactual economy, I split every couple into two singles, reconstruct all inputs from the data, and recalibrate the migration elasticity to match the same population response, so that individuals differ only by gender. The two economies agree on the magnitude and location of the shock: the decline in manufacturing differs by less than 1%, the aggregate welfare gain differs by about 2%, and state-level welfare gains are highly correlated (0.96). They differ sharply, however, in how women’s labor supply responds. A wife decides whether to work together with her husband, and the shock that raises her wage also raises his, making the household less reliant on her earnings. As a result, female labor supply is less elastic when couples are present: it is 50.7% more elastic in the single-agent economy, which generates an 18.2% larger employment gain, primarily because women’s employment responds 36.5% more.

Because couples change how women’s labor supply responds, joint decisions also shape the gains of singles, who make no joint decisions at all. With fewer women entering the labor market, female wages bear more of the adjustment: single women supply labor to a tighter market, and single men to a weaker one. In the single-agent economy, single women gain 17.0% less, single men gain 5.7% more, and married women gain 13.7% more. Joint decision-making also makes welfare gains more unequal across states: they are 26.8% less dispersed in the single-agent economy. Thus, a model without couples captures how large the shock is and where it lands, but not who bears it at each location.

Lastly, I conduct three counterfactual exercises that shut down, one at a time, the migration, participation, and gender-wage responses, and find that the ranking of these margins depends on whether one looks at the size of the welfare gain or at its distribution. Migration matters most for distribution. Eliminating it changes aggregate welfare only slightly (+1%), but reduces the employment response by 12% and makes state-level welfare 61% more unequal, with losses in the exposed states that were losing population and gains in the destinations, whose gains are no longer diluted by arrivals. Among couples, the losses fall on those whose husbands initially work in manufacturing, whereas unexposed couples gain because exposed households no longer enter their labor markets, where they would otherwise depress wages. The added-worker effect does not fill this gap: wives step in when their husbands’ earnings fall relative to their own, but shutting down migration affects husbands’ and wives’ wages in a similar way, so wives have no new reason to work.

Wives’ participation, by contrast, is insurance that every couple holds: damping it is the only exercise that lowers aggregate welfare. When I dampen the participation response, couples migrate more, the employment response falls by 24%, aggregate welfare declines

by 4.6%, and every state loses. All couples lose, including those in which neither spouse was initially exposed, and couples in which only the husband was exposed bear the largest loss (10.2%). For couples, spatial and intra-household insurance are therefore asymmetric substitutes: couples migrate more when I dampen wives' participation, but wives do not work more when I shut down migration.

Finally, the gender wage gap accounts for nearly all of the shock's gendered incidence. When I hold women's wages nearly fixed relative to men's, wives' participation response falls by about 63%, the manufacturing decline shifts from women to men, and most of the gap between the gains of single women and single men closes: single men gain 18.8% more, while single women gain 10.4% less. Aggregate welfare rises by 5.8%, and every type of couple gains, with the most exposed gaining the most.

Taken together, these results show that abstracting from couples does not change the size of a local shock or where it lands, but it does change who bears it and how unequally the gains are distributed across states. Because wives decide whether to work jointly with their husbands, taking into account how the shock affects both spouses' earnings, couples change how female labor supply responds to the shock, and this response shapes wages and welfare for singles as well. For couples, wives' participation and migration are not interchangeable: the first insures every couple, whereas the second insures only those directly exposed and makes welfare gains more equal across states. Thus, understanding the incidence of local shocks requires treating the household, and not only the worker, as the unit that decides where to live and work.

Related Literature. This paper contributes to three strands of the literature. First, it contributes to the dynamic spatial general-equilibrium literature on trade and migration responses to local shocks (Redding and Rossi-Hansberg, 2017; Desmet et al., 2018; Monte et al., 2018; Allen and Donaldson, 2020, 2022; Donald et al., 2025; Desmet and Parro, 2025; Pellegrina and Sotelo, 2025). It is particularly closely related to Caliendo et al. (2019), which emphasizes the interaction between migration, productivity, and the spatial distribution of economic activity under the China Trade Shock, and to Rodríguez-Clare et al. (2026), who extend Caliendo et al. (2019) to separate mobility across sectors within a location from mobility across locations. My contribution is to bring the household into this class of models, with joint location choices and wives' labor supply decisions. Doing so leaves the aggregate effects of the shock largely unchanged, but it reveals how they are distributed: who bears the shock, how unequally the gains are spread across states, and how joint decisions spill over to singles through the labor market. It also shows that a single-agent model disciplined by the same migration evidence makes individuals look more mobile than they are.

Second, the paper contributes to the literature that studies how dual-earner households respond to shocks, highlighting the roles of risk sharing within couples (Attanasio et al., 2005; Pilossoph and Wee, 2021; Calvo et al., 2024) and co-location frictions (Guler et al., 2012; Flabbi and Mabli, 2018; Foerster and Ulbricht, 2023; Jayachandran et al., 2024; Yan, 2025).¹ I bring these insights to the study of local shocks. In an aggregate shock, couples are always better off than singles because they can reallocate labor within the household, and because the shock is aggregate, the joint location friction does not bind. In the presence of a local shock, however, couples face a trade-off between lower mobility and an additional margin of adjustment. My contribution is to quantify this trade-off in general equilibrium and to show that the two margins are asymmetric substitutes that insure different households: wives' participation insures every couple, whereas migration insures only exposed couples.

Third, the paper contributes to the literature on the China Trade Shock and its effects on US local labor markets (Autor et al., 2013, 2014, 2016, 2019, 2021, 2025; Acemoglu et al., 2016; Pierce and Schott, 2016; Greenland et al., 2019; Caliendo et al., 2019; Faber et al., 2022; Borusyak et al., 2026). This literature has documented substantial heterogeneity in the local effects of rising competition from Chinese imports.² My contribution to this literature is twofold. First, using the PSID, I provide new empirical results on the effects of the shock in the US by measuring individual-level exposure and comparing individuals within the same commuting zone. Second, to the best of my knowledge, this is the first paper to provide empirical evidence on within-household responses to the shock in the US. In this respect, the paper is closely related to Parey et al. (2025), who, using UK data, show that household responses to trade shocks can differ sharply by gender within the couple, finding an added-worker effect when wives are exposed. In the US, by contrast, wives increase their participation when their husbands are exposed, but husbands do not increase theirs when their wives are exposed.

The rest of the paper proceeds as follows. Section 2 presents the empirical analysis, detailing the approach and presenting the main empirical findings. Section 3 presents the theoretical model. Section 4 brings the model to the data, and Section 5 presents the results

¹There is an extensive literature on family migration and household adjustment. Starting with Mincer (1978), this literature has emphasized that migration is often a joint decision and that such decisions can have unequal consequences within the household. Subsequent work has documented the role of “trailing wives” and the labor market costs of family migration for women (Costa and Kahn, 2000; Cooke et al., 2009; Benson, 2014; Blau and Kahn, 2017; Fadlon et al., 2020; Braun et al., 2021; Le Barbanchon et al., 2021; Venator, 2026, 2024). Although beyond the scope of this paper, the outcomes of the “trailing wives” under a trade shock remain an open research question.

²In particular, Autor et al. (2013) and subsequent papers document a significant decrease in manufacturing employment in US commuting zones with larger manufacturing shares, with long-lasting consequences that include increases in political polarization (Autor et al., 2020), child and adult poverty, and even mortality among young men (Autor et al., 2019).

of the quantitative exercise. Finally, Section 6 concludes.

2 Empirical Analysis

This section presents the empirical analysis of how couples respond to the China Trade Shock. I first describe the data, define exposure at the individual and couple levels, and outline the empirical design. I then document three empirical facts concerning labor market outcomes, migration responses, and intra-household labor supply adjustments. Together, these findings motivate the theoretical framework that I develop in the next section.

2.1 Empirical Design

2.1.1 Data Source

The primary data source for the empirical analysis is the Panel Study of Income Dynamics (PSID), which follows individuals and their households over time. Most studies of the China Trade Shock use commuting zones as the unit of observation and rely on cross-sectional data. As a result, they cannot distinguish how exposure varies within households or track individuals across time and space. Because the PSID follows both spouses within the same household over time, it allows me to address both limitations.

The PSID has three additional advantages that make it particularly well suited to this analysis. First, the restricted geocoded version records household addresses at the census-tract level in every survey wave.³ This information allows me to follow individuals across local labor markets, compare individuals within the same market, and study migration between markets. Following the previous literature, I define local labor markets as US commuting zones (CZs).

Second, the PSID allows me to construct marital status from household composition and each person’s relationship to the household head in every period. This approach does not rely exclusively on current legal marital status, which may understate the number of couples by excluding cohabitation or overstate it by including individuals who are separated but not divorced. Given the PSID’s sampling structure, I restrict the sample to three marital-status categories:⁴

1. “Married”: Individuals in their first union who are legally married or cohabiting with a different-sex partner

³The PSID staff has previously disclosed all results; access to the restricted data is under Contract 921.

⁴I exclude remarried individuals from the sample. I classify an individual who separated after a cohabitation as “divorced”, even if their legal marital status is single.

2. “Divorced”: Individuals who are legally divorced or separated, or whose cohabiting union ended
3. “Single”: Individuals whom I never observe as “married” to another individual

I then pool single and divorced individuals into an unmarried category because neither group makes joint household decisions with a current partner.⁵

Third, the PSID contains detailed information on labor market outcomes, allowing me to examine employment, unemployment, and labor force participation. For employed individuals, I observe average weekly hours, hourly wages, annual labor income, and several additional outcomes. I also observe total household income for couples.⁶

I use all PSID waves conducted in odd-numbered years between 1991 and 2013.⁷ I restrict the sample to individuals aged 20–55 who lived in the mainland US and were present in the 1999 wave. The final sample contains 4,731 individuals and 38,634 observations across 12 waves.

2.1.2 Individuals Exposed to the China Trade Shock

The China Trade Shock refers to the increase in competition from Chinese manufactured goods in the US following China’s accession to the WTO in 2001 and continuing through 2012 (Autor et al., 2013, 2021). I identify exposed individuals in two steps. First, following Autor et al. (2021) (henceforth ADH), I define the growth in Chinese import competition faced by US manufacturing industry j between 2000 and 2012 as:

$$\Delta IC_j = 100 \times \frac{\Delta M_j^{\text{China,US}}}{Y_{j,2000} + M_{j,2000} - X_{j,2000}},$$

where $\Delta M_j^{\text{China,US}}$ is the change in US imports from China in industry j between 2000 and 2012, $Y_{j,2000}$ denotes industry j ’s shipments, and $M_{j,2000} - X_{j,2000}$ denotes its net imports in 2000. A positive value of ΔIC_j therefore indicates that US manufacturing industry j experienced increased competition from Chinese imports between 2000 and 2012.

To isolate the foreign supply-driven component of US imports from China, I follow ADH and measure industry-level growth in Chinese exports to eight other high-income countries.⁸

⁵Assuming that divorced individuals do not make joint decisions follows previous literature studying life after divorce (Weiss and Willis, 1993; Del Boca and Flinn, 1995; Flinn, 2000; Reynoso, 2024; Didier and Lopez, 2026).

⁶I convert all monetary variables to 2017 dollars using the Personal Consumption Expenditures Price Index (PCE) from the US Bureau of Economic Analysis.

⁷Because the PSID was conducted annually until 1997 and biennially afterward, I consider only odd years to construct a biennial sample before and after 1997, which avoids double-counting migration.

⁸Autor et al. (2013) and their follow-up papers consider Australia, Denmark, Finland, Germany, Japan,

I also use three-year-lagged values of Y_j , M_j , and X_j rather than base-year values. The resulting measure of import competition is

$$\widehat{\Delta IC}_j = 100 \times \frac{\Delta M_j^{\text{China, Other Countries}}}{Y_{j,1997} + M_{j,1997} - X_{j,1997}}.$$

A positive value of this measure indicates that industry j experienced increased competition from Chinese imports in other high-income countries. I therefore classify the corresponding US industry j as negatively exposed to the China Trade Shock.

Second, I identify the industry in which each individual i was employed in 1999 and define the indicator Exposed_i as follows:

$$\text{Exposed}_i = \begin{cases} 1 & \text{if } i \text{ worked in 1999 in industry } j \text{ with } \widehat{\Delta IC}_j > 0 \\ 0 & \text{otherwise.} \end{cases} \quad (1)$$

For the analysis of within-household mechanisms, I further restrict the sample to couples. I divide couples into three groups: those in which neither spouse worked in an exposed industry in 1999, those in which the husband worked in an exposed industry, and those in which the wife worked in an exposed industry. I define couple h as exposed through the husband if

$$\text{Exposed Husband}_h = \begin{cases} 1 & \text{if he worked in 1999 in industry } j \text{ with } \widehat{\Delta IC}_j > 0 \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

and as exposed through the wife if

$$\text{Exposed Wife}_h = \begin{cases} 1 & \text{if she worked in 1999 in industry } j \text{ with } \widehat{\Delta IC}_j > 0 \\ 0 & \text{otherwise.} \end{cases} \quad (3)$$

2.1.3 Summary Statistics

Table 1 reports summary statistics for the pooled 1991–1999 sample. The first column describes the full sample; the second pools single and divorced individuals; and the third and fourth report statistics for husbands and wives, respectively. Married individuals constitute approximately 60% of the sample, single individuals 24%, and divorced individuals the remainder.

Unmarried individuals are slightly younger and less likely to be college graduates or white than married individuals. The average individual is 37 years old; two-fifths of the sample are

New Zealand, Spain, and Switzerland. They select these countries based on the availability of data for the full sample period.

women, one-third are college graduates, and 86% are white. Half of the sample have children under 18, including nearly one-quarter of unmarried individuals. The youngest child in the household is, on average, 4 years old and tends to be younger among unmarried individuals, consistent with the younger average age of unmarried individuals.

Married and unmarried individuals live in similarly exposed CZs but differ sharply in how often they migrate. According to ADH’s measure, one in ten individuals lives in a CZ in the top two deciles of exposure to Chinese import competition. Exposure does not vary meaningfully by marital status, suggesting little spatial sorting along this dimension. The average migration rate is 8%, with unmarried individuals migrating at nearly twice the rate of married individuals.

In 1999, 15% of the sample worked in an industry exposed to competition from Chinese imports, with substantial heterogeneity across groups. Exposure rates were 14% among unmarried individuals, 19% among husbands, and 10% among wives. At the household level, 20% of couples had at least one exposed spouse. As noted by Autor et al. (2013, 2021), the period of the China Trade Shock also featured two related economic forces: increasing automation and broader globalization. Following Autor and Dorn (2013), I calculate routine-task intensity and offshorability indices for the occupations represented in the sample. Slightly less than one-quarter of observations correspond to individuals employed in routine-task-intensive occupations, while approximately one in ten workers is employed in an offshorable occupation. Offshorability varies substantially by gender, reflecting the occupational sorting of men and women. I control for both occupational characteristics in the empirical analysis.

Because the sample consists of prime-age individuals, employment and labor force participation rates are high, at 91% and 95%, but the composition of non-employment differs by marital status. Among non-employed individuals, unemployment is the most common status for unmarried individuals and husbands, whereas non-participation in the labor force is most common among wives. This pattern is consistent with unmarried individuals relying more heavily on reemployment after job loss, whereas married individuals can adjust labor supply within the household, thereby drawing on intra-household insurance.

The average individual works 40 hours per week, earns \$19 per hour in 2017 USD, and receives approximately \$42,000 in annual labor income. Husbands work the most hours, earn the highest hourly wages, and consequently receive the highest annual labor income; wives have the lowest values for all three outcomes. Married and unmarried individuals work similar hours on average, but married individuals earn approximately 20% higher hourly wages and therefore receive more labor income per capita. Although selection into marriage likely explains part of this gap, this pattern is also consistent with couples using intra-household

Table 1: Summary Statistics Before the China Trade Shock

	(1)	(2)	(3)	(4)
	All	Unmarried	Couples	
			Husbands	Wives
Panel A: Distribution by "Marital Status"				
Share of Single Individuals	0.24			
Share of Married Individuals	0.59			
Share of Divorced Individuals	0.16			
Panel B: Demographic Characteristics				
Age ⁺	36.86	34.69	38.92	37.19
Share of Female Individuals	0.42	0.55		
Share of College Graduates	0.33	0.29	0.37	0.32
Share of White Individuals	0.86	0.75	0.93	0.93
Share of Individuals with Kids	0.50	0.23		0.69
Age of Youngest Kid ⁺	3.67	1.75		4.98
Share in Highly Exposed CZs ⁺⁺	0.11	0.12		0.11
Migration Rate	0.08	0.11		0.06
Panel C: Labor Market Outcomes				
Share of Exposed Individuals	0.15	0.14	0.19	0.10
Share in Routine Occupations	0.23	0.22	0.23	0.26
Share in Offshorable Occupations	0.12	0.14	0.08	0.16
Share Employed	0.91	0.89	0.97	0.85
Share Unemployed	0.03	0.06	0.02	0.02
Share Out of the Labor Force	0.05	0.05	0.01	0.13
Weekly Hours ⁺	40.80	39.89	45.34	33.60
Hourly Wage ⁺	19.54	16.98	23.70	16.45
Labor Income ⁺	42671.70	35953.36	57073.74	27617.31
Number of Observations	16,645	4,343	6,151	6,151

Notes. The unmarried category pools single and divorced individuals. Monetary values are expressed in 2017 USD using the PCE price index. ⁺: Indicates the sample mean. ⁺⁺: Highly exposed CZs are those in the top 20% of exposure according to ADH.

labor supply adjustments to allocate resources more effectively.

2.1.4 Difference-in-Differences Design

To estimate the overall effect of exposure to the China Trade Shock, I use the following event-study specification for individual i in commuting zone c and year t :

$$\begin{aligned} \text{Outcome}_{i,c,t} = & \alpha \cdot \mathbb{1}[\text{Exposed}_i = 1] + \sum_{\tau \neq 1999} \beta_\tau \cdot \mathbb{1}[\text{Exposed}_i = 1] \cdot \mathbb{1}[t = \tau] \\ & + X'\delta + \gamma_t + \gamma_c + \varepsilon_{i,c,t} \end{aligned} \quad (4)$$

The coefficients of interest, β_τ , measure the difference in outcomes between exposed and non-exposed individuals in year τ , relative to the corresponding difference in 1999. The control vector includes age and indicators for being female, holding a college degree, having children younger than 18, working in manufacturing in 1999, working in a routine occupation, and working in an offshorable occupation. The specification also includes commuting-zone and year fixed effects. I weight all regressions by PSID sampling weights and two-way cluster standard errors at the individual and industry levels. To summarize the average effect during the shock, I also estimate a difference-in-differences version of Equation 4 that replaces the year-specific interactions with a single interaction:

$$\begin{aligned} \text{Outcome}_{i,c,t} = & \beta_1 \cdot \mathbb{1}[\text{Exposed}_i = 1] + \beta_2 \cdot \mathbb{1}[\text{Exposed}_i = 1] \cdot \mathbb{1}[t \geq 2001] \\ & + X'\delta + \gamma_c + \gamma_t + \varepsilon_{i,c,t}. \end{aligned} \quad (5)$$

To examine joint household decision-making, I estimate the following specifications for spouse s in couple h , commuting zone c , and year t . First, I focus on whether the husband is exposed:

$$\begin{aligned} \text{Outcome}_{s,h,c,t} = & \beta_1 \cdot \mathbb{1}[\text{Exposed Husband}_h = 1] + \beta_2 \cdot \mathbb{1}[\text{Exposed Husband}_h = 1] \cdot \mathbb{1}[t \geq 2001] \\ & + X'\delta + \gamma_c + \gamma_t + \varepsilon_{s,h,c,t} \end{aligned} \quad (6)$$

and then on whether the wife is exposed:

$$\begin{aligned} \text{Outcome}_{s,h,c,t} = & \beta_1 \cdot \mathbb{1}[\text{Exposed Wife}_h = 1] + \beta_2 \cdot \mathbb{1}[\text{Exposed Wife}_h = 1] \cdot \mathbb{1}[t \geq 2001] \\ & + X'\delta + \gamma_c + \gamma_t + \varepsilon_{s,h,c,t}. \end{aligned} \quad (7)$$

In these specifications, the coefficient of interest is β_2 , which captures the post-2001 difference in outcomes between couples with an exposed spouse and those without, relative to the corresponding pre-2001 difference.⁹ I estimate both specifications on the same sample of

⁹For this analysis, I drop 4% of couple-year observations across the full 1991–2013 panel in which both spouses were working in exposed industries in 1999, because I cannot measure whether one spouse’s job was more important to the couple than the other’s.

couples, so the comparison group in Equation 6 includes couples in which only the wife is exposed, and vice versa. Both equations include the controls used in Equation 4. For controls measured at the individual level, I include separate values for each spouse, such as the husband's and wife's ages.

I two-way cluster standard errors by the exposed spouse's industry and by household or individual, depending on how the outcome is measured. The industry dimension refers to the husband's industry in Equation 6 and the wife's in Equation 7. I cluster at the household level for outcomes measured at the household level, such as out-migration and household income, and at the individual level for outcomes measured for each spouse, such as employment and labor force participation.

2.2 Empirical Facts

The three facts proceed from individual-level effects to within-household adjustment. The first establishes the labor market effects of the shock and examines whether exposed individuals respond by leaving their commuting zone. The second shows that couples migrate in response only when the husband is the exposed spouse. The third examines wives' labor supply responses, a margin of adjustment that couples can use alongside or instead of migration. Across all three analyses, I identify the effects by comparing individuals whose exposure differs by their 1999 industry, holding commuting zone and year fixed. The relevant comparison is therefore between exposed and non-exposed individuals facing the same local labor market, rather than between local labor markets with different levels of exposure.

Fact 1: Exposure to the China Trade Shock worsened labor market outcomes and increased out-migration.

Figure 1 presents estimates of Equation 4 for employment, unemployment, labor force participation, weekly hours worked, hourly wages, and annual labor income. Table A.2 reports estimates from the corresponding difference-in-differences specification in Equation 5, which summarizes the average effect during the China Trade Shock.

Panel A of Figure 1 shows that, during the first seven years of the shock, exposed individuals were less likely to be employed than non-exposed individuals. On average, exposure was associated with a 6.3-percentage-point (7%) decline in the probability of employment. After 2009, the point estimates remain close to their 2007 level, but the confidence intervals widen, and the estimates are no longer statistically significant. Most of the decline in employment between 2001 and 2007 reflects individuals leaving the labor force (Panel C) rather than entering unemployment (Panel B). Between 2001 and 2013, exposed individuals were,

on average, 4% less likely to participate in the labor force than non-exposed individuals.

Weekly hours worked follow a pattern similar to employment (Panel D): individuals employed in exposed industries in 1999 worked 5% fewer hours per week during the shock than individuals employed in other industries that year. Hourly wages exhibit a different pattern (Panel E). Although the estimate is positive in 2001, the hourly wages of exposed individuals declined through the final year of the sample. On average, exposed individuals earned 10% lower hourly wages during the shock than non-exposed individuals. Combined with the reduction in hours worked, this decline resulted in 11% lower annual labor income among exposed individuals (Panel F).

My employment estimates are similar in magnitude to ADH’s when I use their continuous measure of import competition. To compare my estimates more directly with the previous literature, I replace the binary exposure measure in Equation 1 with the continuous import-competition measure used by ADH (Table A.3).¹⁰ The average decadalized growth in Chinese import competition implies a 0.5-percentage-point average decline in the probability of employment among exposed individuals relative to non-exposed individuals between 2001 and 2013. For comparison, converting ADH’s cumulative estimate for the employment-to-population ratio between 2001 and 2019 into an evenly distributed effect and retaining only odd-numbered years, as in the PSID sample, implies an average decline of 0.55 percentage points.

These findings are consistent with previous evidence on the adverse effects of the China Trade Shock in the US (Autor et al., 2013, 2014, 2016, 2021; Acemoglu et al., 2016; Pierce and Schott, 2016). Despite using a different dataset, sample restrictions, and empirical approach, I similarly find a negative employment response among individuals working in exposed industries.

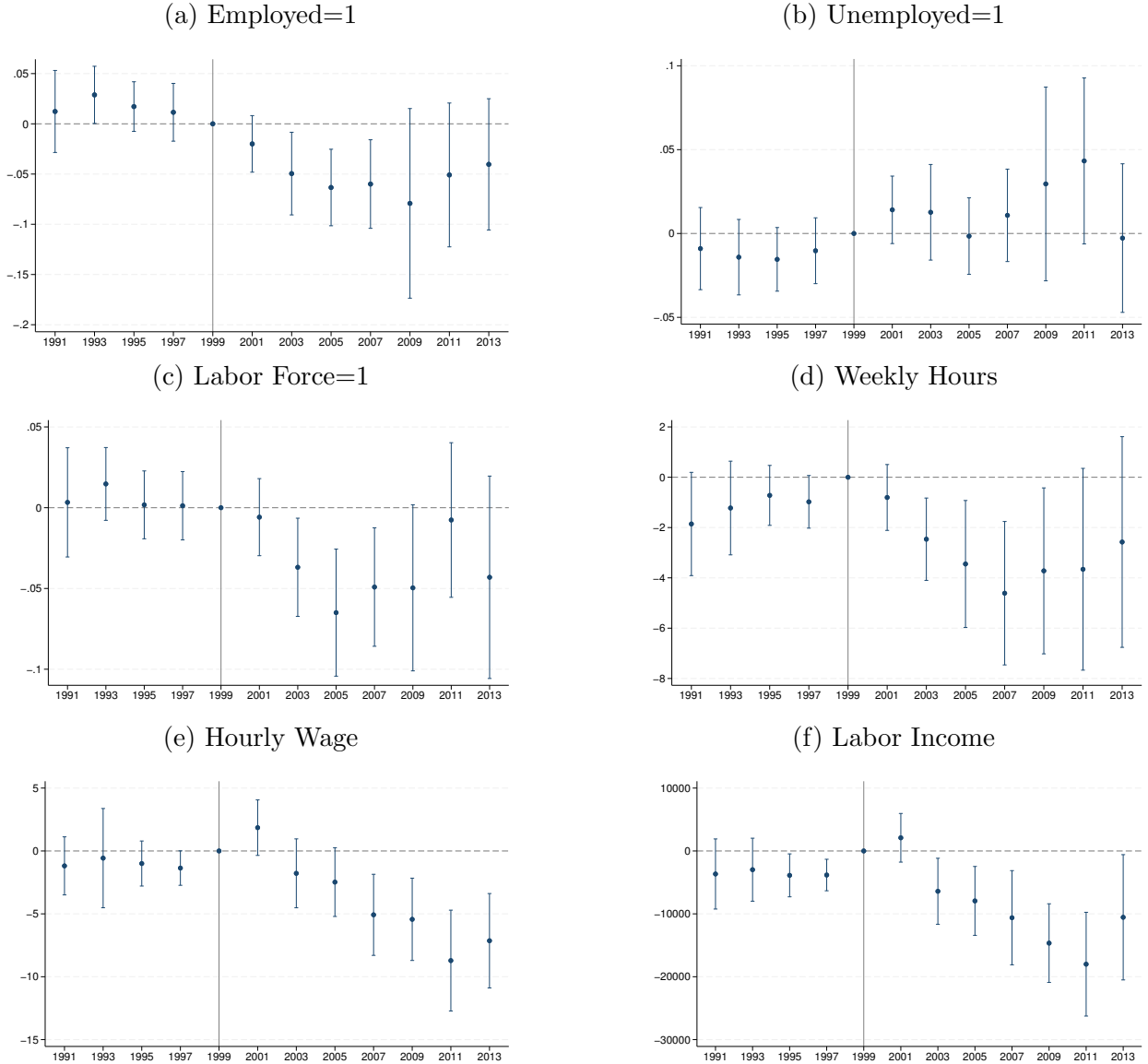
Given the deterioration in labor market outcomes among individuals employed in industries exposed to Chinese import competition, particularly the decline in employment during the first years of the shock, I next examine whether exposed individuals moved to other CZs in search of better opportunities. For each individual i , I define out-migration in year t as follows:

$$\text{Out-Migration}_{i,t} = \begin{cases} 1 & \text{if } CZ_{i,t-2} \neq CZ_{i,t} \\ 0 & \text{if } CZ_{i,t-2} = CZ_{i,t}. \end{cases}$$

During the shock, exposed individuals became increasingly likely to leave their CZ. Figure 2 presents estimates of Equation 4 using out-migration as the outcome: the estimated difference rose by approximately 2 percentage points every two years, reaching 6 percentage

¹⁰One potential concern is that, given the small PSID sample, the particular set of CZs in my sample may drive the estimates. To address this concern, Appendix A.1 replicates ADH’s findings using the same CZs included in my sample, ruling out this alternative explanation.

Figure 1: Exposure to the China Trade Shock and Labor Market Outcomes



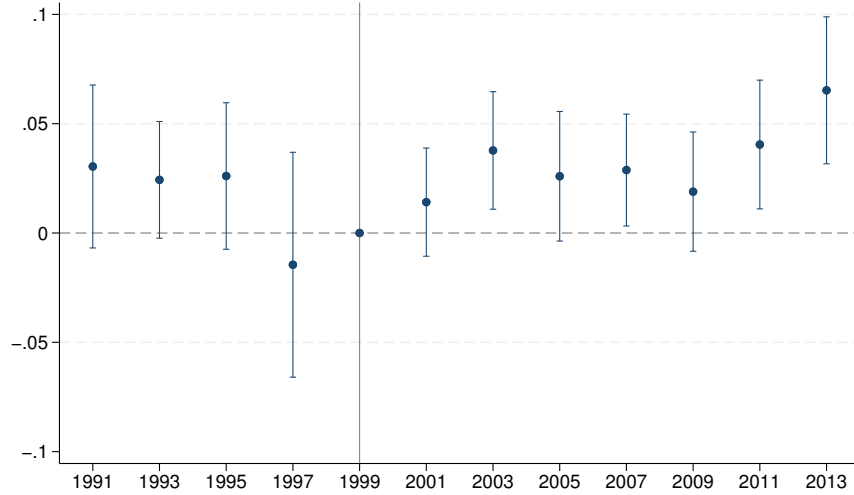
Notes. Each panel reports estimates from Equation 4, using the variable named in the panel title as the outcome. Monetary outcomes are expressed in 2017 USD using the PCE price index and include zero values. Robust standard errors are two-way clustered at the industry and individual levels. The bars indicate 95% confidence intervals.

points in 2013. Most of the response emerges with a lag of seven to ten years, consistent with Greenland et al. (2019)'s evidence on the effects of the China Trade Shock on internal migration in the US. On average, exposed individuals were 4.7 percentage points more likely to out-migrate than non-exposed individuals within the same commuting zone, as reported in Table A.4.

The out-migration response is consistent with ADH's evidence of a statistically significant

decline in the population aged 25 to 39 and a statistically insignificant decline in the overall population of more exposed commuting zones. Moreover, Borusyak et al. (2026) argue that shift-share measures of industry-level shocks may attenuate estimated migration responses when they abstract from workers’ outside options, as in ADH. Accordingly, the individual-level out-migration response I estimate should be larger than their aggregate estimate.

Figure 2: Exposure to the China Trade Shock and Out-Migration Responses



Notes. Robust standard errors are two-way clustered by industry and individual. The bars indicate 95% confidence intervals.

Fact 2: Couples out-migrate only when the husband is exposed.

To examine how couples’ migration responses differ from those of individuals, I estimate Equations 6 and 7 using out-migration as the outcome at the household level.¹¹ Column 1 of Table 2 shows that couples in which the husband was exposed to the shock were 5 percentage points more likely to leave their commuting zone than couples in which the husband was not exposed.¹² By contrast, exposure of the wife produces no statistically significant migration response, and the point estimate in Column 2 is negative. Although this asymmetry may initially appear counterintuitive, it is consistent with evidence that husbands’ employment

¹¹The PSID records only the census tract of the household in which the interview takes place, which it always treats as the family’s primary household. As a result, if only one spouse migrates, the PSID does not record the couple as migrating. Instead, it records the couple as “married, spouse absent,” and the respondent reports for both spouses. According to PSID staff, this case is very unlikely. My estimates are therefore a lower bound on couples’ migration responses, as they only account for couples who migrate together.

¹²The negative coefficient in the first row of Table 2 reflects the relatively favorable employment conditions in those manufacturing industries before the shock, which made couples less likely to migrate if the husband was employed there.

plays a larger role in family migration decisions (Mincer, 1978; Costa and Kahn, 2000; Cooke et al., 2009; Benson, 2014; Blau and Kahn, 2017; Fadlon et al., 2020; Braun et al., 2021; Le Barbanchon et al., 2021; Venator, 2026, 2024).

I can rule out two alternative explanations for this asymmetry: changes in family composition and different migration responses among parents or among women. First, the asymmetry could be driven by changes in family composition. If, for example, couples divorce when the wife is exposed but not when the husband is exposed, this would bias my findings. In Appendix A.2, I study the effect of the trade shock on the probabilities of marriage, divorce, and fertility and find no effect on any of them. Second, the asymmetry could reflect nothing more than different migration responses among parents or among exposed women. In Appendix A.3, I expand Equation 4 to include a triple interaction for having children and for being a woman. In both cases, I cannot reject that parents or women responded in the same way as non-parents or men.

Table 2: Exposure to the China Trade Shock and Couples' Out-Migration Responses

	(1)	(2)
	Out-Migration=1	
Exposed Husband=1	-0.121***	
	(0.043)	
(Exposed Husband=1)x(Post=1)	0.050**	
	(0.022)	
Exposed Wife=1		0.064
		(0.074)
(Exposed Wife=1)x(Post=1)		-0.010
		(0.028)
Observations	13,695	13,695
R^2	0.10	0.09
Baseline Mean	0.05	0.05

Notes. Out-migration is measured at the household level. Robust standard errors are two-way clustered by household and the husband's industry in Column 1, and by household and the wife's industry in Column 2. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Fact 3: Wives' labor force participation falls when they are directly exposed and rises when husbands are exposed.

Couples have an additional margin of adjustment, the so-called “added-worker effect” (Lundberg, 1985), which refers to a spouse who specializes in home production entering the labor market when the other spouse loses their job, to offset the household’s income loss. I therefore study labor market responses to the shock, distinguishing between direct and indirect exposure. When wives are directly exposed to the shock, they are 13% less likely to be employed and 14% less likely to be in the labor force than non-exposed wives (Panel A of Table 3). However, when wives are indirectly exposed because their husbands worked in an exposed industry, they are 5.4% more likely to be in the labor force relative to the wives of non-exposed husbands (Panel B of Table 3). This provides suggestive evidence of the added-worker effect among wives indirectly exposed to the China Trade Shock through their husbands.

Husbands, by contrast, show no added-worker effect. When husbands are indirectly exposed to the shock, they have the same probability of employment as non-exposed husbands, while earning a lower hourly wage (Table A.5), suggesting a lower reservation wage when indirectly exposed.

Table 3: Wife’s Exposure to the China Trade Shock and Labor Market Outcomes

	Wife Employed=1	Wife Labor Force=1
Exposed Wife=1	0.041 (0.099)	0.016 (0.110)
(Exposed Wife=1)x(Post=1)	-0.099** (0.045)	-0.110** (0.048)
Exposed Husband=1	0.027 (0.077)	0.027 (0.068)
(Exposed Husband=1)x(Post=1)	0.041*** (0.013)	0.042*** (0.013)
Observations	14,083	14,083
Baseline Mean	0.77	0.78

Notes. Robust standard errors are two-way clustered at the wife’s industry and individual levels in Panel A and at the husband’s industry and individual levels in Panel B. * for $p < .10$, ** for $p < 0.05$, *** for $p < 0.01$.

The absence of an added-worker effect among husbands contrasts with the findings of Parey et al. (2025) for England and Wales. They show that when English husbands are indirectly exposed to the trade shock, older husbands increase their labor force participation and move into self-employment. They find a positive, though not statistically significant, response among indirectly exposed wives. The difference in the results for husbands could be driven by the fact that 93% of husbands in their sample are in the labor force, compared with 99% in my sample (Table 1). As a result, husbands in my sample face a ceiling effect that hinders identification of an added-worker effect for husbands.

Household income of exposed couples did not differ significantly from that of non-exposed couples during the shock, regardless of which spouse was exposed. To provide suggestive evidence on whether these asymmetric responses insure couples against the shock, I estimate Equations 6 and 7 using a measure of household income from the PSID. Columns 1 and 2 of Table A.6 report the estimates. This suggests that, when the husband is exposed, the combination of out-migration and the wife’s labor force participation responses helps stabilize household income during the shock.

When the wife is exposed, stable household income reflects a smaller loss rather than insurance through migration or added labor supply. These couples do not migrate, the wife’s employment falls, and the husband’s employment does not change (Tables 3 and A.5). Before the shock, however, wives earned roughly half as much labor income as husbands (Table 1), so the loss of the wife’s earnings is small relative to total household income. The point estimate in Column 2 of Table A.6 is negative, consistent with this smaller loss, but imprecisely estimated. The smaller loss also helps explain why these couples do not move: the income at stake is lower than it would be when the husband, the household’s main earner, is exposed to the shock. Household income is also an incomplete measure of welfare, because it does not capture changes in home production, leisure, or the costs of moving. I quantify the welfare effects of both margins with the model developed in the next section.

The empirical evidence points to three central patterns that, together, suggest a trade-off for couples between moving and adjusting labor supply within the household. First, exposure to the China Trade Shock worsened labor market outcomes for directly exposed individuals, who responded by leaving their commuting zones. Second, among couples, this migration response was concentrated in households where the husband was exposed. Third, I find evidence of an added-worker effect during the trade shock, with wives’ labor force participation responding differently depending on which spouse was directly exposed to the shock. On the one hand, joint location decisions make moving harder, so couples move only when the household’s breadwinner is affected. On the other hand, they can reallocate labor within the household, making it more bearable to stay in the same location.

However, the empirical evidence cannot inform us about which margin of adjustment is more important, why, or what the general equilibrium effects of abstracting from couples are. In the next section, I develop a framework in which households jointly decide on location and labor supply, within a quantitative spatial and trade model.

3 Dynamic Spatial Model of Trade and Migration with Heterogeneous Households

This section develops a dynamic spatial and trade model to rationalize the three empirical facts documented in Section 2 and to quantify their welfare and spatial implications. The model extends the framework of Caliendo et al. (2019) (henceforth CDP) by incorporating heterogeneous households and joint decision-making. In the model, households rather than individuals are the unit that moves: a couple jointly chooses where to live and each spouse's sector, and then decides whether the wife works in the market.

The economy consists of L locations, indexed by i as origins and j as destinations, populated by singles and couples. In each location, a population mass N is born at $t = 0$, comprising $N/2$ men and $N/2$ women. A fraction $\bar{s}_i$ of individuals are single, while the remaining fraction, $1 - \bar{s}_i$, belong to heterosexual couples. Time is discrete and indexed by $t \in \{0, 1, 2, \dots\}$.

The model includes three household types, indexed by h : single men ($h = s_m$), single women ($h = s_f$), and couples ($h = c$). All households are forward-looking and respond to local conditions by choosing their place of residence and sector of employment. Couples make these choices jointly. After relocating, they draw a home-production productivity shock β , which, together with the spouses' relative wage, determines whether the wife participates in market work.

A central feature of the model is the interaction between migration and married women's labor force participation. Together, these two decisions shape how sectoral productivity shocks affect the spatial distribution of individuals and output, because men and women are exposed to different sectors.

3.1 Environment

The environment consists of three components: technology and trade, preferences and mobility, and the wife's participation decision. First, technology and trade determine equilibrium wages in each region-sector cell. Relative to CDP, labor is a CES composite of male and female inputs, implying that sectoral shocks have differential effects on men's and women's

wages. Second, preferences and mobility determine households' residential and employment choices. Couples jointly choose a location and a pair of sectors rather than making these decisions sequentially. Finally, conditional on relocation, the couple chooses whether the wife works in the market or in home production.

3.1.1 Technology

In each period t , every region i contains K productive sectors, indexed by $k, n \in \{1, \dots, K\}$. Let $A_{i,k,t} > 0$ denote Hicks-neutral productivity in sector k of region i . Let $\psi_{k,m}, \psi_{k,f} > 0$ denote sector-specific gender-intensity parameters that are common across locations and satisfy $\psi_{k,m} + \psi_{k,f} = 1$.

A representative intermediate-good firm in sector k of origin i combines male labor, $M_{i,k,t}$, and female labor, $F_{i,k,t}$, into an effective labor aggregate using a CES production function with elasticity of substitution $\sigma > 1$:

$$\mathcal{L}_{i,k,t} = \left(\psi_{k,m} M_{i,k,t}^{(\sigma-1)/\sigma} + \psi_{k,f} F_{i,k,t}^{(\sigma-1)/\sigma} \right)^{\sigma/(\sigma-1)}. \quad (8)$$

Output is then given by

$$Y_{i,k,t} = A_{i,k,t} \mathcal{L}_{i,k,t}.$$

The sectoral intermediate good is sold at producer price $p_{i,k,t}$. Shipping the good from origin-sector pair ik to destination-sector pair jn entails iceberg trade costs $\tau_{ik,jn} \geq 1$. Thus, the delivered price paid by sector n in destination region j is $p_{ik,jn,t} = p_{i,k,t} \tau_{ik,jn}$. Under perfect competition and constant returns to scale, the producer price equals the marginal cost of producing the sectoral intermediate good:

$$p_{i,k,t} = \frac{\bar{w}_{i,k,t}}{A_{i,k,t}} \implies \bar{w}_{i,k,t} = p_{i,k,t} A_{i,k,t}.$$

Here, $\bar{w}_{i,k,t}$ is the CES wage index, which gives the cost-minimizing unit cost of effective labor:

$$\bar{w}_{i,k,t} = \left(\psi_{k,m}^\sigma (w_{i,k,t}^M)^{1-\sigma} + \psi_{k,f}^\sigma (w_{i,k,t}^F)^{1-\sigma} \right)^{\frac{1}{1-\sigma}}.$$

Here, $w_{i,k,t}^M$ and $w_{i,k,t}^F$ denote the average wages paid to men and women, respectively, employed in sector k of location i at time t .

The first-order conditions of the cost-minimization problem yield optimal demands for

male and female labor as functions of the effective labor aggregate and relative wages:

$$M_{i,k,t} = \psi_{k,m}^\sigma \left(\frac{w_{i,k,t}^M}{\bar{w}_{i,k,t}} \right)^{-\sigma} \mathcal{L}_{i,k,t}, \quad (9)$$

$$F_{i,k,t} = \psi_{k,f}^\sigma \left(\frac{w_{i,k,t}^F}{\bar{w}_{i,k,t}} \right)^{-\sigma} \mathcal{L}_{i,k,t}. \quad (10)$$

Inverting the labor demand functions in Equations 9 and 10 yields the equilibrium gender-specific factor prices within each region-sector pair:

$$w_{i,k,t}^M = \bar{w}_{i,k,t} \psi_{k,m} \left(\frac{\mathcal{L}_{i,k,t}}{M_{i,k,t}} \right)^{1/\sigma}, \quad w_{i,k,t}^F = \bar{w}_{i,k,t} \psi_{k,f} \left(\frac{\mathcal{L}_{i,k,t}}{F_{i,k,t}} \right)^{1/\sigma}. \quad (11)$$

In each destination region j and sector n , a representative sectoral final-good firm combines goods from all origin-sector pairs into a sector- n composite using a CES aggregator with elasticity of substitution $\eta_n > 1$. Let $Q_{j,n,t}$ denote the quantity of this composite produced at time t . The corresponding unit-cost function is

$$P_{j,n,t} = \left(\sum_{i=1}^L \sum_{k=1}^K p_{ik,jn,t}^{1-\eta_n} \right)^{1/(1-\eta_n)}. \quad (12)$$

The implied share of expenditure by sector n in destination region j on goods from sector k in origin region i is

$$\pi_{ik,jn,t} = \left(\frac{p_{ik,jn,t}}{P_{j,n,t}} \right)^{1-\eta_n}.$$

At the regional level, a representative final-good firm combines sectoral composites into a regional consumption good using a Cobb-Douglas aggregator with expenditure shares $\iota_k > 0$, where $\sum_{k=1}^K \iota_k = 1$:

$$Q_{j,t} = \prod_{k=1}^K \left(\frac{Q_{j,k,t}}{\iota_k} \right)^{\iota_k}$$

where $Q_{j,t}$ denotes the quantity of the regional final good produced in region j at time t . The associated regional price index is

$$P_{j,t} = \prod_{k=1}^K P_{j,k,t}^{\iota_k}. \quad (13)$$

Cobb-Douglas aggregation implies that expenditure on sector n is a constant share of total expenditure in region j : $P_{j,n,t} Q_{j,n,t} = \iota_n P_{j,t} Q_{j,t}$.

Each location j also contains a non-employment sector, indexed by 0, that is available to single men, single women, and married men. Married women do not choose sector 0; instead,

they choose a productive sector and subsequently decide whether to work in the market or engage in home production. A wife who stays at home therefore keeps a potential sector, whose wage enters the couple's participation decision. The non-employment sector produces no output. An individual who chooses sector 0 in region j receives an exogenous nominal benefit b_j , which is constant over time:

$$w_{j,0,t}^M = w_{j,0,t}^F = b_j.$$

3.1.2 Preferences

Mobility decisions. The household's mobility problem is dynamic: each household $h \in \{s_m, s_f, c\}$ is forward-looking and discounts the future with discount factor $\delta \in (0, 1)$. At the start of every period, households begin in a particular location-sector position and must decide whether to switch sectors and whether to stay or move. Every household occupies a single location, so spouses always live together, but the sector component differs by household type: for single individuals, it is one sector in $\{0, 1, \dots, K\}$, while for couples, it is the pair formed by the husband's and wife's sectors. The husband's current and destination sectors satisfy $k_m, n_m \in \{0, 1, \dots, K\}$, while the wife's current and destination market sectors satisfy $k_f, n_f \in \{1, \dots, K\}$.

For convenience, I collapse a household's location-sector position into a single *bundle*. I write the origin (current) bundle as $\mathbf{o} \equiv (i, k_m, k_f)$, the destination (next-period) bundle as $\mathbf{d} \equiv (j, n_m, n_f)$, and a generic bundle summed over as $\mathbf{z} \equiv (\ell, q_m, q_f)$. For single men, a bundle is (i, k_m) ; for single women, (i, k_f) ; and for couples, (i, k_m, k_f) . The same symbols $\mathbf{o}, \mathbf{d}, \mathbf{z}$ denote the bundle for every household type, and I unbundle them into components only where a formula refers to a particular location or sector.

Let $\mathcal{S}_h$ denote the set of feasible bundles for household type h , and define

$$\mathcal{S}_h(j) \equiv \{\mathbf{d} \in \mathcal{S}_h : \text{the location of } \mathbf{d} \text{ is } j\}$$

as the set of feasible bundles available to type h that are located in j . For singles, $\mathcal{S}_h(j)$ contains the $K + 1$ sectors available in j ; for couples, it contains the $(K + 1) \times K$ sector pairs available in j .

To decide their bundle for the next period, each household h in current bundle $\mathbf{o}$ draws two idiosyncratic preference shocks: a *location* shock $\varepsilon_{\mathbf{o}j,t}^{h,L}$ for each destination location j , and a *sector* shock $\varepsilon_{\mathbf{o}\mathbf{d},t}^{h,S}$ for each destination sector bundle within that location. The two enter additively, so the total preference shock attached to destination bundle $\mathbf{d}$, located in

j , is

$$\varepsilon_{\mathbf{o},t}^h = \varepsilon_{\mathbf{o},t}^{h,L} + \varepsilon_{\mathbf{o},t}^{h,S}. \quad (14)$$

Following Rodríguez-Clare et al. (2026), I assume that the vector $\varepsilon_{\mathbf{o},t}^h \equiv (\varepsilon_{\mathbf{o},t}^{h,d})_{d \in \mathcal{S}_h}$ is i.i.d. over time and households and has a nested Gumbel joint distribution,

$$F(\varepsilon_{\mathbf{o},t}^h) = \exp \left(- \sum_{j=1}^L \left(\sum_{d \in \mathcal{S}_h(j)} \exp(-\varepsilon_{\mathbf{o},t}^{h,d}/v) \right)^{v/\vartheta} \right), \quad (15)$$

with $0 < v \leq \vartheta$.¹³ The parameter v governs the dispersion of preferences across sector bundles within a location, so $1/v$ determines substitution across sectors. Similarly, ϑ governs preference dispersion across locations, so $1/\vartheta$ determines substitution across locations. The restriction $v \leq \vartheta$ implies that the sectoral elasticity $1/v$ is at least as large as the locational elasticity $1/\vartheta$. Thus, households substitute at least as readily across sectors within a location as they do across locations.¹⁴

A household moving from origin bundle $\mathbf{o}$ to destination bundle $\mathbf{d}$ incurs mobility cost $\mu_{\mathbf{o}\mathbf{d}} \geq 0$, with $\mu_{\mathbf{o}\mathbf{o}} = 0$. The lifetime utility of a household of type h occupying bundle $\mathbf{o}$ at time t is

$$v_{\mathbf{o},t}^h = \log(W_{\mathbf{o},t}^h) + \max_{d \in \mathcal{S}_h} \{ \delta \mathbb{E}(v_{\mathbf{d},t+1}^h) - \mu_{\mathbf{o}\mathbf{d}} + \varepsilon_{\mathbf{o},t}^h \},$$

where $W_{\mathbf{o},t}^h$ denotes the period- t indirect utility of household type h in current bundle $\mathbf{o}$. For singles $h \in \{s_m, s_f\}$, $W_{\mathbf{o},t}^{s_m} = w_{i,k,t}^M/P_{i,t}$ and $W_{\mathbf{o},t}^{s_f} = w_{i,k,t}^F/P_{i,t}$, as in CDP but allowing for gender-specific wages. For couples, $W_{\mathbf{o},t}^c$ also depends on the wife's market-work decision, as I describe below.

Let $V_{\mathbf{o},t}^h \equiv \mathbb{E}(v_{\mathbf{o},t}^h)$. Then, using the properties of the nested Gumbel distribution (Appendix B.1), I can write the value function as

$$V_{\mathbf{o},t}^h = \log(W_{\mathbf{o},t}^h) + \vartheta \log \left(\sum_{\ell=1}^L \left(\sum_{z \in \mathcal{S}_h(\ell)} \exp[\delta V_{\mathbf{z},t+1}^h - \mu_{\mathbf{o}\mathbf{z}}]^{1/v} \right)^{v/\vartheta} \right). \quad (16)$$

I decompose the choice probability into a location choice and a conditional sector-bundle choice. Let $\Omega_{\mathbf{o},t}^h \equiv \sum_{z \in \mathcal{S}_h(j)} \exp[\delta V_{\mathbf{z},t+1}^h - \mu_{\mathbf{o}\mathbf{z}}]^{1/v}$. The share of type- h households in bundle

¹³Rodríguez-Clare et al. (2026) extend Caliendo et al. (2019) by allowing the elasticities of mobility across sectors within a region and across regions to differ, adding a margin that better matches the empirical evidence in Autor et al. (2013). In this setting, the extension distinguishes between mobility across sectors and mobility across regions, with the latter being the relevant margin for couples.

¹⁴When $v = \vartheta$, Equation 15 reduces to the product of independent Gumbel distributions with scale ϑ .

$\mathbf{o}$ that choose location j is

$$\lambda_{\mathbf{o}j,t}^h = \frac{(\Omega_{\mathbf{o}j,t}^h)^{v/\vartheta}}{\sum_{\ell=1}^L (\Omega_{\mathbf{o}\ell,t}^h)^{v/\vartheta}}. \quad (17)$$

Conditional on choosing location j , the share that selects destination bundle $\mathbf{d} \in \mathcal{S}_h(j)$ is

$$\lambda_{\mathbf{od}|j,t}^h = \frac{\exp [\delta V_{\mathbf{d},t+1}^h - \mu_{\mathbf{od}}]^{1/v}}{\Omega_{\mathbf{o}j,t}^h}. \quad (18)$$

The unconditional share of type- h households in bundle $\mathbf{o}$ that choose destination bundle $\mathbf{d} \in \mathcal{S}_h(j)$ is therefore

$$\lambda_{\mathbf{od},t}^h = \lambda_{\mathbf{od}|j,t}^h \lambda_{\mathbf{o}j,t}^h. \quad (19)$$

Thus, the location elasticity $1/\vartheta$ governs the response of location shares to each location's inclusive value, whereas the sectoral elasticity $1/v$ governs the response of within-location shares to the value of each sector bundle.

Couple's problem. After relocating, a couple decides whether the wife works in the market or stays at home, based on the spouses' relative wage and a home-production productivity draw. Each couple draws an independently and identically distributed home-production productivity shock $\beta \sim \text{Pareto}(\chi, \kappa)$, with support $\beta \geq \chi > 0$ and shape parameter $\kappa > 0$.¹⁵ The couple then jointly chooses the wife's labor-supply status $e_{\mathbf{d},t} \in \{0, 1\}$, where $e_{\mathbf{d},t} = 1$ indicates that the wife stays at home, along with private consumption levels $c_{\mathbf{d},f,t}$ and $c_{\mathbf{d},m,t}$ and public-good expenditure $q_{\mathbf{d},t}$. When the wife stays at home, the productivity shock increases the effective level of the household public good, $Q_{\mathbf{d},t} = \beta^{e_{\mathbf{d},t}} q_{\mathbf{d},t}$.

Conditional on choosing bundle $\mathbf{d}$, the couple therefore solves

$$\begin{aligned} \max_{c_{\mathbf{d},f,t}, c_{\mathbf{d},m,t}, q_{\mathbf{d},t}, e_{\mathbf{d},t} \in \{0,1\}} \quad & (Q_{\mathbf{d},t}^\gamma c_{\mathbf{d},f,t}^{1-\gamma})^\alpha (Q_{\mathbf{d},t}^\gamma c_{\mathbf{d},m,t}^{1-\gamma})^{1-\alpha} \\ \text{s.t.} \quad & P_{j,t} (c_{\mathbf{d},f,t} + c_{\mathbf{d},m,t} + q_{\mathbf{d},t}) = \underbrace{w_{j,n_m,t}^M + w_{j,n_f,t}^F (1 - e_{\mathbf{d},t})}_{I(e)}, \end{aligned} \quad (20)$$

$$Q_{\mathbf{d},t} = \beta^{e_{\mathbf{d},t}} q_{\mathbf{d},t},$$

with $0 < \alpha < 1$ and $0 < \gamma < 1$, where α is the wife's bargaining weight and γ is the expenditure share of the public good.

Following Chiappori and Mazzocco (2017), the optimal allocations conditional on $\mathbf{d}$ and

¹⁵The home-production productivity shock captures the non-monetary value generated when the wife engages in home production (Becker, 1965).

$e_{\mathbf{d},t}$ are

$$\begin{aligned} c_{\mathbf{d},f,t}^*(e) &= \alpha(1-\gamma)\frac{I(e)}{P_{j,t}}, \\ c_{\mathbf{d},m,t}^*(e) &= (1-\alpha)(1-\gamma)\frac{I(e)}{P_{j,t}}, \\ q_{\mathbf{d},t}^*(e) &= \gamma\frac{I(e)}{P_{j,t}} \implies Q_{\mathbf{d},t}^*(e) = \beta^e \gamma \frac{I(e)}{P_{j,t}} \end{aligned} \tag{21}$$

The wife works in the market when the gain in household income from her earnings outweighs the gain in the household public good from her home production. To find the decision rule for $e_{\mathbf{d},t}$, let $\mathcal{A} \equiv \alpha^{\alpha(1-\gamma)}(1-\alpha)^{(1-\alpha)(1-\gamma)}\gamma^\gamma(1-\gamma)^{1-\gamma}$, and define the sector-specific female-to-male wage ratio faced by a couple in $\mathbf{d}$ as

$$r_{\mathbf{d},t} \equiv \frac{w_{j,n_f,t}^F}{w_{j,n_m,t}^M}.$$

Then, the couple's static indirect utility conditional on choosing $\mathbf{d}$ and drawing β is

$$\mathcal{U}_{\mathbf{d},t}(\beta) = \mathcal{A} \frac{w_{j,n_m,t}^M}{P_{j,t}} \begin{cases} (1+r_{\mathbf{d},t}) & \text{if } e_{\mathbf{d},t} = 0 \\ \beta^\gamma & \text{if } e_{\mathbf{d},t} = 1 \end{cases}$$

Hence, the wife works in the market if and only if

$$\beta \leq \beta_{\mathbf{d},t}^*, \quad \beta_{\mathbf{d},t}^* = (1+r_{\mathbf{d},t})^{1/\gamma}.$$

The market-work probability is therefore

$$\rho_{\mathbf{d},t} \equiv \Pr(e_{\mathbf{d},t} = 0) = 1 - \left(\frac{\chi}{\beta_{\mathbf{d},t}^*} \right)^\kappa = 1 - \chi^\kappa (1+r_{\mathbf{d},t})^{-\kappa/\gamma}. \tag{22}$$

An interior participation margin in bundle $\mathbf{d}$ requires $0 < \rho_{\mathbf{d},t} < 1$. The upper bound, $\rho_{\mathbf{d},t} < 1$, holds for any $\chi > 0$, whereas the lower bound, $\rho_{\mathbf{d},t} > 0$, requires the lower support bound to lie below the market-work threshold: $\chi < (1+r_{\mathbf{d},t})^{1/\gamma}$. Moreover, because the home-production productivity shock enters the couple's problem as $Q_{\mathbf{d},t} = \beta^{e_{\mathbf{d},t}} q_{\mathbf{d},t}$, the model imposes $\chi \geq 1$, ensuring that staying at home never reduces the effective level of the public good. For this condition to hold in every location-sector-time cell, the relevant restriction is $1 \leq \chi < \min_{\mathbf{d},t} (1+r_{\mathbf{d},t})^{1/\gamma}$.

Given the wife's labor-supply rule, the ex ante expected utility from choosing $\mathbf{d}$ at time t is

$$W_{\mathbf{d},t}^c \equiv \mathbb{E}[\mathcal{U}_{\mathbf{d},t}(\beta)] = \mathcal{A} \frac{w_{j,n_m,t}^M}{P_{j,t}} \left[(1+r_{\mathbf{d},t}) F_\beta(\beta_{\mathbf{d},t}^*) + \int_{\beta_{\mathbf{d},t}^*}^{\infty} \beta^\gamma f_\beta(\beta) d\beta \right]$$

$$= \frac{w_{j,n_m,t}^M}{P_{j,t}} \cdot \underbrace{\mathcal{A} \left[(1 + r_{d,t}) + \chi^\kappa \frac{\gamma}{\kappa - \gamma} (1 + r_{d,t})^{1-\kappa/\gamma} \right]}_{B_{d,t}},$$

where I can express $B_{d,t}$ entirely in terms of $\rho_{d,t}$ as $B_{d,t} = \mathcal{A} \chi^\gamma (1 - \rho_{d,t})^{-\gamma/\kappa} \frac{\kappa/\gamma - \rho_{d,t}}{\kappa/\gamma - 1}$. This term captures the quality-adjusted value of destination-sector bundle $\mathbf{d}$ for couples. Conditional on choosing $\mathbf{d}$, the couple's payoff combines the husband's sector-specific wage with the expected gain from the wife's endogenous labor-supply decision, summarized by $B_{d,t}$. Finiteness of the expected value requires $\kappa > \gamma$.

Comparative statics of the market-work probability. The probability that the wife works in the market in Equation 22 depends only on the relative wage r , the parameters of the Pareto distribution, χ and κ , and the public-good expenditure share, γ . It does not depend on the level of either spouse's wage.

Participation rises with the wife's wage relative to her husband's:

$$\frac{\partial \rho}{\partial r} = \frac{\kappa}{\gamma} \cdot \frac{1 - \rho}{1 + r} > 0. \quad (23)$$

A higher relative wage raises the opportunity cost of staying home. Thus, the threshold β^* rises, and some draws that used to justify staying home no longer do, so more wives work.

Participation falls with the lower bound of home productivity, χ , and rises with the shape parameter, κ :

$$\frac{\partial \rho}{\partial \chi} = -\frac{\kappa}{\chi} (1 - \rho) < 0, \quad \frac{\partial \rho}{\partial \kappa} = -\frac{(1 - \rho) \log(1 - \rho)}{\kappa} > 0. \quad (24)$$

Raising the lower bound χ scales the whole distribution of home productivity upward while leaving the threshold unchanged, so every couple gains more when she stays at home. More couples draw above β^* , so participation falls. Because χ is a scale parameter, the effect is proportional: the elasticity of $1 - \rho$ with respect to χ is κ . By contrast, a larger shape parameter concentrates the draws near χ and thins the right tail. Staying home takes an exceptionally good draw, one above β^* , and those draws become rarer, so more wives work.

Participation falls with the public-good share, γ :

$$\frac{\partial \rho}{\partial \gamma} = -\frac{\kappa(1 - \rho)}{\gamma^2} \log(1 + r) < 0. \quad (25)$$

A higher public-good share raises the utility from any given β , since β^γ grows with γ when $\beta > 1$. The threshold $(1 + r)^{1/\gamma}$ therefore falls: a smaller home-productivity draw now matches the income gain from working. More couples choose home production. The effect is proportional to $\log(1 + r)$ because $\log \beta^* = \log(1 + r)/\gamma$, so a change in γ moves the threshold

more when the income gain from working is larger.

Indirect utility. The indirect utility relevant for household h 's dynamic migration problem in Equation 16 is¹⁶

$$W_{o,t}^h = \begin{cases} \frac{w_{i,k,t}^M}{P_{i,t}} & \text{if } h = s_m \\ \frac{w_{i,k,t}^F}{P_{i,t}} & \text{if } h = s_f \\ \frac{B_{o,t} w_{i,k_m,t}^M}{P_{i,t}} & \text{if } h = c \end{cases}$$

Here, $P_{i,t}$ denotes the regional aggregate consumption price index defined in Equation 13.

3.2 Equilibrium

The economic geography at time t , denoted by Γ_t , consists of the time-varying exogenous sectoral productivities and the time-invariant shares of single individuals born in each region, migration costs, iceberg trade costs, and non-employment benefits: $\Gamma_t = \{A_{i,k,t}, \bar{s}_i, \mu_{od}, \tau_{ik,jn}, b_j\}_{i,j=1, k,n=1}^{L,K}$. I collect the remaining time-invariant parameters in the vector $\bar{\Gamma} = \{N, L, K, \delta, \chi, \kappa, \alpha, \gamma, \sigma, \nu, \vartheta, \{\psi_{k,m}, \psi_{k,f}, \eta_k, \iota_k\}_{k=1}^K\}$.

Let $N_{d,t}^{sm}$ and $N_{d,t}^{sf}$ denote, respectively, the masses of single men and single women living in location j and working in sector $n \in \{0, \dots, K\}$ at time t . Let $N_{d,t}^c$ denote the mass of couples living in location j in which the husband is assigned to sector $n_m \in \{0, \dots, K\}$ and the wife is assigned to productive market sector $n_f \in \{1, \dots, K\}$. The supplies of male and female labor to productive sectors $n = 1, \dots, K$ are therefore

$$M_{j,n,t} = N_{j,n,t}^{sm} + \sum_{n_f=1}^K N_{j,n,n_f,t}^c, \quad F_{j,n,t} = N_{j,n,t}^{sf} + \sum_{n_m=0}^K \rho_{j,n_m,n,t} N_{j,n_m,n,t}^c. \quad (26)$$

These expressions make the components of a couple's bundle explicit. The husband's sector is the second argument, and the wife's sector is the third. Accordingly, male labor supply in sector n sums over the wife's sector, whereas female labor supply in sector n sums over the husband's sector and weights each cell by the wife's probability of participation. Thus, female labor supply responds to wages through both mobility and participation, whereas male labor supply responds only through mobility.

Let $N_t = \{N_{d,t}^{sm}, N_{d,t}^{sf}, N_{d,t}^c\}$ denote the complete distribution of household masses across locations and sectoral bundles at time t . Because the problems of intermediate-good

¹⁶Singles have the same preferences over private consumption and the household good as spouses, $\gamma \log Q + (1-\gamma) \log c$, but do not share Q with a spouse. Their indirect utility is therefore $\log(w/P)$ up to the constant $\gamma \log \gamma + (1-\gamma) \log(1-\gamma)$, which cancels from all welfare changes. This formulation assumes that welfare is comparable across marital statuses; the money-metric index below makes this comparison explicit.

firms, final-good firms, and married women's labor supply are static, I define the *temporary equilibrium* as follows.

Definition 1. Given $(N_t, \Gamma_t, \bar{\Gamma})$, a **temporary equilibrium** is a vector of productive-sector wages $\{w_{j,n,t}^M, w_{j,n,t}^F\}$, sectoral prices $\{p_{j,n,t}, P_{j,n,t}\}$, regional price indices $\{P_{j,t}\}$, and allocations $\{\rho_{d,t}, M_{j,n,t}, F_{j,n,t}, Y_{j,n,t}, Q_{j,n,t}, Q_{j,t}, \pi_{ik,jn,t}\}$ such that, for every productive region-sector pair jn and period t :

- (i) Intermediate-good firms maximize profits subject to the CES production technology, and wages satisfy (11);
- (ii) Final-good firms maximize profits subject to sectoral CES and regional Cobb-Douglas aggregation. Sectoral price indices satisfy (12), and regional price indices satisfy (13);
- (iii) Couples optimally choose the wife's labor supply, and participation probabilities satisfy (22) for every sector pair (n_m, n_f) ;
- (iv) Labor markets clear: $p_{j,n,t} Y_{j,n,t} = w_{j,n,t}^M M_{j,n,t} + w_{j,n,t}^F F_{j,n,t}$;
- (v) Goods markets clear under balanced trade: $p_{i,k,t} Y_{i,k,t} = \sum_{j=1}^L \sum_{n=1}^K \pi_{ik,jn,t} \iota_n P_{j,t} Q_{j,t}$.

Given the path of exogenous fundamentals and an initial distribution of households across location-sector bundles, denoted by $N_0 = \{\{N_{o,0}^{sm}\}, \{N_{o,0}^{sf}\}, \{N_{o,0}^c\}\}$, I can define the model's *sequential competitive equilibrium*. Using the two-stage choice probabilities in Equations 17 and 18, the population of each household type evolves according to

$$N_{d,t+1}^h = \sum_{o \in \mathcal{S}_h} \lambda_{od|j,t}^h \lambda_{oj,t}^h N_{o,t}^h, \quad d \in \mathcal{S}_h(j), \quad \forall h. \quad (27)$$

Definition 2. Given $(N_0, \Gamma_t, \bar{\Gamma})$, a **sequential competitive equilibrium** is a sequence of productive-sector wages $\{w_{j,n,t}^M, w_{j,n,t}^F\}_{t=0}^\infty$, prices $\{p_{j,n,t}, P_{j,n,t}, P_{j,t}\}_{t=0}^\infty$, values $\{V_{o,t}^h\}_{t=0}^\infty$, and allocations $\{\rho_{d,t}, \pi_{ik,jn,t}, \lambda_{oj,t}^h, \lambda_{od|j,t}^h, N_{o,t}^h, M_{j,n,t}, F_{j,n,t}, Y_{j,n,t}, Q_{j,n,t}, Q_{j,t}\}_{t=0}^\infty$ for every household type h , such that, in each period t :

- (i) The temporary equilibrium holds;
- (ii) Couples and single individuals solve their respective migration problems: value functions satisfy (16), while location and conditional sectoral choice probabilities satisfy (17) and (18);
- (iii) The population of each household type evolves according to (27);
- (iv) Productive-sector male and female labor supplies satisfy (26).

3.3 Welfare Analysis of a Sectoral Productivity Shock

To quantify welfare, I combine Equation 16 with $\mu_{oo} = 0$ to express the expected lifetime utility of household type h in bundle $\mathbf{o}$ at time t as

$$\begin{aligned} V_{\mathbf{o},t}^h &= \log(W_{\mathbf{o},t}^h) + \delta V_{\mathbf{o},t+1}^h - \vartheta \log(\lambda_{\mathbf{o}i,t}^h) - v \log(\lambda_{\mathbf{o}\mathbf{o}|i,t}^h) \\ &= \sum_{s=t}^{\infty} \delta^{s-t} \log \left(\frac{W_{\mathbf{o},s}^h}{(\lambda_{\mathbf{o}i,s}^h)^{\vartheta} (\lambda_{\mathbf{o}\mathbf{o}|i,s}^h)^v} \right). \end{aligned} \quad (28)$$

Here, $\lambda_{\mathbf{o}i,t}^h$ denotes the probability of remaining in the origin location, and $\lambda_{\mathbf{o}\mathbf{o}|i,t}^h$ denotes the probability of remaining in the origin sector bundle, conditional on remaining in the origin location. Appendix B.1 derives this recursion.

I define the no-shock economy by an alternative sequence of sectoral productivities $\{A_{j,n,t}\}_{t \geq 1}$. Following CDP's timing assumption, the change is unanticipated at $t = 0$, and households fully learn the new productivity path at $t = 1$. Let $\hat{x}_{t+1} = \dot{x}'_{t+1}/\dot{x}_{t+1}$ denote the ratio of changes over time in the no-shock and with-shock equilibria, where $\dot{x}_{t+1} \equiv x_{t+1}/x_t$, and let $\tilde{x}_t \equiv x'_t/x_t = \prod_{s=1}^t \hat{x}_s$ denote the corresponding ratio of levels, under the normalization that the value of x is the same in both economies in $t = 0$, such that $x'_0 = x_0$.

I use Dynamic Hat Algebra¹⁷ to compute expected lifetime utility in the with-shock economy, $V_{\mathbf{o},0}^h$, and in the no-shock economy, $V_{\mathbf{o},0}'^h$. I define the consumption compensating variation for household type h in bundle $\mathbf{o}$ at time $t = 0$, denoted by $\zeta_{\mathbf{o}}^h$, implicitly by

$$V_{\mathbf{o},0}'^h = V_{\mathbf{o},0}^h + \sum_{s=0}^{\infty} \delta^s \log(\zeta_{\mathbf{o}}^h) \iff \log(\zeta_{\mathbf{o}}^h) = (1 - \delta)(V_{\mathbf{o},0}'^h - V_{\mathbf{o},0}^h).$$

Because the temporary equilibrium and migration shares are identical across the two economies at $t = 0$, the welfare change in consumption-equivalent units is

$$Z_{\mathbf{o}}^h \equiv \log(\zeta_{\mathbf{o}}^h) = \sum_{s=1}^{\infty} \delta^s \log \left(\frac{\hat{W}_{\mathbf{o},s}^h}{(\hat{\lambda}_{\mathbf{o}i,s}^h)^{\vartheta} (\hat{\lambda}_{\mathbf{o}\mathbf{o}|i,s}^h)^v} \right), \quad (29)$$

where

$$\hat{W}_{\mathbf{o},s}^h = \begin{cases} \hat{w}_{i,k_m,s}^M / \hat{P}_{i,s} & \text{if } \mathbf{o} = (i, k_m), h = s_m, \\ \hat{w}_{i,k_f,s}^F / \hat{P}_{i,s} & \text{if } \mathbf{o} = (i, k_f), h = s_f, \\ \hat{B}_{\mathbf{o},s} \hat{w}_{i,k_m,s}^M / \hat{P}_{i,s} & \text{if } \mathbf{o} = (i, k_m, k_f), h = c. \end{cases}$$

Equation 29 decomposes the welfare change into changes in real income, which for couples

¹⁷See Appendix C for more details.

include the change in the value of the wife's participation option, $\hat{B}_{o,s}$, and changes in the option value of moving, captured by the staying probabilities. Conditional on the observed staying shares, a larger ϑ , corresponding to a lower migration elasticity, increases the welfare weight assigned to changes in the probability of remaining in the origin location. Similarly, a larger v increases the weight assigned to changes in the probability of remaining in the origin sector bundle conditional on staying in that location. Thus, declines in these probabilities enter the welfare measure with weights ϑ and v , respectively, where $\vartheta \geq v$.

I aggregate welfare changes across initial location-sector bundles to obtain the aggregate welfare effect of a sectoral productivity shock:

$$Z = \underbrace{\sum_{i=1}^L \sum_{k=0}^K N_{o,0}^{s_m} Z_o^{s_m}}_{Z^{sm}} + \underbrace{\sum_{i=1}^L \sum_{k=0}^K N_{o,0}^{s_f} Z_o^{s_f}}_{Z^{sf}} + \underbrace{\sum_{i=1}^L \sum_{k_m=0}^K \sum_{k_f=1}^K N_{o,0}^c Z_o^c}_{Z^c}. \quad (30)$$

Money Metric Welfare Index. The aggregate in Equation 30 adds welfare changes measured in utility units, which are not directly comparable between singles and spouses. For couples, \$1 of market expenditure can generate more than \$1 in consumption-equivalent value due to economies of scale within marriage. Thus, a 1% gain for a spouse and a 1% gain for a single person are percentages of different bases. Following Chiappori and Meghir (2015), I construct a Money Metric Welfare Index (MMWI) for singles and couples to obtain a comparable money-metric measure of welfare changes for the quantitative analysis.

The MMWI measures each person's welfare as the income they would need to attain their utility level if they had to purchase the household public good at its full market price. For a single, this income is the real wage, w/P , because a single already pays the full price of the public good. Thus, the MMWI for a spouse can be understood as the income they would need to attain the utility level they get as married if they were single.

Constructing the MMWI for a spouse requires three intermediate objects. I first define the Conditional Sharing Rule (CSR), denoted by ζ , which allocates the income remaining after expenditure on the public good between the spouses. Let α_g represent the bargaining weight of the spouse of gender g , such that $\alpha_f = \alpha$ and $\alpha_m = 1 - \alpha$. Given the functional form of the couple's utility, the CSR corresponds to a fixed share of income after paying for the public good, where the share reflects each spouse's bargaining weight, so that $\zeta_g = \alpha_g(1 - \gamma)I(e)$. Then, I define the Lindahl prices, which correspond to the share of the cost of one unit of Q , $P_Q \equiv P\beta^{-e}$, paid by each spouse. Because both spouses value Q identically, each bears a share of its cost equal to their bargaining weight, $P_g = \alpha_g P_Q$. Next, I combine the CSR and Lindahl prices to compute the Generalized Sharing Rule (GSR), which adds to each spouse's private expenditure the value of the public good at their Lindahl price,

$\zeta_g^* = \tilde{\zeta}_g + P_g Q^* = \alpha_g I(e)$. Thus, a wife who remains at home does not earn a wage, but is allocated resources equivalent to αw^M , a share of her husband's wage.

The MMWI of each spouse rescales their GSR by the ratio of the public good's market price to their Lindahl price, raised to the power γ . Let $\bar{P}_Q = P$ denote the market price of the public good, that is, its price without the boost from home production. Then, the MMWI of spouse g , denoted by m_g , corresponds to

$$m_g = \underbrace{\zeta_g^*}_{\alpha_g I(e)} \cdot \underbrace{\bar{P}_Q^\gamma}_{P^\gamma} \cdot \underbrace{P_g^{-\gamma}}_{(\alpha_g P \beta^{-e})^{-\gamma}} = \alpha_g^{1-\gamma} \beta^{e\gamma} I(e). \quad (31)$$

Because couples choose their bundle before β is realized, I weight welfare changes below by the ex ante index, $\bar{m}_g = \alpha_g^{1-\gamma} \mathbb{E}[\beta^{e\gamma} I(e)]$, which moves with the wife's participation probability.

Summing across spouses, $m_f + m_m = (\alpha^{1-\gamma} + (1-\alpha)^{1-\gamma}) \beta^{e\gamma} I(e)$, where $\alpha^{1-\gamma} + (1-\alpha)^{1-\gamma} > 1$. Thus, the spouses would jointly require more income than the household possesses if each had to purchase the public good separately, and $\alpha^{1-\gamma} + (1-\alpha)^{1-\gamma}$ measures the economies of scale within marriage in money terms. Hence, I now measure welfare for couples in the same units as wages and prices, so I can compare it with that of singles.

I aggregate welfare changes across household types by weighting each person's consumption-equivalent change by their money metric in the initial period, $\bar{Z}^s = \bar{Z} + \text{Cov}(m, Z)/\bar{m}$, where $\bar{Z}$ is the unweighted per-person average and the covariance is taken across individuals. To a first-order approximation, mZ is the income that compensates a person for the shock, so $\bar{Z}^s$ is the aggregate money gain as a share of aggregate money-metric income. The covariance term is positive when the gains accrue to individuals with larger money metrics, such as spouses, and negative when they accrue to those with smaller money metrics, such as singles. Because each spouse's expected utility is a constant multiple of the couple's (Equation B.1), both spouses inherit the couple's consumption-equivalent change, Z_o^c ; only their weights differ. Appendix B.2 derives the index step by step.

The model's mechanisms map directly onto the three empirical facts. Sectoral productivity shocks transmitted through trade lower wages in exposed region-sector cells and lead workers to leave them, consistent with Fact 1. Because men and women are distributed differently across sectors, these wage changes differ by gender. Couples jointly choose a single location, but spouses enter this decision asymmetrically: the husband's wage $w_{j,n_m,t}^M$ scales the couple's entire payoff, whereas the wife's wage enters only through the relative wage $r_{d,t}$ in $B_{d,t}$. This asymmetry provides the mechanism underlying Fact 2. Finally, the participation threshold $\beta_{d,t}^*$ generates the adjustment margin in Fact 3, with the direction observed in the data: a decline in the husband's sectoral wage raises $r_{d,t}$, increases the threshold,

and induces more married women to enter market work, whereas a decline in the wife’s own sectoral wage lowers $r_{d,t}$ and has the opposite effect. Section 4 disciplines the model using the data.

4 Bringing the Model to the Data

This section applies the model developed above to the data to quantify the China Trade Shock in an economy with heterogeneous households. It first describes the quantitative environment, distinguishing parameters drawn from the literature from inputs observed directly in the data. It then presents the calibration of preference dispersion across sectors and locations, as well as the couple-specific parameters, followed by the method for quantifying the China Trade Shock.

4.1 Quantitative Environment

The quantitative environment follows CDP. The economy comprises 87 locations: the 50 US states, with the District of Columbia included in Virginia, and 37 foreign trade partners, consisting of China, the 35 largest US trading partners, and the Rest of the World. It includes 22 productive sectors, comprising 12 manufacturing sectors, retail, construction, and eight service sectors, as well as a non-employment sector. Each model period corresponds to one quarter, and I solve the model for 200 quarters beginning in 2000Q1. I choose to adopt CDP’s quantitative environment because it provides a trade block that has already been disciplined by the data, ensuring that the productivity and trade fundamentals are not constructed specifically for this analysis. One caveat is the difference in timing: the empirical analysis in Section 2 follows ADH’s 2001–2012 window (Autor et al., 2021), whereas the quantitative exercise retains the earlier 2000–2007 window (Autor et al., 2013).

Because the trade side is taken directly from CDP, the quantitative exercise uses their full production block rather than the simplified version presented in Section 3. Intermediate-good firms use both labor and intermediate inputs from other sectors, and total factor productivity combines the Hicks-neutral component introduced in Section 3 with a variety-specific component. Implementing the household extension therefore requires replacing the labor input demanded by intermediate-good firms with the effective labor aggregate in Equation 8. I take all other inputs to the trade block directly from CDP.¹⁸

¹⁸Sectoral trade elasticities $\{\eta_n\}$ are from Caliendo and Parro (2015); bilateral trade shares $\{\pi_{ik,jn,t}\}$, region-sector output and expenditure, and the path of sectoral productivity changes in China are from CDP; final-good expenditure shares $\{\iota_n\}$ come from the Bureau of Economic Analysis (BEA) input-output tables.

δ and σ from previous literature. I set the discount factor to $\delta = 0.99$, following CDP, and take the within-sector elasticity of substitution between male and female labor, σ , from Acemoglu et al. (2004) rather than estimating it directly. This choice is motivated by the identification requirements implied by the CES technology in Section 3:

$$\log \left(\frac{w_{i,k,t}^M}{w_{i,k,t}^F} \right) = \log \left(\frac{\psi_{k,m}}{\psi_{k,f}} \right) - \frac{1}{\sigma} \log \left(\frac{M_{i,k,t}}{F_{i,k,t}} \right).$$

Identifying σ requires exogenous variation in the relative supply of male and female labor; without such variation, the estimated coefficient would be biased. Acemoglu et al. (2004) estimate an analogous relationship (their Equation 11), using the mobilization of men during World War II as an exogenous shift in women’s relative labor supply. Accordingly, I set σ to 1.4, the lower bound of their estimated range. This conservative choice reflects the fact that the cohorts examined in their study were almost entirely retired by 2000Q1, when the quantitative model begins. Since then, as more women entered the labor force, they specialized in occupations where they had a comparative advantage, so men and women sort differently across occupations and are less substitutable within a sector (Ngai and Petrongolo, 2017; Cortes et al., 2023; Bhalotra et al., 2025).

Directly observed inputs. I construct household-side inputs from US microdata. I measure the initial distribution of households across states and sector pairs, $\{N_{o,0}^{s,m}, N_{o,0}^{s,f}, N_{o,0}^c\}$,¹⁹ and the state-level share of singles, $\bar{s}_i$, using the 5% IPUMS Census 2000. I measure male and female employment, wages, and wives’ labor-force participation using the 2000 Census and the American Community Survey (ACS) for 2001–2007. I recover the sector-specific gender-intensity parameters $\{\psi_{k,m}, \psi_{k,f}\}$ from the hours worked by gender in each sector. I measure regional non-employment income, b_j , using state unemployment-insurance benefits from the US Department of Labor and hold it fixed at its 2000 value. Table 4 summarizes these inputs and their sources.²⁰

One discrepancy between the model and the data remains: the model assigns every wife to a productive sector, but the data do not reveal the potential sector of a non-employed wife. In the model, couples first choose the wife’s productive sector and then determine

¹⁹The quantitative exercise holds household composition fixed: marriage, divorce, fertility, and population growth are excluded, so couples remain together throughout the transition. Figure A.4 provides supporting evidence for this assumption.

²⁰Two conventions determine the interpretation of the participation margin. First, non-employment refers to being *out of the labor force*, rather than unemployed. Including unemployed individuals would roughly double the measured flows into non-employment and would confound the wife’s participation decision with the husband’s job loss. Second, following the convention in this literature, an unemployed worker retains the sector of their most recent job. Thus, a positive sector index does not necessarily indicate current employment, and I always measure participation using an explicit labor-force indicator.

Table 4: Observed Quantitative Inputs

Object	Description	Source
$\bar{s}_i$	State-level share of singles	IPUMS Census 2000
N_i	State-level population	IPUMS Census 2000 and ACS 2001–2007
$N_{o,0}^{s,m}, N_{o,0}^{s,f}$	Initial populations of single men and women by state $\times$ sector	IPUMS Census/ACS
$N_{o,0}^c$	Initial population of couples by state $\times$ husband-wife sector pair	IPUMS Census/ACS
$M_{o,t}, F_{o,t}$	Male and female employment by state $\times$ sector	IPUMS Census/ACS
$\psi_{k,m}, \psi_{k,f}$	Sector-specific gender intensities	Hours worked by gender, IPUMS Census/ACS
b_j	State-level non-employment in- come	US Department of Labor unemployment-insurance bene- fits, fixed at the 2000 value

Notes. The sample follows CDP and includes individuals aged 25–65 across 50 states, excluding widows. Non-employment refers to being out of the labor force rather than being unemployed. Sectors are classified according to CDP’s 22-sector partition.

her participation status. A naive approach would assign non-employed wives to productive sectors in proportion to the sectoral distribution of employed wives. This approach creates a selection problem: sectors employing larger shares of wives would also receive larger shares of non-employed wives, even though wives assigned to those sectors are precisely the least likely to be out of the labor force. To recover each non-employed wife’s potential sector, I use the model to derive a selection correction.²¹ The implied mass of non-employed married women in state j and sector n is

$$NonF_{j,n,t}^c = \underbrace{F_{j,n,t}^c}_{\text{observed}} \cdot \underbrace{\frac{1 - \rho_{j,n,t}}{\rho_{j,n,t}}}_{\text{the model's odds of non-employment}}.$$

Scaling observed employment by $1/\rho_{j,n,t}$ corrects for selection and assigns non-employed wives to their potential productive sectors.

The final data inputs are the initial migration matrices. Because I solve the model using Dynamic Hat Algebra, the time-invariant mobility costs μ_{od} cancel from the recursion

²¹See Appendix D.1 for details.

governing the choice probabilities (Proposition C.2). The recursion therefore requires only two initial objects for each household type: the location shares $\lambda_{oj,0}^h$ defined in Equation 17 and the within-location shares $\lambda_{od|j,0}^h$ defined in Equation 18.

Following CDP, I construct the initial unconditional transition probability $\lambda_{od,0}^h$ from Equation 19 as the product of an interstate migration probability from the ACS and a sectoral transition probability from the Current Population Survey (CPS). For interstate movers, I impute the sectoral transition probability using the destination state’s sectoral transition matrix:

$$\lambda_{(i,k)(j,n),0}^h = \underbrace{f_{ij}^h}_{\text{ACS}} \times \underbrace{P_j^h(n | k)}_{\text{CPS}}.$$

This construction presents two challenges: the surveys differ in frequency, and CPS cells become sparse once split by household type. First, the ACS is annual, whereas the CPS is monthly. I address this mismatch following CDP. Second, to address the sparse cells, I temporarily aggregate the 12 manufacturing sectors into a single sector, and do the same for the service sectors. I then disaggregate the resulting transition probabilities using national gender-specific sorting patterns. Appendix D.2 documents each step in the construction of the migration matrices.

4.2 Dispersion of Preferences across Sectors and Locations

The model features two dispersion parameters: v , taken from the literature, and ϑ , which I calibrate. The parameter v governs substitution across sectors within a location (Equation 18), whereas ϑ governs substitution across locations (Equation 17).

CDP’s dispersion parameter maps directly onto v , because it is estimated only from sectoral transitions within a location. In fact, following Artuc et al. (2010), they estimate their dispersion parameter using sectoral transitions among men in the March Current Population Survey (CPS) from 1975 to 2000. Because the CPS follows addresses, respondents who change residence are removed from the sample. The matched panel therefore records sectoral transitions only among individuals who remain in the same location and contains no transitions across locations. Consequently, the estimated parameter corresponds to applying Artuc et al. (2010)’s method to the within-location choice in Equation 18, which is precisely the role of v in the present model. The only distinction is that continuation values V^h are household-type specific, reflecting the economies of scale associated with marriage. Because the model does not impose differential rates of sectoral substitution across household types, I set v to CDP’s quarterly value of 5.34 for all household types.

I calibrate ϑ by indirect inference to match the population response to import competition

in Autor et al. (2013), following Rodríguez-Clare et al. (2026). Autor et al. (2013) regress the change in the log of the working-age population of a commuting zone on its exposure to Chinese imports and obtain a coefficient of -0.050 . I estimate the same regression using model-generated state-level data. The dependent variable is the percentage change in state population between 2000 and 2007, scaled to a decade as in Autor et al. (2013). The regressor is the state’s exposure to Chinese imports over 2000–2007, predicted using the Autor et al. (2013) instrument and rescaled to the mean of their commuting-zone measure so that the model and empirical coefficients have the same units.²² The calibrated value, $\vartheta = 10.67$, generates a coefficient of -0.051 , compared with the target of -0.050 . Because $v < \vartheta$, households substitute more readily across sectors within a location than across locations.

Accounting for couples lowers the migration elasticity needed to match the same population response. Calibrating ϑ in an otherwise identical economy composed entirely of single individuals, in which each individual independently chooses a location and sector, yields a migration elasticity that is 5% higher under the same target and value of v . Given the model’s solution timing, this difference provides a lower bound on the quantitative importance of accounting for couples.²³

4.3 Calibrating the Couple’s Maximization Problem

Four parameters govern the couple’s maximization problem and have no counterpart in CDP because CDP’s economy has no couples. They are the wife’s weight in the household objective α , the public-good expenditure share γ , and the lower bound and shape of the home-production shock, χ and κ . Because I solve the couple’s problem following the three-stage formulation in Chiappori and Mazzocco (2017), I identify them *sequentially*, each from a distinct piece of variation: α and γ from household expenditure shares, the slope κ/γ from the China Trade Shock, and χ from the level of married women’s participation. Table 5 summarizes the resulting values; Appendix D.3 describes the construction in detail.

I calibrate α and γ to exactly match observed expenditure shares. Because the couple’s utility function in Equation 20 is Cobb–Douglas, α corresponds to the wife’s share of total private consumption,²⁴ while γ corresponds to the share of public-good expenditure in total

²²I restrict the model sample to the 48 states in the Autor et al. (2013) estimation sample, excluding Alaska and Hawaii. Using first-stage fitted exposure makes the ordinary least squares coefficient in the model the counterpart of the two-stage least squares coefficient in the data.

²³In this quantification, the with-shock economy is almost entirely data over the shock period: I replace its first 29 quarters, starting in 2000Q1, with observed data. Thus, differences in migration elasticities arise almost entirely through the no-shock economy. If I solved both the with-shock and no-shock economies from $t = 0$, the difference in migration elasticities between the household and the single-agent model would be larger.

²⁴The quantification holds α fixed, thereby ruling out shock-induced renegotiation of intrahousehold bar-

household expenditure:

$$\alpha = \frac{c_f^*}{c_f^* + c_m^*}, \quad \gamma = \frac{q^*}{q^* + c_f^* + c_m^*}.$$

I measure both expenditure shares using the 2000–2007 Consumer Expenditure Interview Survey. I measure the public-good share γ directly as the share of total expenditure devoted to market inputs into home production.²⁵ Estimating the bargaining weight α requires an additional step because the observed assignable private good, adult apparel, represents only a subset of private consumption. I therefore recover the spouses’ resource shares using the method of Browning et al. (2013), which relies on expenditures by *singles*. The resulting values are $\alpha = 0.47$ and $\gamma = 0.49$.²⁶

I calibrate κ and χ using the wife’s participation rule in Equation 22, which is log-linear in the relative wage. Expressing the equation in terms of the non-participation probability and taking logs yields

$$\log(1 - \rho(r)) = \underbrace{\kappa \log \chi}_{\text{intercept}} - \underbrace{(\kappa/\gamma)}_{\text{slope}} \log(1 + r). \quad (32)$$

Direct estimation of Equation 32 is complicated by endogeneity and sample selection. First, the relative wage r faced by a couple is an equilibrium outcome. Second, I observe relative wage changes only for couples in which the wife participates in market work, creating a truncation problem.

To address the endogeneity of r , I use the empirical results in Section 2 to provide exogenous variation through the China Trade Shock. When the husband is exposed, the shock directly affects his wage but not his wife’s wage, influencing her participation decision through the relative wage r , as required by the exclusion restriction underlying Equation 32. The same shock simulated in the quantitative exercise therefore disciplines the model’s participation margin. To address truncation, the change in $1 + r$ is evaluated at the relative wage of all couples, not only those in which the wife works (Appendix D.3.2).

Because Equation 32 is linear, I identify its slope using the following Wald ratio:

$$\frac{\kappa}{\gamma} = - \frac{\Delta \log(1 - \rho)}{\Delta \log(1 + r)},$$

gaining power. It also abstracts from the intensive margin of the wife’s labor supply: the wife either participates in market work or does not. The latter restriction appears empirically innocuous because, alongside the participation responses documented in Fact 3, wives’ hours exhibit no statistically significant response to either spouse’s exposure. These estimates are available upon request.

²⁵The home-production goods comprise shelter, utilities, food at home, and market substitutes for home time, such as domestic services and appliance repair.

²⁶Appendix D compares these estimates with those in the previous literature and discusses potential sources of discrepancy.

where I measure both changes within the same empirical experiment. Wives of exposed husbands increase their labor force participation by 4.2 percentage points from a baseline of 0.78 (Panel b of Table 3), implying $\Delta \log(1 - \rho) = -0.212$. The husband's exposure increases the relative wage r by 12.2 log points, corresponding to a 4.8 log-point increase in $1 + r$ (Table D.1). Together, these estimates imply

$$\frac{\kappa}{\gamma} = -\frac{-0.212}{0.048} = 4.44 \quad \implies \quad \kappa = \gamma \times \frac{\kappa}{\gamma} = 2.17.$$

The resulting calibration satisfies the model's theoretical restriction, $\kappa > \gamma$.

Conditional on the estimated slope, I calibrate χ to match the *level* of married women's labor force participation. Specifically, I choose χ so that the model's mass-weighted average participation rate equals its empirical counterpart in $t_0 = 2000Q1$. The resulting value, $\chi = 1.92$, satisfies the support restriction $\chi \geq 1$ and yields an interior participation margin in every cell. The calibration therefore uses two participation rates, each for a different purpose: I identify the slope from the PSID experiment using that sample's baseline participation rate of 0.78, whereas χ anchors the level to the model population's mass-weighted participation rate of 0.621.

The calibrated parameters imply that married women's labor force participation responds roughly one-for-one to the female wage. Differentiating $\rho = 1 - \chi^\kappa(1 + r)^{-\kappa/\gamma}$ with respect to the female wage and evaluating the resulting expression at the model's anchor, $\bar{\rho} = 0.621$ and $r_0 = 0.607$, yields

$$e = \frac{\partial \log \rho}{\partial \log w_f} = \frac{\kappa}{\gamma} \cdot \frac{1 - \bar{\rho}}{\bar{\rho}} \cdot \frac{r_0}{1 + r_0} = 1.02$$

Thus, a 1% increase in the female wage raises married women's labor force participation by 1.02%. This elasticity lies within the range for married women surveyed by Blundell and MaCurdy (1999) and exceeds the ≈ 0.25 – 0.30 aggregate extensive-margin elasticity preferred by Chetty et al. (2011), which is not specific to married women.

Table 5: Calibrated Household Parameters

Object	Name	Value
α	Wife's weight in the household objective	0.47
γ	Public-good expenditure share	0.49
κ/γ	Slope of the participation rule	4.44
κ	Pareto shape parameter for home productivity	2.17
χ	Pareto lower bound for home productivity	1.92

Notes. See Appendix D for details.

4.4 Quantifying the China Trade Shock

To quantify the effects of the China Trade Shock, I need to define the shock within the model. Following CDP, I represent the China Trade Shock as an increase in total factor productivity in *Chinese* manufacturing sectors and calibrate the path of Chinese sectoral productivity so that the resulting growth in US imports from China reproduces the first stage in Autor et al. (2013).²⁷ The model therefore defines the shock at the industry level, consistent with the reduced-form evidence in Section 2.

I quantify the shock by comparing the two economies defined for productivity shocks in Section 3.3. As in CDP, the *with-shock* economy includes the observed growth in Chinese productivity, whereas the *no-shock* economy eliminates this growth. Formally, $\dot{A}'_{\text{China},k,t} = 1$ for every manufacturing sector k and period t in the no-shock economy.

All results reported below are differences between the with-shock and no-shock economies. Accordingly, a positive employment effect indicates that the China Trade Shock increased employment relative to a world without the shock, while a positive welfare effect indicates that the relevant household type was better off.

5 Quantitative Consequences of the Shock Under Joint Decision-Making

In this section, I quantify how the China Trade Shock affected each household type and how much it matters to abstract away from couples. First, I examine the effects of the China Trade Shock on manufacturing employment, total employment, population changes, participation responses, and welfare. Second, I compare the baseline economy with a single-agent economy in which every couple is split into two singles. Third, I perform three counterfactual exercises to quantify the importance of each margin of adjustment by shutting down, one by one, the response of migration, participation, and the gender wage gap.

5.1 Quantitative Effects of the China Trade Shock

The China Trade Shock, as expected by the previous reduced-form evidence, reduces aggregate manufacturing employment and lowers it for every group (Panel A of Figure 3). The decline observed over the first 15 years does not reverse in the long run (Table 6). There are important gender differences in the manufacturing decline: women’s decline is about half

²⁷The shock is a foreign supply shock that is transmitted through the trade block as a demand shock to US producers: cheaper Chinese imports reduce both the prices and quantities at which US manufacturers sell in domestic and third-country markets.

that of men’s and varies little by marital status. Married men experience the largest manufacturing decline, -0.41 p.p. by $t = 199$, compared with -0.29 p.p. for single men. When manufacturing decline is measured as manufacturing jobs per capita, a metric comparable to CDP, both quantification exercises give similar declines (-0.21 vs -0.23 p.p.).²⁸

Rather than pushing workers into non-employment, the shock raises the employment rate of every group, with a particularly large increase for single women (Panel B of Figure 3). The reason is that the shock is a boom for workers. The increase in productivity in Chinese manufactured goods implies a cheaper consumption basket in the US, thereby reducing the regional price index. The relative real wage differential between working in productive sectors and non-employment then increases, incentivizing single women and all men to work more. In fact, services grow by more than manufacturing shrinks ($+0.25$ vs -0.21 jobs per capita; Panel C of Figure 3).

Married men experience the largest manufacturing losses yet gain the least in employment: because most of them are already working, most of their adjustment comes from reallocation across sectors rather than exiting the workforce. For married women, the employment gain is the added-worker effect, because her participation is the couple’s decision: the shock hits male-intensive manufacturing, raising $r = w_f/w_m$ and drawing wives into the labor force, as shown in Equation 23. Wives respond the most when only their husband starts in manufacturing (Table 7).

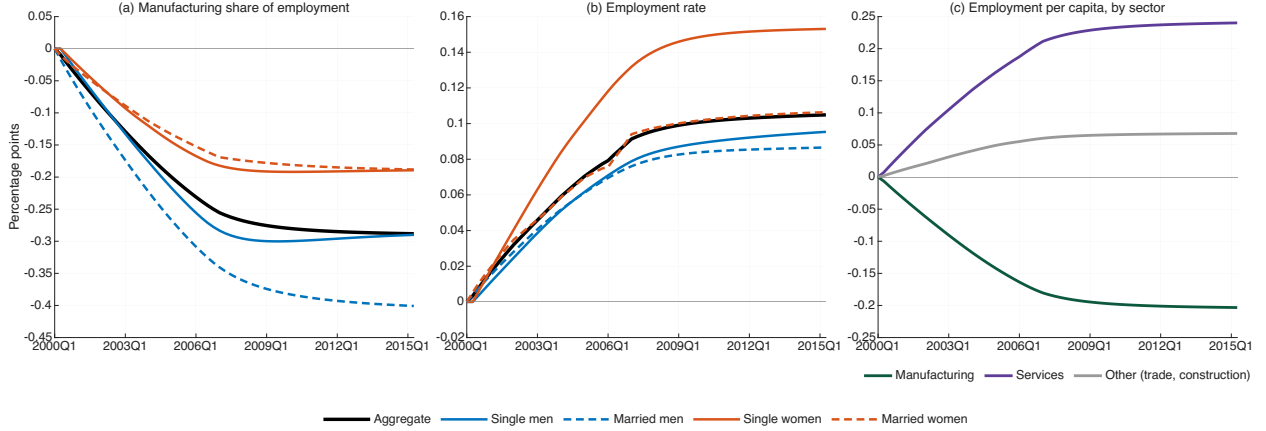
In Figure 4, I compare the long-run decline in manufacturing employment, the change in population, and the change in married women’s labor force participation across states, and find that the states hit hardest by the shock tend to lose population and to see larger increases in married women’s participation. Across states, the correlation between population change and manufacturing share loss is -0.58 (Panel A), indicating that the states hit hardest by the shock tend to be also the ones that lose population. I map the three outcomes in Figure E.2.

Married women’s participation increases in every state, and it tends to increase the most where manufacturing falls the most. Across states, the correlation between manufacturing share loss and married women’s participation response is $+0.51$ (Panel B): wives go to work where their husbands’ jobs went. At the same time, the correlation between participation response and population change is -0.44 , implying that wives work more in states where people leave.

Lower prices, together with the increase in the employment rate, out-migration, and

²⁸This decline is also almost the same in every economy I study below: in the single-agent economy and in the three counterfactual exercises, manufacturing jobs per capita fall within 2.6% of the baseline (Table E.1). Couples redistribute the shock, but they do not change the extent to which manufacturing employment falls.

Figure 3: Employment Effects of the China Trade Shock



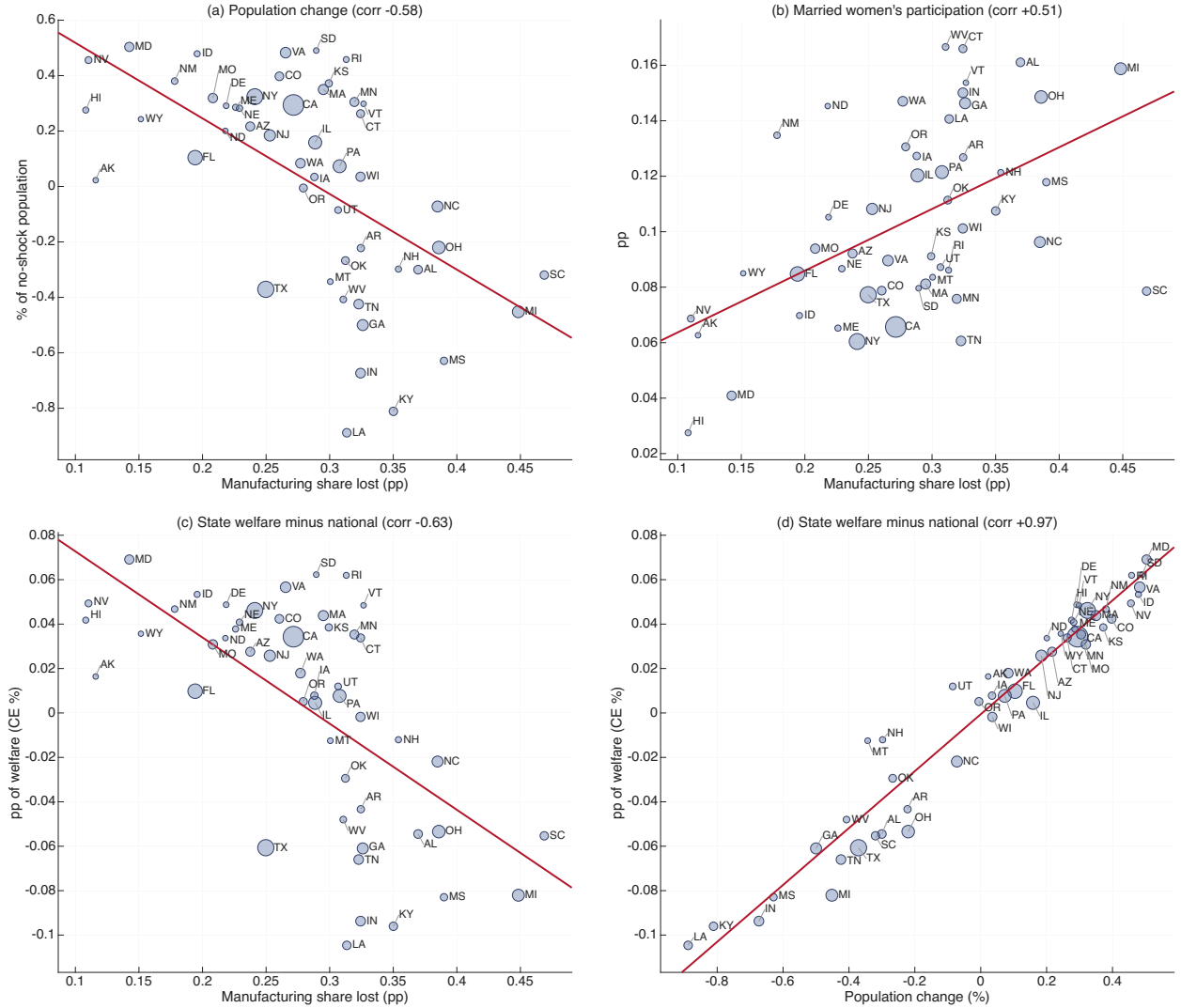
Notes. Responses to the shock (with-shock minus no-shock economy) in the baseline economy over the first 62 quarters (2000Q1–2015Q2). Panel A is the manufacturing share of employment, in p.p. of employment; Panel B is the employment rate, in p.p. of population; Panel C is employment per capita by sector, in p.p. of population, where manufacturing covers the 12 manufacturing sectors and other covers wholesale and retail trade and construction. Long-run values ($t = 199$) are reported in Table 6.

participation responses, translate into welfare gains for all groups, with the average individual in the US experiencing a 0.18% increase in welfare, as measured by the Money Metric Welfare Index (MMWI).²⁹ These gains, however, differ across household types (Table 6): single women gain the most, at 0.23, compared with single men at 0.16 and couples at 0.17; that is, single women gain 33.9% more than the average individual, and single men 10.1% less. Single women are concentrated in expanding sectors, are underrepresented in the 12 manufacturing sectors affected by the shock, and do not face joint location decisions. Couples fall in between the two single groups, but their gains are much closer to those of single men. The reason is that a couple's indirect utility is $\frac{Bw^M}{P}$, so the husband's wage drives the household's gain, while the wife's contribution operates only through her participation.

Within couples, the welfare gain decreases monotonically with exposure and is ordered by the husband's exposure (Table 7). When neither spouse starts in manufacturing, couples have the highest welfare gain (0.18), followed by couples in which only the wife does (10.3% less), then couples in which only the husband does (22.1% less), and finally couples in which both do (30.4% less). Since his wage enters utility directly and hers only through the expected value of her productivity, his exposure reduces household income, whereas hers affects the

²⁹One potential concern would be that incorporating couples also changes welfare in the rest of the world. In Figure E.1, I compare the welfare gains for the other 36 regions (35 countries and the rest of the world; China excluded) in my model and in CDP. The two series correlate at 0.997, and the same 30 of 36 regions gain in both; the largest difference is in the UK, which gains 0.039 p.p. less in my model. Welfare in the rest of the world is also the same in every economy I study below (Table E.1).

Figure 4: The China Trade Shock across States



Notes. Each dot is a state, sized by its population in 2000Q1. All outcomes are long-run responses ($t = 199$) in the baseline economy. Manufacturing share lost is the decline in manufacturing's share of the state's employment, in p.p.; population change is the change in the state's population as a percentage of its no-shock level; married women's participation is the change in their participation rate, in p.p., averaged over the couples living in the state; state welfare minus national is the state's money-metric welfare gain minus the national gain (0.175), in p.p.. Lines are OLS fits; correlations are unweighted across the 50 states.

option value of her working.

In Figure E.3, I compare welfare gains across states and find that the geography of gains is nearly identical across household types. The states that gain the least are those that lose the most population: Louisiana, Kentucky, Indiana, and Mississippi are the four states furthest below their national average and the four that lose the most population. Across states,

Table 6: Long-Run Effects of the China Trade Shock by Household Type

	Aggregate	Single men	Single women	Married men	Married women
Population share (%)	100.0	11.7	15.0	36.7	36.7
Manuf. share of employment (pp)	-0.293	-0.287	-0.188	-0.412	-0.192
Employment rate (pp)	0.112	0.106	0.158	0.096	0.111
Migration response (pp)	+0.0167	+0.0206	+0.0115	+0.0259	+0.0053
Welfare (% CE)	0.175	0.157	0.234	0.170	0.170
Welfare relative to agg. (%)	—	-10.1	+33.9	-2.9	-2.9

Notes. Responses at $t = 199$ (with-shock minus no-shock economy), except population shares (2000Q1) and welfare. The manufacturing share is in p.p. of employment, and the employment rate is in p.p. of population. The migration response is the difference between the shares of manufacturing and non-manufacturing workers who have changed state at least once by $t = 199$, in p.p.; individuals are tagged by their sector at $t = 1$, married men by the husband's sector, and married women by the wife's (potential) sector. Welfare is the money-metric consumption-equivalent gain, weighting each person by their money metric at 2000Q1: the wage for a single person and $\alpha_g^{1-\gamma} B w_m$ for a spouse. Married men and women carry the couple's welfare. Welfare relative to aggregate is $100 \times (G_g/G - 1)$.

Table 7: Effects of the China Trade Shock on Couples by Exposure

	Share of couples (%)	Participation response (pp)	Migration response (pp)	Welfare (% CE)	Welfare relative to Neither (%)
Both in manuf.	3.1	0.103	+0.0078	0.125	-30.4
Husband only	13.7	0.105	+0.0141	0.140	-22.1
Wife only	6.6	0.101	-0.0062	0.161	-10.3
Neither	76.6	0.100	-0.0109	0.180	—
All couples	100.0	0.101	-0.0066	0.170	-5.5

Notes. Couples are tagged by both spouses' sectors at $t = 1$; a non-working wife is tagged by her potential sector. The wife's participation response is the change in the quarterly participation rate at $t = 199$, in p.p., for the couples that started in each cell. The couple's migration response is the change in the share of couples in the cell that have changed state at least once by $t = 199$, measured in p.p.. Welfare is the money-metric consumption-equivalent gain of both spouses. Welfare relative to Neither is $100 \times (G_{\text{cell}}/G_{\text{Neither}} - 1)$.

welfare relative to the national gain correlates at +0.97 with the population change, and at -0.63 with the manufacturing share lost (Panels C and D of Figure 4). The states that gain most are those with the least manufacturing to lose. Couples experience the greatest spatial

inequality in welfare, with a coefficient of variation of 0.31, compared with 0.23 for single men and 0.16 for single women. This reflects the joint-location friction: a couple cannot take one job in one state and the other in another.

5.2 Couples versus a Single-Agent Economy

To quantify the importance of abstracting away from couples, I compare the previous results with the effect of the shock in a counterfactual world in which everyone is single, which I call the *single-agent economy*.

To build the single-agent economy, I split every couple into two singles, reconstruct all inputs from the data, and recalibrate the migration elasticity to match the same moment as in the baseline economy. I then solve the model again under this structure. In Tables 8 and 9, I compare the manufacturing decline, employment rate, and welfare effects in both economies; all percentage changes are relative to the baseline economy. The aggregate decline in manufacturing employment is nearly identical in both economies (a difference of 0.9%), so both economies face the same shock.

Even though the manufacturing decline is the same, the single-agent economy generates 18.2% larger employment gains than the baseline economy (Table 8), primarily because women’s employment responds 36.5% more. This difference arises because female labor supply is 50.7% more elastic in the single-agent economy (0.682 vs 0.452). A wife’s labor supply decision depends on the relative wage $r = w_f/w_m$, rather than on her own wage alone, so her employment response is dampened whenever her husband’s wage moves with hers.

Although the two economies generate similar aggregate welfare gains, the single-agent economy’s is 2.1% larger; they differ in how those gains are distributed. In the single-agent economy, married women gain 13.7% more than in the baseline economy, whereas married men gain 3.3% less (Table 9). Because a wife’s decision to participate depends on her wage relative to her husband’s, she responds to the shock partly through his exposure and cannot fully capture the expansion in her own sector; moreover, both spouses share a single household-level gain.

Couples also redistribute gains among singles, who make no joint decisions: in the single-agent economy, single women gain 17.0% less and single men 5.7% more than in the baseline economy. Because female employment responds 36.5% more in the single-agent economy, wages absorb less of the shock here. Specifically, in the single-agent economy, the female wage response is 10.7% smaller, the male wage response is 9.4% larger, and the relative wage rises by 38.6% less. In the baseline economy, single women therefore supply labor to a tighter

Table 8: Employment Effects in the Baseline and in the Single-Agent Economy

	Baseline	Single-agent	% change vs baseline
<i>Panel A. Manufacturing share of employment (pp)</i>			
Aggregate	−0.293	−0.291	−0.9
Men	−0.383	−0.380	−0.9
Women	−0.191	−0.186	−2.2
<i>Panel B. Employment rate (pp)</i>			
Aggregate	0.112	0.132	+18.2
Men	0.099	0.092	−6.5
Women	0.124	0.170	+36.5
<i>Panel C. Mechanism</i>			
Female labor-supply elasticity	0.452	0.682	+50.7
Female wage response (%)	0.405	0.362	−10.7
Male wage response (%)	0.236	0.258	+9.4
Relative wage r response (%)	0.169	0.104	−38.6

Notes. Responses at $t = 199$. Panel A is in p.p. of employment and Panel B in p.p. of population. % change vs baseline is $100 \times (\text{single-agent/baseline} - 1)$. In Panel C, wage responses are employment-weighted log changes in real wages ($\times 100$); the female labor-supply elasticity is the log female employment response over the log female real-wage response; the relative wage response is the female minus the male wage response.

market, while single men supply labor to a weaker one.

Joint decision-making leaves the geography of welfare gains almost unchanged but makes it more unequal. The spatial distribution of welfare gains is nearly identical across the two economies (Figure E.4), with a cross-state correlation of +0.96. However, welfare gains are 26.8% less dispersed in the single-agent economy (Table 9): the coefficient of variation is 0.28 in the baseline economy, compared with 0.20 in the single-agent economy. The baseline economy generates the largest relative losses in states that lose population, including Louisiana, Ohio, Texas, and Georgia.

Taken together, the results indicate that incorporating couples changes *who* gains from the China Trade Shock far more than it changes *how much* the economy gains. In both the baseline and the single-agent economies, the shock is a boom: manufacturing's share of employment falls by about 0.29 p.p., workers are drawn from non-employment into the expanding service sectors, and aggregate welfare rises by almost the same amount in both. The two economies also agree on the geography, with state welfare correlated at +0.96, but

Table 9: Welfare Effects in the Baseline and in the Single-Agent Economy

	Baseline	Single-agent	% change vs baseline
<i>Panel A. Welfare (% CE)</i>			
Aggregate	0.175	0.178	+2.1
Single men	0.157	0.166	+5.7
Single women	0.234	0.194	-17.0
Married men	0.170	0.164	-3.3
Married women	0.170	0.193	+13.7
<i>Panel B. Spatial</i>			
CV of state welfare	0.278	0.204	-26.8
Cross-state correlation of welfare		0.961	

Notes. Money-metric consumption-equivalent gains (%). Each row is the same people in both economies. The single-agent economy dissolves every couple into two singles, so its married men and women are the baseline’s spouses and its single men and women are the baseline’s singles; both economies weight each person by their baseline money metric at 2000Q1. Weighting the single-agent economy by its own wages instead, its aggregate gain is 0.177 (+1.2%). In Panel B, CV is the population-weighted coefficient of variation of state welfare gains in each economy, and the correlation is across the 50 states.

they disagree on the incidence. Because the shock falls on male-intensive manufacturing, it raises the relative wage $r = w_f/w_m$, incentivizing wives to enter the labor force, most where manufacturing contracted most (a cross-state correlation of +0.51), and most in couples where only the husband is exposed.

Through general equilibrium effects, this added-worker response creates a redistribution between single men and single women. Given that married women respond only to relative wages, female labor supply is less elastic than in the single-agent economy (0.452 vs 0.682), and the female wage therefore does more of the adjusting. Single women sell into a scarcer female labor market and gain more than they would in a world without couples; single men gain less.

5.3 Counterfactual Exercises

In this subsection, I perform three counterfactual exercises to quantify the importance of each margin of adjustment, shutting down, one at a time, the migration, participation, and gender-wage responses. In the first one, I set $1/\vartheta \rightarrow 0$ to shut down migration responses. In the second one, I set $\kappa/\gamma \rightarrow 1$ and recalibrate χ accordingly to dampen participation

responses. In the last one, I set $1/\sigma \rightarrow 0$ to reduce gender-wage responses. In Table E.2, I summarize the exact value used in each exercise. Each exercise is a full re-solve of the model with the corresponding parameter changed, while all other parameters (except χ in the second exercise) are held at their baseline values. I report the three exercises side by side in Tables 10 and 11, where each effect is a percentage change relative to its value in the baseline economy, except the spatial statistics in Panel B of Table 11, which are levels.

Table 10: Effects of the China Trade Shock under the Three Counterfactual Exercises

	Baseline (level)	No migration	No participation % change vs baseline	No gender-wage
<i>Panel A. Employment, $t = 199$ (pp)</i>				
Manufacturing share of employment				
Aggregate	-0.293	-2.2	-2.9	+0.7
Men	-0.383	-2.5	+2.9	+4.7
Women	-0.191	-1.7	-16.3	-5.0
Employment rate				
Aggregate	0.112	-11.7	-24.3	-13.3
Men	0.099	-19.6	-8.9	+28.1
Women	0.124	-5.8	-35.7	-44.1
<i>Panel B. Married women's participation response, $t = 199$ (pp)</i>				
All couples	0.101	-3.4	-59.2	-63.3
Both in manufacturing	0.103	-3.0	-59.5	-62.5
Husband only	0.105	-2.8	-59.6	-61.4
Wife only	0.101	-3.4	-59.3	-63.6
Neither	0.100	-3.5	-59.2	-63.7
<i>Panel C. Migration response, manufacturing – other, ever moved by $t = 199$ (pp)</i>				
Single men	+0.0206	—	+1.8	-2.1
Single women	+0.0115	—	-16.0	-3.3
Couples, by husband's sector	+0.0259	—	+3.7	-3.3
Couples, by wife's sector	+0.0053	—	+6.7	-5.1

Notes. The first column reports the baseline economy's response; the other columns report $100 \times$ (exercise/baseline – 1). A dash marks a margin that is shut down. Panel A: responses at $t = 199$, in p.p. of employment (manufacturing share) and of population (employment rate). Panel B: married women's quarterly participation rate at $t = 199$, by the couple's exposure at $t = 1$. Panel C: difference between the shares of manufacturing and non-manufacturing workers who have changed state at least once by $t = 199$; couples are tagged by each spouse's sector at $t = 1$, and a non-working wife by her potential sector.

Table 11: Welfare Effects of the China Trade Shock under the Three Counterfactual Exercises

	Baseline (level)	No migration	No participation % change vs baseline	No gender-wage
<i>Panel A. Welfare (% CE)</i>				
Aggregate	0.175	+1.0	−4.6	+5.8
Single men	0.157	+2.9	−2.4	+18.8
Single women	0.234	+0.9	+8.8	−10.4
Couples	0.170	+0.7	−7.2	+7.1
Both in manufacturing	0.125	−2.7	−7.6	+9.4
Husband only	0.140	−3.4	−10.2	+8.4
Wife only	0.161	+0.7	−5.0	+7.5
Neither	0.180	+1.6	−6.9	+6.7
<i>Panel B. Spatial (levels)</i>				
CV of state welfare	0.278	0.448	0.289	0.261
Corr.	—	0.984	−0.254	0.009

Notes. The first column reports the baseline economy’s response; in Panel A, the other columns report $100 \times (\text{exercise/baseline} - 1)$, while in Panel B they report levels. A dash marks a margin that is shut down. Panel A: money-metric consumption-equivalent gains, weighting everyone by their baseline money metric at 2000Q1 in every economy; in the second exercise, recalibrating χ and κ roughly doubles B , so on its own money metrics, its aggregate gain falls by 6.9%. Panel B: CV is the population-weighted coefficient of variation of state welfare gains; the correlation is between the change in state welfare, in p.p., and the state’s population change in the baseline economy.

5.3.1 Counterfactual 1: No migration response ($\vartheta \rightarrow \infty$)

Because $1/\vartheta$ is the migration elasticity in the model, setting it to zero shuts down migration responses to the shock. Tables 10 and 11 compare employment effects, the participation response of married women, and welfare effects in the baseline and in the counterfactual economies.³⁰ In both cases, the decline in manufacturing employment is similar, but the employment effect is 11.7% smaller in the counterfactual economy, 19.6% smaller for men and 5.8% smaller for women. Given that migration across state lines is not allowed, individuals cannot access better employment opportunities elsewhere.

Participation does not compensate for the lack of spatial mobility: the participation response of married women is 3.4% smaller in the counterfactual economy. Although this result may seem counterintuitive, it is driven by how couples decide on her participation.

³⁰Table E.3 provides the results of counterfactual 1 in detail.

Given that ρ is homogeneous of degree zero in wages, it depends only on $r = w_f/w_m$ and not on the level, and this counterfactual moves both spouses' wages by nearly the same amount, leaving r intact to the third decimal. The participation response falls the least for couples in which the husband is exposed (-2.8% , -3.0%) and the most for unexposed couples (-3.5%).

Shutting down migration barely moves aggregate welfare ($+1.0\%$; Table 11), even though the employment response is 11.7% smaller. Moving is how people reach jobs, not how they capture gains, and since sector switching is still allowed, a displaced worker still has their primary adjustment margin. For couples, the near-null difference hides heterogeneity by exposure. Shutting the margin costs couples in which the husband starts in manufacturing 2.7% and 3.4% of their gains (both spouses and husband only, respectively), and leaves wife-only and unexposed couples *better* off by 0.7% and 1.6% . In fact, unexposed couples benefit because they do not have to bear the costs of mobility decisions made by exposed households. When the exposed relocate, they disproportionately move to areas where the unexposed already live; their arrival increases local labor supply and pushes local wages down. With migration shut down, nobody arrives, and the unexposed keep a tighter local market. Because unexposed couples are most of the population, their small gain accounts for almost all of the aggregate change (Figure E.5).

5.3.2 Counterfactual 2: No participation response ($\kappa/\gamma \rightarrow 1$)

To dampen the participation response, I lower κ/γ , which scales the response of the participation probability to $\ln(1+r)$:

$$\frac{\partial \rho}{\partial \ln(1+r)} = \frac{\kappa}{\gamma}(1-\rho),$$

so, once the participation *level* is held fixed, κ/γ is the only free multiplier on the response. The model requires $\kappa > \gamma$, so the most the response can be damped is $\kappa_0/\gamma = 4.44\times$; at $\kappa = 0.65$, the response is damped $3.39\times$. In equilibrium, married women's employment response falls by a factor of 2.59 . I also recalibrate χ to keep the anchor participation rate unchanged.

Damping participation has the largest employment effect of the three counterfactual exercises (Table 10): while the manufacturing decline is only 3% smaller in the counterfactual, the employment response is 24.3% smaller. If wives cannot respond to the shock by increasing their participation, the overall employment effect is lower: the participation response of married women falls by 59.2% , and their manufacturing decline is 33.9% smaller, while single women's is 20.9% larger (Table E.4).

While in the previous counterfactual couples did not substitute participation for migra-

tion, in this counterfactual they substitute migration for participation. The difference in migration rates between couples in manufacturing and other sectors increases by 3.7% for the husband’s sector and by 6.7% for the wife’s sector (Panel C of Table 10).³¹ The substitution therefore runs one way only: couples answer a damped participation margin by moving more, but they cannot answer a closed spatial margin by working more. Exposed single women, by contrast, are less likely to migrate in the counterfactual economy because married women cannot enter the market, allowing single women to take full advantage of higher wages in booming sectors.

In terms of welfare gains (Table 11), the aggregate gain falls by 4.6% while single women gain 8.8% more and couples’ gains fall by 7.2%. It is the only one of the three exercises that *widens* the single-women-minus-single-men welfare gap. It works through the same mechanism as the single-agent comparison, pushed further. Damping κ/γ takes married women further out of the labor market; female labor gets scarcer still, and the single women who are free to respond capture the wage.

Among couples, the losses support the added-worker mechanism: they are largest when only the husband is exposed (-10.2%), because a wife whose husband is exposed and who is not herself exposed has room to respond, and damping the response is what takes it away. When she is in manufacturing too, her own wage is falling, and there is less of a response to dampen. Even unexposed couples gain 6.9% less, unlike in the previous counterfactual. The added-worker margin is insurance every couple holds; the spatial margin is insurance only the exposed use.

5.3.3 Counterfactual 3: No gender-wage response ($\sigma \rightarrow \infty$)

The third exercise holds the gender wage ratio nearly fixed. Men and women enter production through a CES nest with elasticity σ , and the wage ratio moves as $\hat{r} = (\hat{M}/\hat{F})^{1/\sigma}$. If $1/\sigma \rightarrow 0$, $\hat{r} \rightarrow 1$, thus shutting down responses through the relative wage. Because the limit cannot be reached numerically, I set $1/\sigma = 0.01$ against a calibrated 0.714, so 1.4% of the gender-wage response survives.

Holding the gender wage ratio fixed shifts the manufacturing decline from women onto men (Table 10). The aggregate decline in manufacturing’s share is only 0.7% larger, while underneath it, men’s declines deepen (single men +6.0%, married men +4.4%) and women’s become shallower, single women -8.5% , married women -3.3% (Table E.5). Since wages can no longer do the adjusting, quantities have to. The shock hits male-intensive manufacturing

³¹Given the results in the empirical analysis, I focus on migration rate differentials. The individual levels are available upon request. Even if, in this exercise, the increase in the migration response differential when she is exposed is almost twice as large as when he is exposed, in the baseline economy couples leave their states 4.9 times more when he is exposed, consistent with the evidence in Fact 2.

and relative male labor falls, so in the baseline economy $r = w_f/w_m$ rises and part of the burden is absorbed by the relative wage and by the wives it pulls in. At $\sigma = 100$, the relative wage barely moves, so the same contraction has to show up as who leaves manufacturing. Holding r fixed also shuts down most of the participation response, which falls by 63.3%, thereby reducing the employment effect for women (-44.1%) and increasing it for men ($+28.1\%$). Finally, the migration response differential of exposed households falls across all groups (Panel C of Table 10).

In terms of welfare (Table 11), the aggregate gain rises by 5.8%; the gains of single men increase by 18.8%, and those of single women decline by 10.4%, in the same direction as the employment effects. In fact, this is the only counterfactual in which single women are worse off than in the baseline economy. The gendered incidence of this shock is carried almost entirely by $r = w_f/w_m$; when this channel is shut down, single women are no longer able to take advantage of the shock. By contrast, all couples gain more than in the baseline economy, and the more exposed ones gain the most ($+9.4\%$ when both spouses start in manufacturing, compared with $+6.7\%$ when neither does), because removing the relative wage movement helps the couples who were absorbing most of it.

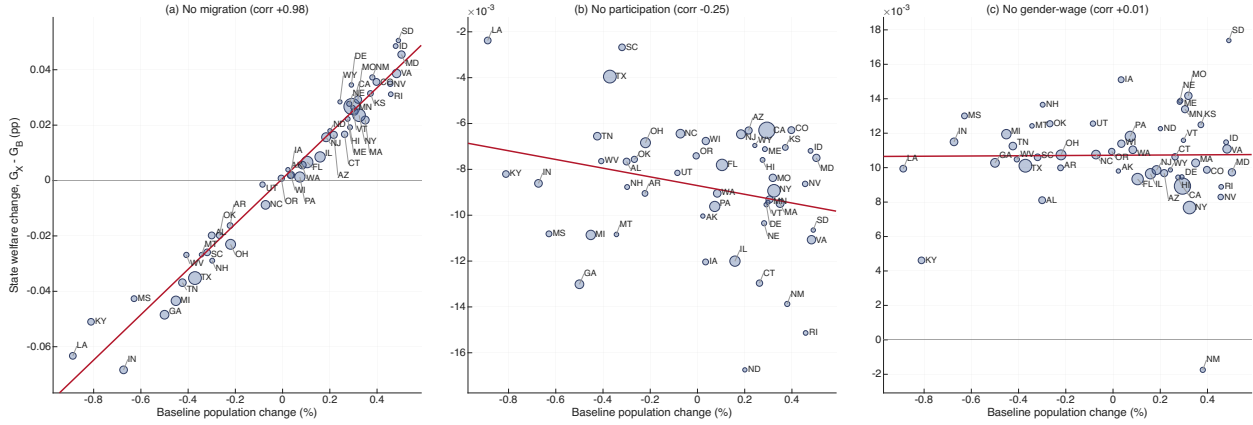
5.3.4 The Three Exercises Together

Looking at the three exercises side by side by household type in Panel A of Table 11 makes the contrast between the two margins of adjustment clear: shutting down migration costs only the couples in which the husband starts in manufacturing, whereas damping participation costs all types of couples. Because unexposed couples are most of the population, they also drive the aggregate change in every exercise (Figure E.5): they account for most of the $+1.0\%$, the -4.6% , and the $+5.8\%$.

The three exercises also affect states very differently: shutting down migration redistributes welfare across states, whereas damping participation and holding the gender wage ratio fixed move every state in the same direction. In Figure 5, I plot the change in each state's welfare relative to the baseline economy, in p.p., against the state's population change in the baseline economy. None of the three exercises changes where manufacturing falls: the largest state-level difference in manufacturing share lost is 0.036 p.p., compared with an average state decline of 0.28 p.p.; thus, they do not move where the shock lands; they move what it costs to be there.

Shutting the spatial response moves aggregate welfare by $+1.0\%$ (Table 11), but this near-null hides large spatial effects: welfare becomes 61% more spatially unequal (the coefficient of variation of state welfare gains rises from 0.28 to 0.45; Panel B of Table 11), and the correlation between a state's welfare change in this exercise and its population response in

Figure 5: State Welfare under the Three Counterfactual Exercises



Notes. Each dot is a state, sized by its population in 2000Q1. The vertical axis is the change in the state's money-metric welfare gain relative to the baseline economy, in p.p.; the horizontal axis is the state's population change in the baseline economy, as a percentage of its no-shock level. Each panel has its own vertical scale. Lines are OLS fits; correlations are unweighted across the 50 states.

the baseline is +0.98 (Figure 5). Migration redistributes gains across states: the losers are the exposed states that were losing population, led by Indiana, Louisiana, and Kentucky; the winners are the destinations that were gaining it, which no longer have their gains diluted by arrivals. Counterfactuals 2 and 3 are nearly uniform by comparison, but in opposite directions, and their state changes are unrelated to population change (correlations of -0.25 and 0.01). Damping participation not only reduces aggregate welfare gains by 4.6% (Table 11), but it also makes every state worse off. Flattening the gender wage response increases aggregate welfare gains by 5.8% (Table 11) and makes every state better off except New Mexico.

Taken together, the margin shut-downs rank the mechanisms, but the ranking depends on whether one looks at the size of the gain or at its distribution. In terms of distribution, migration is the dominant margin: shutting the spatial response moves aggregate welfare by only 1%, yet it lowers the employment response by 12% and makes state welfare 61% more unequal across states. Within couples, the loss falls on those whose husbands start in manufacturing, so mobility is insurance only for the exposed. However, given that the participation rule depends only on the ratio of the spouses' wages, and closing the spatial margin moves both wages by almost the same amount, participation does not rise to fill the gap.

In terms of size, damping participation is the one experiment that lowers the aggregate gain (-4.6%) and cuts the employment response the most (-24%). It hurts every couple, including unexposed ones, which shows that the added-worker margin is insurance that every

couple holds, not just the exposed. The substitution runs one way only: when participation is damped, couples move more, but when migration is shut, they do not work more.

Finally, the gender-wage channel accounts for nearly all of the gendered incidence: single men gain 18.8% more, single women 10.4% less, and the manufacturing decline shifts from women to men.

6 Conclusion

Workers migrate less in response to local shocks than theory predicts, and the conventional explanation is that moving is costly. This paper argues that the explanation also lies in how spatial models of local labor markets represent households: they typically assume that each worker decides for themselves where to live, yet most adults live in couples. Couples must weigh two labor-market exposures against each other, which makes them less likely to move, but they also have an insurance margin that singles lack: when one spouse's earnings fall, the other can work more. I show that accounting for couples changes who gains from local shocks far more than how much the economy gains.

Using PSID data to compare individuals who were and were not exposed to the China Trade Shock within the same commuting zone, I document three facts. First, exposed individuals experienced worse labor-market outcomes and were more likely to leave their commuting zones. Second, couples responded asymmetrically: out-migration increased when the husband was exposed but did not increase when the wife was exposed. Third, wives' labor supply exhibited a similar asymmetry, rising when the husband was exposed—an added-worker effect—and falling sharply when the wife herself was exposed. Thus, couples mitigate the shock through both spatial and intra-household insurance, and which margin they use depends on which spouse is hit.

To quantify what joint decision-making implies for the economy as a whole, I extend Caliendo et al. (2019) to incorporate heterogeneous households: single men, single women, and couples. Couples jointly choose a location and a pair of sectors, with substitution across sectors and locations occurring at different rates, as in Rodríguez-Clare et al. (2026). After choosing where to live and work, they decide whether the wife works in the market or at home. The couple's problem therefore introduces two margins that do not exist for singles and that operate in opposite directions: joint location choice makes moving more difficult, whereas endogenous labor-force participation makes staying more bearable. When I split every couple into two singles and calibrate to the same population response, the resulting economy implies a higher migration elasticity: ignoring couples makes individuals appear more mobile than they are.

The quantitative exercise shows that the trade shock is, overall, a boom for US workers: cheaper imports lower the price index, employment rates rise across all household types, and the average individual gains 0.18% in consumption-equivalent terms. Manufacturing's share of employment falls by 0.29 p.p., and individuals adjust through both migration and increased labor-force participation. The states experiencing the largest manufacturing-employment losses also lose the most population and exhibit the largest increases in married women's labor supply. Moreover, participation increases most in the states with the largest population losses.

Accounting for couples changes who gains from the shock far more than how much the economy gains. Single women gain the most, single men the least, and couples fall in between, with gains ordered by the husband's exposure, because his earnings are the household's main source of income. When I compare these results with those from a single-agent economy in which every couple is split into two singles, the two economies agree on the decline in manufacturing employment, the aggregate welfare gain, and its geography, but they differ in how women's labor supply responds. A wife decides whether to work together with her husband, and the shock that raises her wage also raises his, making the household less reliant on her earnings. As a result, fewer women enter the labor market, and female wages bear more of the adjustment. Joint decisions therefore shape the gains of singles, who make no joint decisions at all: in the single-agent economy, single women would gain 17.0% less, and single men 5.7% more. Joint decision-making also makes welfare gains more unequal across states, since spouses cannot take jobs in different states.

For couples, spatial and intra-household insurance are asymmetric substitutes. When I dampen the participation response, couples move more; when I shut down migration, however, wives do not work more. Wives step in when their husbands' earnings fall relative to their own, but when migration is shut down, both spouses' wages fall in similar proportions, so wives have no new reason to work. The two margins also insure different households. Participation provides insurance to every couple: dampening this response reduces the gains of all couples, including those with no direct exposure, and lowers aggregate welfare by 4.6%. Migration, by contrast, insures only exposed households and matters most for how the gains are distributed. Shutting it down has little effect on aggregate welfare, but it makes welfare across states 61% more unequal, with the losses concentrated among couples whose husbands initially work in manufacturing.

The policy implication is that aggregate gains, and even their geography, provide an incomplete picture of who benefits from the shock. Trade adjustment assistance is designed around individuals, targeting workers or industries that are exposed to trade shocks. Yet the binding mobility constraint operates at the household level, and a transfer that replaces

a displaced husband's earnings does not restore the option his family has lost. Policies that reduce the cost of moving jointly or increase the destination's value to the spouse who is not driving the move would help alleviate this constraint. Moreover, because couples change how women's labor supply responds, the shock's incidence reaches single workers who were never employed in an exposed industry.

The direction of the gender asymmetry in my results is a feature of *this* shock rather than a general property of the model. Because the China Trade Shock affects manufacturing, a male-intensive sector, it exposes husbands more than wives and draws wives into the labor market. Indeed, the gender wage gap accounts for nearly all of the shock's gendered incidence: holding women's wages nearly fixed relative to men's eliminates about 63% of wives' participation response and closes most of the gap between single women's and single men's gains. A shock to a female-intensive sector would reverse the asymmetry, although its welfare consequences need not be a mirror image. Thus, this paper opens a broader research agenda on how the economy adjusts to local shocks and who bears them when the household, not only the worker, decides where to live and work.

References

- Acemoglu, D., Autor, D., Dorn, D., Hanson, G., and Price, B. (2016). Import Competition and the Great US Employment Sag of the 2000s. *Journal of Labor Economics*, 34(S1):S141–S198.
- Acemoglu, D., Autor, D., and Lyle, D. (2004). Women, War, and Wages: The Effect of Female Labor Supply on the Wage Structure at Midcentury. *Journal of Political Economy*, 112(3):497–551.
- Allen, T. and Arkolakis, C. (2025). Chapter 1- Quantitative Regional Economics. In Donaldson, D. and Redding, S. J., editors, *Handbook of Regional and Urban Economics*, volume 6 of *Handbook of Regional and Urban Economics*, pages 1–72. Elsevier.
- Allen, T. and Donaldson, D. (2020). Persistence and Path Dependence in the Spatial Economy. Working Paper 28059, National Bureau of Economic Research.
- Allen, T. and Donaldson, D. (2022). Persistence and Path Dependence: A Primer. *Regional Science and Urban Economics*, 94:103724.
- Anderson, L., Washington, C., Gryn, T., and Kreider, R. M. (2024). Households and Families: 2020. Technical Report C2020BR-10, U.S. Census Bureau.
- Artuc, E., Chaudhuri, S., and McLaren, J. (2010). Trade Shocks and Labor Adjustment: A Structural Empirical Approach. *American Economic Review*, 100(3):1008–1045.
- Attanasio, O., Low, H., and Sánchez-Marcos, V. (2005). Female Labor Supply as Insurance Against Idiosyncratic Risk. *Journal of the European Economic Association*, 3(2-3):755–764.
- Autor, D. and Dorn, D. (2013). The Growth of Low-Skill Service Jobs and the Polarization of the US Labor Market. *American Economic Review*, 103(5):1553–1597.
- Autor, D., Dorn, D., and Hanson, G. (2013). The China Syndrome: Local Labor Market Effects of Import Competition in the United States. *American Economic Review*, 103(6):2121–2168.
- Autor, D., Dorn, D., and Hanson, G. (2016). The China Shock: Learning from Labor-Market Adjustment to Large Changes in Trade. *Annual Review of Economics*, 8(1):205–240.
- Autor, D., Dorn, D., and Hanson, G. (2019). When Work Disappears: Manufacturing Decline and the Falling Marriage Market Value of Young Men. *American Economic Review: Insights*, 1(2):161–178.
- Autor, D., Dorn, D., Hanson, G., Jones, M. R., and Setzler, B. (2025). Places Versus People: The Ins and Outs of Labor Market Adjustment to Globalization. In *Handbook of Labor Economics*, volume 6, pages 549–653. Elsevier.

- Autor, D., Dorn, D., Hanson, G., and Majlesi, K. (2020). Importing Political Polarization? The Electoral Consequences of Rising Trade Exposure. *American Economic Review*, 110(10):3139–3183.
- Autor, D., Dorn, D., Hanson, G., and Song, J. (2014). Trade Adjustment: Worker-Level Evidence. *The Quarterly Journal of Economics*, 129(4):1799–1860.
- Autor, D., Dorn, D., and Hanson, G. H. (2021). On the Persistence of the China Shock. Working Paper 29401, National Bureau of Economic Research.
- Becker, G. (1965). A Theory of the Allocation of Time. *The Economic Journal*, 75(299):493–517.
- Benson, A. (2014). Rethinking The Two-Body Problem: The Segregation of Women into Geographically Dispersed Occupations. *Demography*, 51:1619–1639.
- Bhalotra, S., Fernández, M., and Wang, F. (2025). The Distribution of the Gender Wage Gap: An Equilibrium Model. IFS Working Papers W25/39, Institute for Fiscal Studies.
- Blau, F. and Kahn, L. (2017). The Gender Wage Gap: Extent, Trends, and Explanations. *Journal of Economic Literature*, 55(3):789–865.
- Blundell, R. and MaCurdy, T. (1999). Labor Supply: A Review of Alternative Approaches. *Handbook of Labor Economics*, 3:1559–1695.
- Borusyak, K., Dix-Carneiro, R., and Kovak, B. (2026). Understanding Migration Responses to Local Shocks.
- Braun, C., Nusbaum, C., and Rupert, P. (2021). Labor Market Dynamics and the Migration Behavior of Married Couples. *Review of Economic Dynamics*, 42:239–263.
- Browning, M., Chiappori, P.-A., and Lewbel, A. (2013). Estimating Consumption Economies of Scale, Adult Equivalence Scales, and Household Bargaining Power. *The Review of Economic Studies*, 80(4):1267–1303.
- Browning, M., Chiappori, P.-A., and Weiss, Y. (2014). *Economics of the Family*. Cambridge University Press.
- Caliendo, L., Dvorkin, M., and Parro, F. (2019). Trade and Labor Market Dynamics: General Equilibrium Analysis of the China Trade Shock. *Econometrica*, 87(3):741–835.
- Caliendo, L. and Parro, F. (2015). Estimates of the Trade and Welfare Effects of NAFTA. *The Review of Economic Studies*, 82(1):1–44.
- Calvo, P., Lindenlaub, I., and Reynoso, A. (2024). Marriage Market and Labour Market Sorting. *The Review of Economic Studies*, 91(6):3316–3361.
- Chetty, R., Guren, A., Manoli, D., and Weber, A. (2011). Are Micro and Macro Labor Supply Elasticities Consistent? A Review of Evidence on the Intensive and Extensive Margins. *American Economic Review*, 101(3):471–475.

- Chiappori, P.-A. and Mazzocco, M. (2017). Static and Intertemporal Household Decisions. *Journal of Economic Literature*, 55(3):985–1045.
- Chiappori, P.-A. and Meghir, C. (2015). Intrahousehold Inequality. In *Handbook of income distribution*, volume 2, pages 1369–1418. Elsevier.
- Cooke, T., Boyle, P., Couch, K., and Feijten, P. (2009). A Longitudinal Analysis of Family Migration and the Gender Gap in Earnings in the United States and Great Britain. *Demography*, 46(1):147–167.
- Cortes, G. M., Jaimovich, N., and Siu, H. E. (2023). The Growing Importance of Social Tasks in High-Paying Occupations: Implications for Sorting. *Journal of Human Resources*, 58(5):1429–1451.
- Costa, D. and Kahn, M. (2000). Power Couples: Changes in the Locational Choice of the College Educated, 1940–1990. *The Quarterly Journal of Economics*, 115(4):1287–1315.
- Del Boca, D. and Flinn, C. J. (1995). Rationalizing Child-Support Decisions. *American Economic Review*, pages 1241–1262.
- Desmet, K., Nagy, D. K., and Rossi-Hansberg, E. (2018). The Geography of Development. *Journal of Political Economy*, 126(3):903–983.
- Desmet, K. and Parro, F. (2025). Chapter 4 - Spatial Dynamics. In Donaldson, D. and Redding, S. J., editors, *Handbook of Regional and Urban Economics*, volume 6 of *Handbook of Regional and Urban Economics*, pages 225–286. Elsevier.
- Didier, R. and Lopez, P. (2026). Child Support and Parental Labor Supply: Evidence from Ecuador. *Mimeo*.
- Dix-Carneiro, R. and Kovak, B. K. (2017). Trade Liberalization and Regional Dynamics. *American Economic Review*, 107(10):2908–2946.
- Donald, E., Fukui, M., and Miyauchi, Y. (2025). Optimal Dynamic Spatial Policy. Working Paper 34290, National Bureau of Economic Research.
- Faber, M., Sarto, A. P., and Tabellini, M. (2022). Local Shocks and Internal Migration: The Disparate Effects of Robots and Chinese Imports in the US. Working Paper 30048, National Bureau of Economic Research.
- Fadlon, I., Lyngse, F. P., and Nielsen, T. H. (2020). Early Career Setbacks and Women’s Career-Family Trade-Off. Working Paper 28245, National Bureau of Economic Research.
- Flabbi, L. and Mabli, J. (2018). Household Search or Individual Search: Does It Matter? *Journal of Labor Economics*, 36(1):1–46.
- Flinn, C. J. (2000). Modes of Interaction Between Divorced Parents. *International Economic Review*, 41(3):545–578.

- Foerster, H. and Ulbricht, R. (2023). The Colocation Friction: Dual-Earner Job Search and Labor Market Outcomes.
- Greenland, A., Lopresti, J., and McHenry, P. (2019). Import Competition and Internal Migration. *Review of Economics and Statistics*, 101(1):44–59.
- Guler, B., Guvenen, F., and Violante, G. (2012). Joint-Search Theory: New Opportunities and New Frictions. *Journal of Monetary Economics*, 59(4):352–369.
- Imbens, G. W. and Rubin, D. B. (2015). *Causal Inference in Statistics, Social, and Biomedical Sciences*. Cambridge University Press.
- Jayachandran, S., Nassal, L., Notowidigdo, M., Paul, M., Sarsons, H., and Sundberg, E. (2024). Moving to Opportunity, Together. Working Paper 32970, National Bureau of Economic Research.
- Le Barbanchon, T., Rathelot, R., and Roulet, A. (2021). Gender Differences in Job Search: Trading Off Commute Against Wage. *The Quarterly Journal of Economics*, 136(1):381–426.
- Lundberg, S. (1985). The Added Worker Effect. *Journal of Labor Economics*, 3(1, Part 1):11–37.
- Mincer, J. (1978). Family Migration Decisions. *Journal of Political Economy*, 86(5):749–773.
- Monte, F., Redding, S. J., and Rossi-Hansberg, E. (2018). Commuting, Migration, and Local Employment Elasticities. *American Economic Review*, 108(12):3855–3890.
- Ngai, L. R. and Petrongolo, B. (2017). Gender Gaps and the Rise of the Service Economy. *American Economic Journal: Macroeconomics*, 9(4):1–44.
- Parey, M., Irastorza-Fadrique, A., and Levell, P. (2025). Household Responses to Trade Shocks. *Journal of International Economics*, 157:104144.
- Pellegrina, H. S. and Sotelo, S. (2025). Migration, Specialization, and Trade: Evidence from Brazil’s March to the West. *Journal of Political Economy*, 133(12):3993–4049.
- Pierce, J. R. and Schott, P. K. (2016). The Surprisingly Swift Decline of US Manufacturing Employment. *American Economic Review*, 106(7):1632–1662.
- Pilosoph, L. and Wee, S. L. (2021). Household Search and the Marital Wage Premium. *American Economic Journal: Macroeconomics*, 13(4):55–109.
- Redding, S. J. and Rossi-Hansberg, E. (2017). Quantitative Spatial Economics. *Annual Review of Economics*, 9:21–58.
- Reynoso, A. (2024). The Impact of Divorce Laws on the Equilibrium in the Marriage Market. *Journal of Political Economy*, 132(12):4155–4204.

- Rodríguez-Clare, A., Ulate, M., and Vasquez, J. P. (2026). Trade with Nominal Rigidities: Understanding the Unemployment and Welfare Effects of the China shock. *Journal of Political Economy*, 134(2):626–664.
- Venator, J. (2024). Dual Earner Migration, Earnings, and Unemployment Insurance.
- Venator, J. (2026). Concentrating on His Career or Hers?: Descriptive Evidence on Occupational Co-agglomeration in Dual-Earner Households. *Labour Economics*, page 102860.
- Voena, A. (2015). Yours, Mine, and Ours: Do Divorce Laws Affect the Intertemporal Behavior of Married Couples? *American Economic Review*, 105(8):2295–2332.
- Weiss, Y. and Willis, R. J. (1993). Transfers Among Divorced Couples: Evidence and Interpretation. *Journal of Labor Economics*, 11(4):629–679.
- Yan, Z. (2025). Households in Motion: Co-location Frictions and Gender Inequality.

Appendix

“It Takes Two to Tango”: Joint Location Choices and Local Shocks

Contents

A	Robustness Exercises	A-2
A.1	The China Trade Shock across Sample Commuting Zones	A-2
A.2	The China Trade Shock and Family Structure	A-4
A.3	Other Migration Drivers	A-6
A.4	Additional Tables	A-9
B	Model Derivations	A-14
B.1	Choice Probabilities and Expected Values	A-14
B.2	The Money-Metric Welfare Index	A-15
C	Dynamic Hat Algebra	A-19
C.1	Solving the Model	A-19
C.2	Solving the No-Shock Equilibrium	A-22
D	Quantification of the Model	A-26
D.1	Assignment of Non-Employed Wives to Their Potential Sector	A-26
D.2	Building the Migration Matrix	A-27
D.3	Calibration of the Couple’s Maximization Problem	A-29
E	Additional Results in the Quantitative Exercises	A-34
E.1	Robustness Exercises	A-34
E.2	Additional Results	A-36

A Robustness Exercises

A.1 The China Trade Shock across Sample Commuting Zones

Given the PSID sample restrictions and the size of the PSID, I include 375 commuting zones out of the 722 that ADH use in their analysis. My results could therefore be driven by which zones are included in the PSID rather than by differential exposure to the China Trade Shock. To assess this concern, I replicate ADH’s main results using the 375 CZs represented in the sample.³²³³

Table A.1 first compares the CZs in the sample with the full set used by ADH. Because the restricted sample is a subset of the full sample, the two groups are not independent, and a conventional difference-in-means test is therefore ill-defined. The final column instead reports the standardized difference, defined as the difference in means between the restricted and full samples divided by the square root of the average of their variances. This measure is expressed in standard-deviation units, so it is comparable across variables measured on different scales. Following Imbens and Rubin (2015), absolute standardized differences below 0.25 indicate comparability. All standardized differences in Table A.1 are 0.13 or smaller, including the difference for the change in import penetration. By this criterion, the CZs in the PSID sample are comparable to the full ADH sample with respect to both the control variables and the China Trade Shock itself.

Next, I estimate ADH’s Equation (2) for the CZs in the sample:

$$\Delta Y_{i,t+h} = \alpha_t + \beta_{1,h} \Delta IP_i^{China,US} + X'_{i,t} \beta_2 + \varepsilon_{i,t+h},$$

where $\Delta Y_{i,t+h}$ denotes the change in outcome Y for CZ i between $t = 2000$ and year $t + h$. As in ADH, $\Delta IP_i^{China,US}$ is instrumented with $\Delta IP_i^{China,Others}$, following the identification logic described in Section 2. The analysis considers ADH’s three main outcome categories: employment, income, and population headcount.

Estimated on the CZs in the PSID sample, ADH’s specification yields responses that closely track ADH’s full-sample estimates, except for the changes in population of younger age groups. Figure A.1 replicates ADH’s employment results for the manufacturing employment-to-population ratio in panel a, the non-manufacturing employment-to-population ratio in panel b, and the total wage-and-salary employment-to-population ratio in panel c. Across all three outcomes, the estimates based on the full set of CZs closely align with those based only on the CZs in the PSID sample.

³²Because the analysis uses the geocoded restricted version of the PSID, the identities of the commuting zones in the sample cannot be disclosed, nor can a map be provided at any geographic level below the state.

³³This exercise uses the replication package provided by David Dorn on his website.

Table A.1: Comparing CZs in the Sample with All CZs

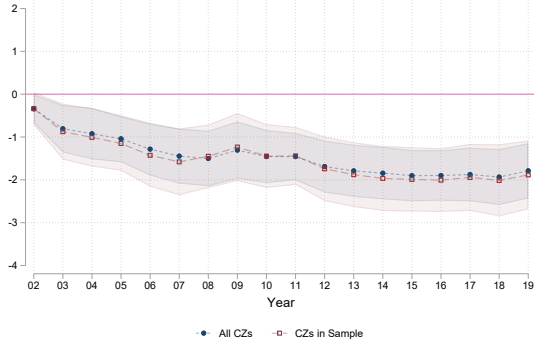
	Full		Restricted			Standardized
	Mean	SD	Mean	SD	Diff	Difference
Control variables (in 2000)						
Manufacturing emp./total emp.	16.18	7.46	15.68	6.91	-0.49	-0.07
Female emp./total emp.	64.38	5.51	64.54	5.21	0.16	0.03
Routine occupation emp./total emp.	31.90	2.37	32.20	2.16	0.30	0.13
Offshorability index	0.00	0.51	0.06	0.48	0.07	0.13
College educated/total pop.	53.61	7.47	54.41	6.88	0.80	0.11
Foreign born/total pop.	14.76	12.81	15.74	12.89	0.97	0.08
Non-white population/total pop.	18.18	10.95	18.81	10.36	0.63	0.06
Population 65+/total pop.	12.37	2.91	12.15	2.81	-0.21	-0.07
Population 40-64/total pop.	30.10	1.87	30.02	1.79	-0.08	-0.04
Population 0-17/total pop.	25.63	2.22	25.65	2.14	0.02	0.01
Δ log population 1970-1990/2	12.31	12.25	12.77	12.35	0.47	0.04
Change in CZ import penetration (2000 to 2012)						
Imports by US	0.89	0.59	0.88	0.55	-0.01	-0.02
Imports by other high-inc. countries	1.22	0.69	1.21	0.64	-0.01	-0.02
Number of Commuting Zones	722		375			

he income responses in the sample CZs also closely track ADH's full-sample estimates. Figure A.2 reports the corresponding results for income: personal income per capita in panel a, government transfers per capita in panel b, and total labor compensation per worker in panel c. The only visible divergence is a somewhat larger increase in government transfers among the sample CZs, although the coefficients are not statistically distinguishable.

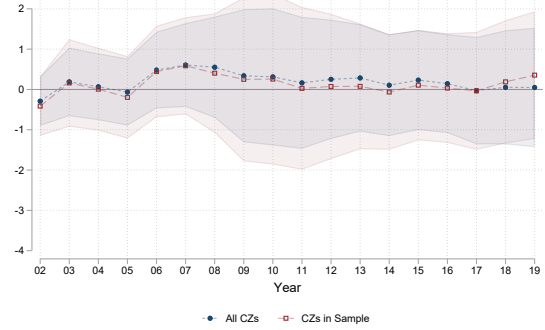
Finally, population headcounts, which ADH use to measure migration responses, are the only outcomes for which the two sets of estimates diverge, and only among younger age groups (Figure A.3). The estimates are similar for the total population and for individuals ages 40–64. However, the population decline among individuals ages 25–39 is larger in the sample CZs, and the estimates indicate a statistically significant decline among individuals ages 18–24 that is absent from ADH's full-sample results. The difference among individuals aged 25–39 is relevant for my sample, whose average age is 37: it suggests that young adults left the sample CZs somewhat more than they left the average CZ. My main estimates, however, compare exposed and non-exposed individuals, so a population decline common to both groups within a CZ does not, by itself, drive the estimated differences.

Figure A.1: China Trade Shock Impact on Employment

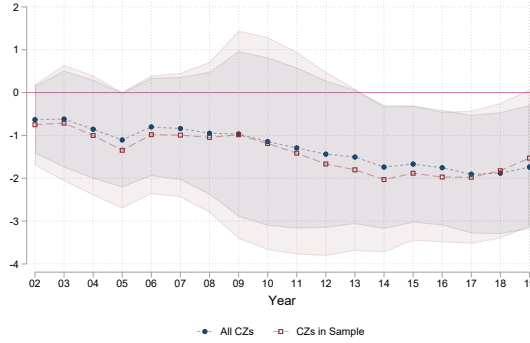
(a) Manufacturing Emp./Working-age pop.



(b) Non-manufacturing Emp./Working-age pop.



(c) Total Wage and Salary Emp./Working-age pop.



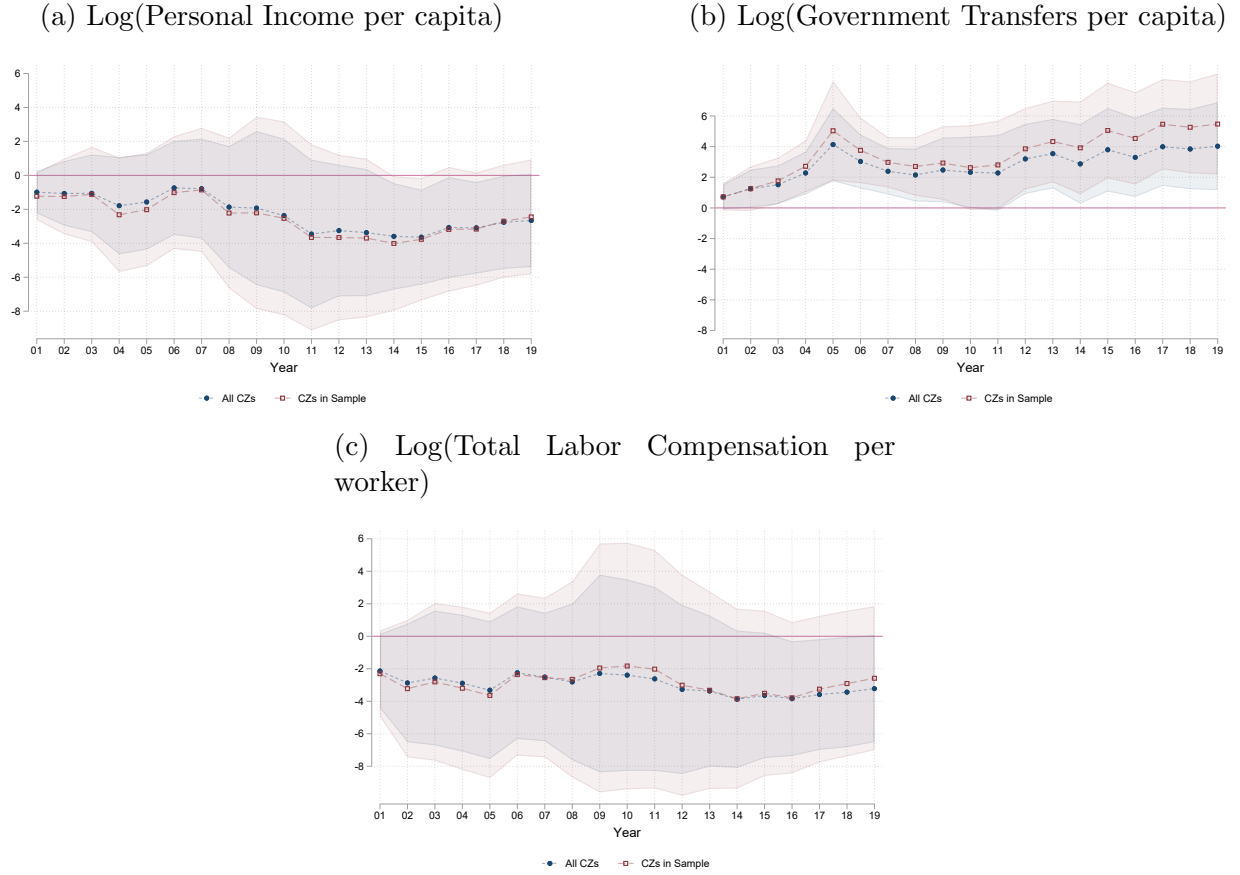
Notes. In each panel, I plot the original ADH result for that outcome variable in blue and, in red, the result when I consider only the CZs in the sample. A CZ is considered in the sample if I can estimate its fixed effect when estimating Equation 5.

Taken together, the two exercises indicate that, although the PSID sample restrictions reduce geographic coverage, they do not appear to select local labor markets in which the China Trade Shock had systematically different effects. The CZs represented in the PSID are comparable to the full ADH sample in terms of observable characteristics, including exposure to the China Trade Shock, and estimating ADH's specification on these CZs produces similar coefficients.

A.2 The China Trade Shock and Family Structure

One potential concern is that the China Trade Shock alters household structure. Such changes could bias the results through two channels: the shock may affect marriage, divorce, or fertility decisions that subsequently shape out-migration responses, and couples who re-

Figure A.2: China Trade Shock Impact on Log Income

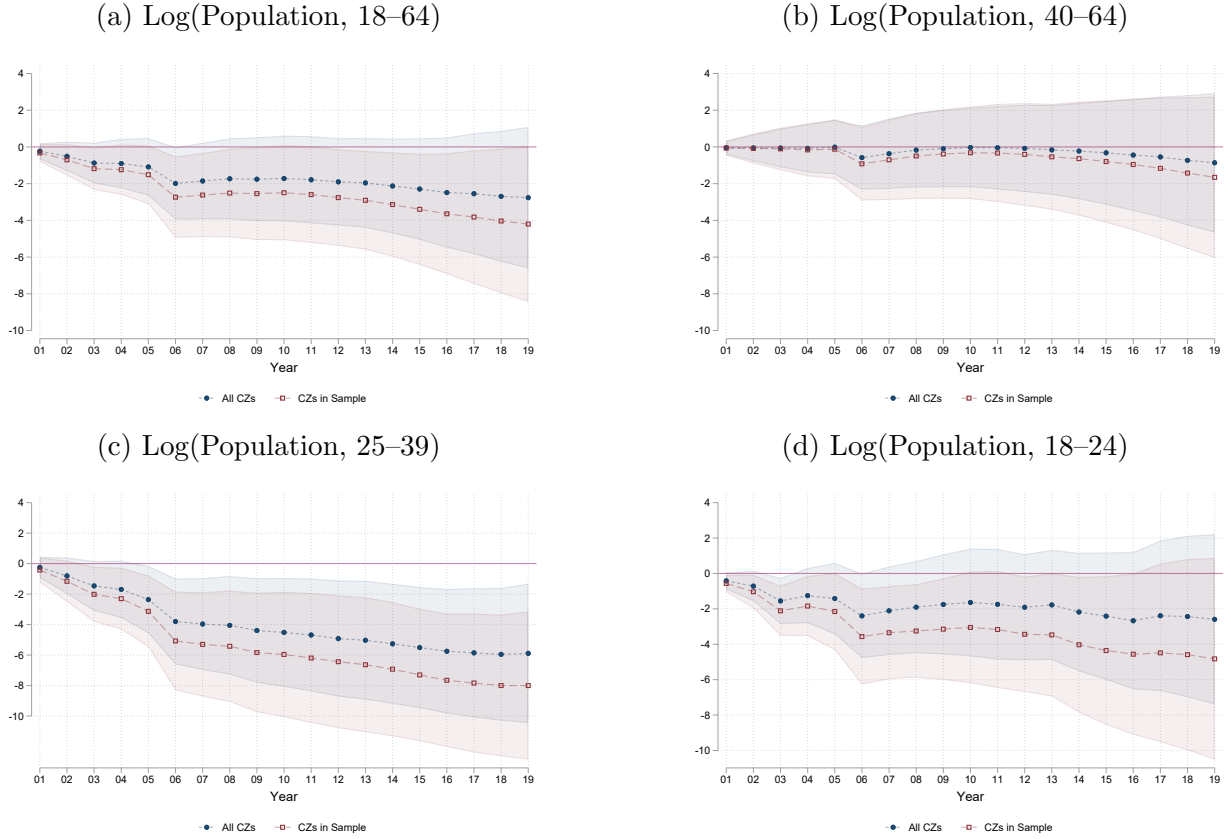


Notes. In each panel, I plot the original ADH result for that outcome variable in blue and, in red, the result when I consider only the CZs in the sample. A CZ is considered in the sample if I can estimate its fixed effect when estimating Equation 5.

main together after the shock may differ systematically from those who divorce. To assess this concern, I use the full sample of individuals and estimate Equation 4 for the probabilities of marrying, divorcing, and having a new child. Figure A.4 shows that, between 2001 and 2013, exposed individuals were no more likely than non-exposed individuals to marry, divorce, or have a child.

At first glance, this result may appear inconsistent with Autor et al. (2019), who find that a one-unit trade shock reduces the share of previously married women by 0.21 p.p. and increases the share of never-married women by 1.2 p.p.. The apparent discrepancy reflects differences in how marital status is measured. In the ACS, marital status records legal status and does not separately identify cohabiting women. Consequently, women who transition from single to cohabiting are classified as married in my data, whereas they are classified as single in Autor et al. (2019). In addition, because the ACS is a repeated cross-section, a

Figure A.3: China Trade Shock Impact on Log Population Headcounts



Notes. In each panel, I plot the original ADH result for that outcome variable in blue and, in red, the result when I consider only the CZs in the sample. A CZ is considered in the sample if I can estimate its fixed effect when estimating Equation 5.

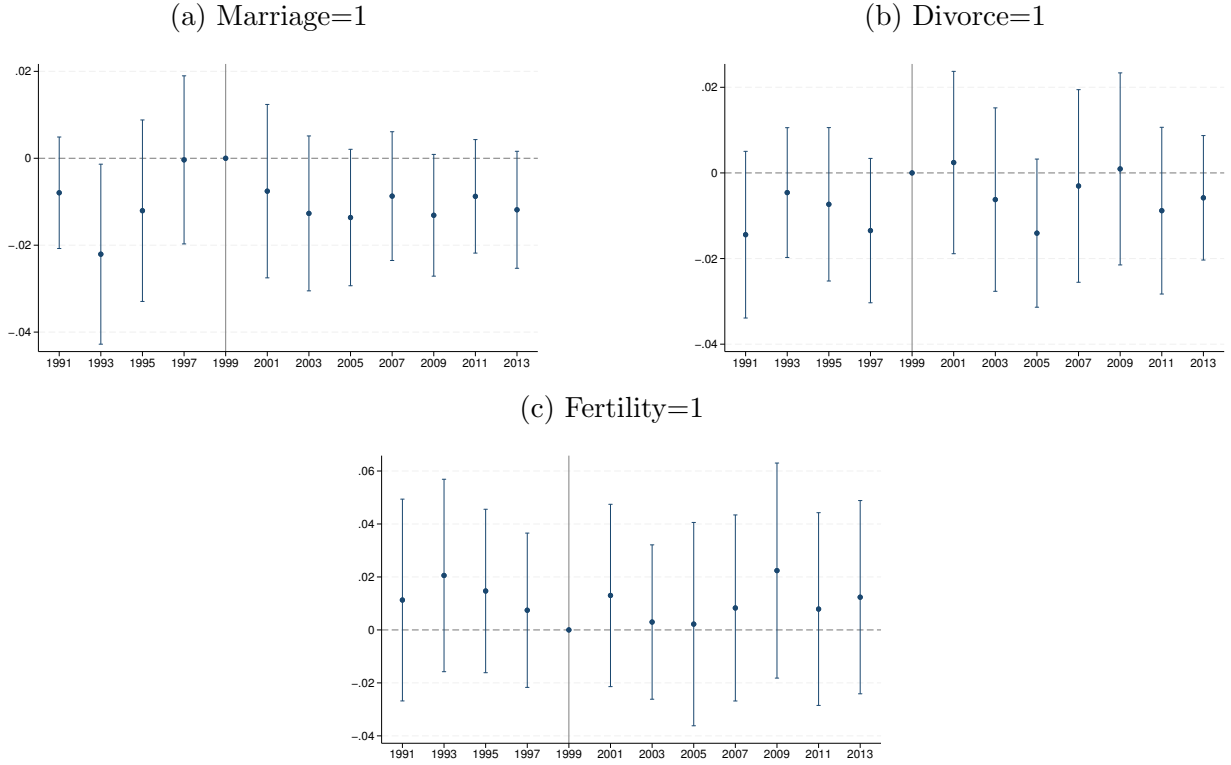
woman's prior marital status cannot be recovered once she is observed as married. Women who transition from divorced to married are therefore classified simply as married in their analysis, whereas they are excluded from my sample.

I also find no robust effect of trade shocks on fertility, in contrast to Autor et al. (2019). One likely explanation is that I allow both men and women to be exposed, whereas they focus exclusively on women. My estimates therefore capture a broader set of household shocks and adjustment margins.

A.3 Other Migration Drivers

The asymmetric migration response within couples could instead reflect two other drivers: the presence of children or gender differences in responses to the shock. First, the behavioral differences I attribute to joint decision-making may reflect the presence of children rather than couple status itself. To address this concern, I extend Equation 4 to a triple-difference

Figure A.4: Exposure to the China Trade Shock and Family Outcomes

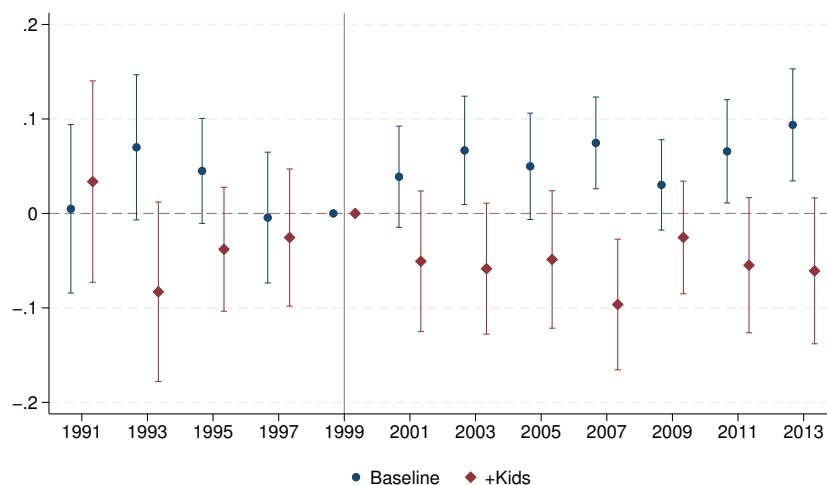


Notes. Each panel plots the estimates of Equation 4 for the outcome named in its title. Robust standard errors are two-way clustered at the industry and individual levels. Confidence intervals are constructed at the 95% level. In the equivalent static difference-in-differences model, the coefficient of interest is not statistically significant for any of the three outcomes.

specification that accounts for whether an individual has a child younger than 18. If individuals with children respond differently from those without children, the coefficient on the triple interaction should be statistically different from zero. Figure A.5 shows that this coefficient is not statistically significant in any year except 2007. Thus, exposure does not generate systematically different out-migration responses among individuals with and without children. Individuals with children nevertheless have a baseline out-migration probability that is 4.2 p.p. lower than that of individuals without children. Therefore, the two groups differ in their intercepts, even though their responses to exposure are similar.

Second, the estimated differences could reflect gender-specific responses to the shock. To address this concern, I extend Equation 4 to a triple-difference specification that accounts for whether the individual is a woman. If women and men respond differently to the shock, the coefficient on the triple interaction should be statistically different from zero. Figure A.6 shows that this coefficient is not statistically significant in any year during the shock. Thus, exposed women do not exhibit a systematically different out-migration response from

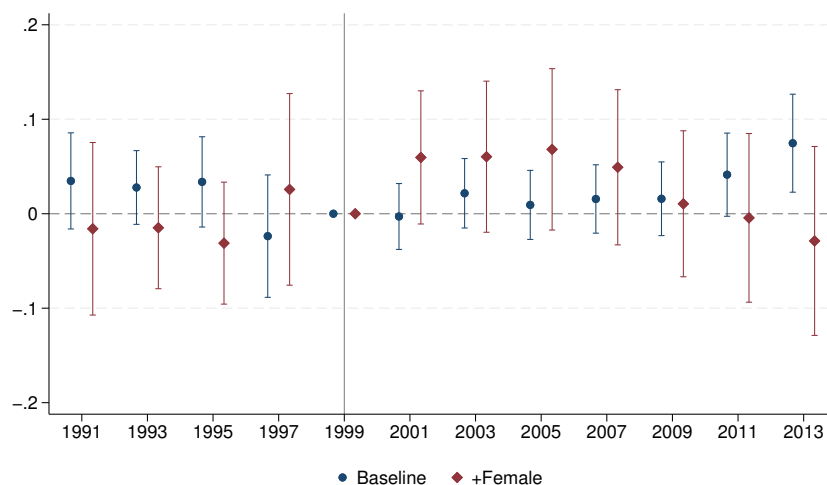
Figure A.5: Exposure to the China Trade Shock and Out-Migration Responses, by Parental Status



Notes. Robust standard errors are two-way clustered at the industry and individual levels. Confidence intervals are constructed at the 95% level. In blue, I plot the coefficient on the interaction between exposure and year. In red, I plot the coefficient on the triple interaction among exposure, year, and an indicator for having a child younger than 18.

exposed men.

Figure A.6: Exposure to the China Trade Shock and Out-Migration Responses, by Gender



Notes. Robust standard errors are two-way clustered at the industry and individual levels. Confidence intervals are constructed at the 95% level. In blue, I plot the coefficient on the interaction between exposure and year. In red, I plot the coefficient on the triple interaction among exposure, year, and an indicator for being a woman.

Taken together, these results indicate that the asymmetric migration response among

couples, depending on which partner is exposed, is driven neither by the presence of children in the household nor by general gender differences in responses to the shock.

A.4 Additional Tables

Table A.2: Exposure to the China Trade Shock and Labor Market Outcomes

	(1)	(2)	(3)
	Employed=1	Unemployed=1	Lab Force=1
Exposed=1	-0.007 (0.019)	-0.002 (0.008)	-0.008 (0.017)
(Exposed=1)x(Post=1)	-0.063*** (0.017)	0.023** (0.010)	-0.039*** (0.012)
Baseline Mean	0.90	0.04	0.94
	(1)	(2)	(3)
	Weekly Hours	Hourly Wage	Labor Income
Exposed=1	-0.454 (1.644)	-0.677 (4.011)	1,978.003 (8,565.422)
(Exposed=1)x(Post=1)	-2.058** (0.889)	-2.581** (1.160)	-5595.397** (2,273.988)
Baseline Mean	40.62	24.26	50,236.60
Observations	38,634	38,634	38,634

Notes. Robust standard errors are two-way clustered at the industry and individual levels.

* for $p < 0.10$, ** for $p < 0.05$, *** for $p < 0.01$.

Table A.3: Binary versus Continuous Exposure to the China Trade Shock

	(1)	(2)	(3)
	Employed=1	Unemployed=1	Labor Force=1
Exposed=1	-0.007 (0.019)	-0.002 (0.008)	-0.008 (0.017)
(Exposed=1)x(Post=1)	-0.063*** (0.017)	0.023** (0.010)	-0.039*** (0.012)
Shock	0.003* (0.002)	-0.000 (0.001)	0.003** (0.001)
(Shock)x(Post=1)	-0.007*** (0.002)	0.003*** (0.001)	-0.003*** (0.001)
Observations	38,634	38,634	38,634
R2	0.08	0.05	0.06
Baseline Mean	0.90	0.04	0.94

Notes. Robust standard errors are two-way clustered at the industry and individual levels.

* for $p < 0.10$, ** for $p < 0.05$, *** for $p < 0.01$.

Table A.4: Exposure to the China Trade Shock and Out-Migration Responses

	(1)
	Out-Migration=1
Exposed=1	-0.064 (0.042)
(Exposed=1)x(Post=1)	0.047** (0.021)
Observations	37,443
R^2	0.09
Baseline Mean	0.07

Notes. Robust standard errors are two-way clustered at the industry and individual levels. * for $p < 0.10$, ** for $p < 0.05$, *** for $p < 0.01$.

Table A.5: Wife's Exposure to the China Trade Shock and Husband's Labor Market Outcomes

	(1)	(2)
	Husband Employed=1	Husband's Hourly Wage
Exposed Wife=1	-0.021 (0.020)	-1.527 (1.355)
(Exposed Wife=1)x(Post=1)	-0.030 (0.026)	-9.349*** (1.557)
Observations	14,083	14,083
R^2	0.10	0.33
Baseline Mean	0.95	30.42

Notes. All monetary values are in 2017 USD, using the PCE index, and include zeros. Robust standard errors are two-way clustered at the wife's industry and individual levels. * for $p < 0.10$, ** for $p < 0.05$, *** for $p < 0.01$.

Table A.6: Couple's Exposure to the China Trade Shock and Household Income Responses

	(1)	(2)
	Household Income	
Exposed Husband=1	7,357.215 (11692.540)	
(Exposed Husband=1)x(Post=1)	3,930.551 (6,330.419)	
Exposed Wife=1		529.105 (12003.901)
(Exposed Wife=1)x(Post=1)		-2348.701 (4,316.531)
Observations	14,083	14,083
R^2	0.35	0.35
Baseline Mean	120226.29	120600.92

Notes. All monetary values are in 2017 USD, using the PCE index, and include zeros. Robust standard errors are two-way clustered at the husband's industry and household levels in Column 1 and the wife's industry and household levels in Column 2. * for $p < 0.10$, ** for $p < 0.05$, *** for $p < 0.01$.

B Model Derivations

This appendix derives the choice probabilities and value function under the nested Gumbel shock and then develops the money-metric measure used to compare welfare across singles and couples.

B.1 Choice Probabilities and Expected Values

The derivation follows Appendix B.2 of Rodríguez-Clare et al. (2026). Fix h , $\mathbf{o}$, and t , and suppress these indices whenever doing so creates no ambiguity. Define $x_{\mathbf{d}} \equiv \delta V_{\mathbf{d},t+1} - \mu_{\mathbf{o}\mathbf{d}}$. The household chooses $\mathbf{d}$ if and only if $x_{\mathbf{d}} + \varepsilon_{\mathbf{d}} \geq x_{\mathbf{z}} + \varepsilon_{\mathbf{z}}$ for every $\mathbf{z}$.

Choice probabilities. Let $G_{\mathbf{d}}(\cdot)$ denote the partial derivative of F in Equation 15 with respect to its $\mathbf{d}$ -th argument. For any $\mathbf{d} \in \mathcal{S}_h(j)$,

$$G_{\mathbf{d}}(\varepsilon) = \frac{1}{\vartheta} \left(\sum_{\mathbf{z} \in \mathcal{S}_h(j)} e^{-\varepsilon_{\mathbf{z}}/\vartheta} \right)^{v/\vartheta-1} e^{-\varepsilon_{\mathbf{d}}/\vartheta} F(\varepsilon).$$

The choice probability for $\mathbf{d}$ is $\int G_{\mathbf{d}}(\varepsilon_{\mathbf{d}}, \{\varepsilon_{\mathbf{d}} + x_{\mathbf{d}} - x_{\mathbf{z}}\}_{\mathbf{z} \neq \mathbf{d}}) d\varepsilon_{\mathbf{d}}$. Evaluating each remaining argument at $\varepsilon_{\mathbf{z}} = \varepsilon_{\mathbf{d}} + x_{\mathbf{d}} - x_{\mathbf{z}}$ yields $e^{-\varepsilon_{\mathbf{z}}/\vartheta} = e^{-\varepsilon_{\mathbf{d}}/\vartheta} e^{(x_{\mathbf{z}} - x_{\mathbf{d}})/\vartheta}$ and therefore

$$\sum_{\mathbf{z} \in \mathcal{S}_h(\ell)} e^{-\varepsilon_{\mathbf{z}}/\vartheta} = e^{-\varepsilon_{\mathbf{d}}/\vartheta} e^{-x_{\mathbf{d}}/\vartheta} \Omega_{\ell}, \quad \Omega_{\ell} \equiv \sum_{\mathbf{z} \in \mathcal{S}_h(\ell)} e^{x_{\mathbf{z}}/\vartheta},$$

so Ω_{ℓ} is shorthand for the inclusive value $\Omega_{\mathbf{o}\ell,t}^h$, where $\Omega_{\mathbf{o}j,t}^h \equiv \sum_{\mathbf{z} \in \mathcal{S}_h(j)} \exp [\delta V_{\mathbf{z},t+1}^h - \mu_{\mathbf{o}\mathbf{z}}]^{1/\vartheta}$. Substituting, and writing $T \equiv e^{-x_{\mathbf{d}}/\vartheta} \sum_{\ell} \Omega_{\ell}^{v/\vartheta}$,

$$\Pr(\mathbf{d}) = \frac{e^{x_{\mathbf{d}}/\vartheta}}{\Omega_j} \cdot \frac{\Omega_j^{v/\vartheta}}{\sum_{\ell} \Omega_{\ell}^{v/\vartheta}} \int_{-\infty}^{\infty} \frac{1}{\vartheta} T e^{-\varepsilon_{\mathbf{d}}/\vartheta} \exp(-T e^{-\varepsilon_{\mathbf{d}}/\vartheta}) d\varepsilon_{\mathbf{d}}.$$

The integrand is the density of a univariate Gumbel distribution with scale ϑ , so the integral equals one. The first fraction corresponds to $\lambda_{\mathbf{o}\mathbf{d}|j,t}^h$ in Equation 18, while the second corresponds to $\lambda_{\mathbf{o}j,t}^h$ in Equation 17.

Expected value. The joint probability that $x_{\mathbf{d}} + \varepsilon_{\mathbf{d}} \leq a$ and that $\mathbf{d}$ is chosen is obtained by integrating the same expression up to $a - x_{\mathbf{d}}$. This yields $\lambda_{\mathbf{o}\mathbf{d},t}^h \exp(-\tilde{T} e^{-a/\vartheta})$, where $\tilde{T} \equiv \sum_{\ell} \Omega_{\ell}^{v/\vartheta}$. Conditional on $\mathbf{d}$ being chosen, the maximized payoff therefore follows a Gumbel distribution with location $\vartheta \log \tilde{T}$ and scale ϑ , implying a mean of $\vartheta \log \tilde{T} + \vartheta \bar{e}$,

where $\bar{e}$ is the Euler–Mascheroni constant. Because this expression does not depend on $\mathbf{d}$,

$$\mathbb{E} \left[\max_{\mathbf{d}} \{x_{\mathbf{d}} + \varepsilon_{\mathbf{d}}\} \right] = \vartheta \log \left(\sum_{\ell=1}^L \Omega_{\mathbf{o}\ell,t}^{v/\vartheta} \right) + \vartheta \bar{e},$$

which, after dropping the constant, delivers Equation 16.

Welfare recursion. From Equation 17 evaluated at the origin location i , $\sum_{\ell} \Omega_{\mathbf{o}\ell,t}^{v/\vartheta} = (\lambda_{\mathbf{o}i,t}^h)^{-1} \Omega_{\mathbf{o}i,t}^{v/\vartheta}$. From Equation 18 evaluated at the origin bundle and $\mu_{\mathbf{o}\mathbf{o}} = 0$, $\Omega_{\mathbf{o}i,t} = (\lambda_{\mathbf{o}\mathbf{o}|i,t}^h)^{-1} \exp(\delta V_{\mathbf{o},t+1}^h)^{1/v}$. Combining,

$$\vartheta \log \left(\sum_{\ell} \Omega_{\mathbf{o}\ell,t}^{v/\vartheta} \right) = \delta V_{\mathbf{o},t+1}^h - \vartheta \log (\lambda_{\mathbf{o}i,t}^h) - v \log (\lambda_{\mathbf{o}\mathbf{o}|i,t}^h),$$

which, substituted into Equation 16 and iterated forward, gives Equation 28.

B.2 The Money-Metric Welfare Index

I follow Chiappori and Meghir (2015) to construct a Money-Metric Welfare Index (MMWI), defined as the income each spouse would require to attain their utility level if they faced the full price of public consumption. For notational simplicity, I suppress the indices $\mathbf{d}$ and t .

Let $u_g \equiv \gamma \log Q + (1 - \gamma) \log c_g$, for $g \in \{f, m\}$, denote the log utility of spouse g . Substituting Equation 21 into u_g , I obtain

$$\begin{aligned} u_f &= \log \frac{I(e)}{P} + e\gamma \log \beta + K + (1 - \gamma) \log \alpha, \\ u_m &= \log \frac{I(e)}{P} + e\gamma \log \beta + K + (1 - \gamma) \log(1 - \alpha), \end{aligned}$$

where $K \equiv \gamma \log \gamma + (1 - \gamma) \log(1 - \gamma)$. In levels, $U_g = \mathcal{A}_g \beta^{e\gamma} I(e)/P$ with $\mathcal{A}_f = e^K \alpha^{1-\gamma}$ and $\mathcal{A}_m = e^K (1 - \alpha)^{1-\gamma}$. Ex ante utility corresponds to

$$\mathbb{E}[U_g] = (\mathcal{A}_g/\mathcal{A}) \mathbb{E}[\mathcal{U}]. \tag{B.1}$$

Thus, each spouse's expected utility is a fixed multiple of the couple's expected utility. The ratio of private consumption is $c_f^*/c_m^* = \alpha/(1 - \alpha)$ and is independent of wages, β , and e . Because the model contains no distribution factors, an increase in the wife's wage raises household income without changing her share of it.

1. Conditional sharing rule (CSR). The CSR allocates between the spouses the income remaining after expenditure on the public good:

$$\tilde{\varsigma}_f = Pc_f^* = \alpha(1 - \gamma)I, \quad \tilde{\varsigma}_m = Pc_m^* = (1 - \alpha)(1 - \gamma)I, \quad \tilde{\varsigma}_f + \tilde{\varsigma}_m = (1 - \gamma)I.$$

The CSR therefore excludes both the γI spent on the public good and the increase in public-good production generated by home production.

2. Lindahl prices. One unit of Q costs the couple $P_Q \equiv P\beta^{-e}$. Thus, when the wife remains at home, home production reduces the effective price of the public good. Each spouse's personalized price equals their marginal willingness to pay for Q , expressed in monetary units:

$$P_g = P \frac{\partial u_g / \partial Q}{\partial u_g / \partial c_g} = \frac{\gamma}{1 - \gamma} \cdot \frac{Pc_g^*}{Q^*} \implies P_f = \alpha P_Q, \quad P_m = (1 - \alpha)P_Q.$$

These personalized prices satisfy the Samuelson condition $P_f + P_m = P_Q$. Because both spouses value Q identically, each bears a share of its cost equal to their bargaining weight.

3. Generalized sharing rule (GSR). The GSR adds each spouse's valuation of public expenditure at their personalized price to their private expenditure:

$$\varsigma_f^* = Pc_f^* + P_f Q^* = \alpha(1 - \gamma)I + \alpha\gamma I = \alpha I, \quad \varsigma_m^* = (1 - \alpha)I.$$

A wife who remains at home ($e = 1$, $I = w^M$) has no labor earnings but is allocated resources equivalent to αw^M . The home-production shock does not directly affect the GSR; instead, it operates through the Lindahl price.

4. Money-Metric Welfare Index (MMWI). The MMWI of spouse g , denoted by m_g , is the income that spouse g would need to attain utility u_g when facing a reference price vector rather than personalized prices. An individual with preferences $Q^\gamma c^{1-\gamma}$ and income m , facing price P for private consumption and $\bar{P}_Q$ for the public good, attains

$$v(m) = \log m - (1 - \gamma) \log P - \gamma \log \bar{P}_Q + K.$$

Equating $v(m_g)$ with u_g and solving for m_g yields the general formula $m_g = \varsigma_g^* (\bar{P}_Q / P_g)^\gamma$.

The remaining question is which reference price $\bar{P}_Q$ to use. If I use the couple's own effective price, $\bar{P}_Q = P\beta^{-e}$, then $m_f = \alpha^{1-\gamma}I$ and $m_m = (1 - \alpha)^{1-\gamma}I$. Under this reference price, the index does not capture home production: for a wife who stays at home, it equals $\alpha^{1-\gamma}w^M$ regardless of her draw of β . I therefore use the market price, $\bar{P}_Q = P$, so that the index explicitly accounts for home production. In this case, $v(m) = \log(m/P) + K$, and

setting $v(m_f) = u_f$ yields

$$m_f = \alpha^{1-\gamma} \beta^{e\gamma} I, \quad m_m = (1 - \alpha)^{1-\gamma} \beta^{e\gamma} I. \quad (\text{B.2})$$

Rewriting the wife's index as $m_f = \alpha I \cdot \alpha^{-\gamma} \cdot \beta^{e\gamma}$ separates three forces. The first term, αI , is her GSR share. The second term, $\alpha^{-\gamma}$, corresponds to the Lindahl-price adjustment. The third, $\beta^{e\gamma}$, captures the value of home production for spouses.

Couples solve the mobility problem in Equation 16 before β is realized. The relevant ex ante index is therefore the certain income $\bar{m}_g$ that allows a person with the same preferences to attain $\mathbb{E}[U_g]$. Because income m yields level utility $e^K m/P$, and because $\mathbb{E}[U_g] = e^K \alpha_g^{1-\gamma} \mathbb{E}[\beta^{e\gamma} I]/P$, with $\alpha_f \equiv \alpha$ and $\alpha_m \equiv 1 - \alpha$,

$$\bar{m}_f = \alpha^{1-\gamma} \Phi w^M, \quad \bar{m}_m = (1 - \alpha)^{1-\gamma} \Phi w^M. \quad (\text{B.3})$$

where $\Phi \equiv \mathbb{E}[\beta^{e\gamma} I]/w^M$.

Summing Equation B.2 across spouses, I can express the gains from the public nature of Q as

$$\frac{m_f + m_m}{\beta^{e\gamma} I} = G(\alpha, \gamma) \equiv \alpha^{1-\gamma} + (1 - \alpha)^{1-\gamma} > 1 \quad \text{whenever } 0 < \gamma < 1.$$

Thus, the spouses would jointly require more income than the household possesses if each had to purchase the public good separately at its full market price.

5. MMWI for singles. The central assumption is that singles have the same preferences as spouses but cannot share the public good with another household member. A single of gender g with wage w^g therefore solves

$$\max \gamma \log Q^g + (1 - \gamma) \log c^g \quad \text{s.t.} \quad P(c^g + Q^g) = w^g.$$

This problem implies $c^g = (1 - \gamma)w^g/P$ and $Q^g = \gamma w^g/P$, with indirect utility $u_s^g = \log \frac{w^g}{P} + K$. The constant K does not affect choices and cancels out in every change in welfare. Thus, the model's $W^{sg} = w^g/P$ is an exact representation of these preferences, and no quantitative result changes. What a single lacks is not the public good itself but its *publicness*: a single person must pay its full price. Accordingly, the MMWI for a single of gender g is

$$m^g = \frac{w^g}{P}.$$

6. Money-Metric-Weighted Aggregate Welfare. Starting from Equation 30, I can express the per-person consumption-equivalent change in utility as

$$\bar{Z} = \frac{\sum N^{s_m} Z^{s_m} + \sum N^{s_f} Z^{s_f} + \sum N^c Z^c}{\sum N^{s_m} + \sum N^{s_f} + \sum N^c}.$$

The measure $\bar{Z}$ averages percentage changes defined relative to different bases: for a single, the base is w/P ; for a spouse, it is a share of the household's full income, together with a shared public good. To express the per-person consumption-equivalent change in monetary units, I therefore construct a money-metric-weighted measure that aggregates across singles and couples.

Because each spouse's expected utility is a constant multiple of the couple's expected utility, the consumption-equivalent change of each spouse is identical to that of the couple when expressed in hat notation. The money-metric-weighted measure is therefore

$$\bar{Z}^s = \frac{\sum m_i Z_i}{\sum m_i} = \bar{Z} + \frac{\text{Cov}(m_i, Z_i)}{\bar{m}}.$$

The covariance term captures whether the gains from the shock accrue to individuals with larger money metrics, such as spouses, or to those with smaller money metrics, such as singles.

C Dynamic Hat Algebra

This appendix develops the Dynamic Hat Algebra used to solve and simulate the model. It expresses the equilibrium entirely in terms of changes over time and ratios across the with-shock and no-shock economies, so that I never have to recover the levels of the fundamentals. The appendix proceeds in two parts. I first state the two propositions that characterize the with-shock economy: one for the temporary equilibrium at a given point in time and another for the full sequential equilibrium, together with the algorithm used to solve them. I then derive the corresponding results for the no-shock economy, defined by an alternative path of sectoral productivities.

C.1 Solving the Model

Notation. Let $\dot{x}_{t+1} = x_{t+1}/x_t$ denote the proportional change in any variable x between t and $t+1$ in the with-shock economy. Let $\hat{x}_{t+1} = \dot{x}'_{t+1}/\dot{x}_{t+1}$, where $\dot{x}'_{t+1} = x'_{t+1}/x'_t$, denote the ratio of the changes in the no-shock economy to those in the with-shock economy. Primes denote levels in the no-shock economy throughout. For every feasible bundle $\mathbf{o}$, define $u_{\mathbf{o},t}^h = \exp(V_{\mathbf{o},t}^h)$, and define regional expenditure as $E_{j,t} \equiv P_{j,t}Q_{j,t}$. Because local sectoral productivities $\{A_{j,n,t}\}$ are the only fundamentals that vary over time, a quantitative application begins with a path of productivity changes $\{\dot{A}_{j,n,t}\}$ and solves the equilibrium system below.

For couples, define the quality-adjusted value of destination $\mathbf{d}$ as

$$B_{\mathbf{d},t} = \mathcal{A} \left[(1 + r_{\mathbf{d},t}) + \chi^\kappa \frac{\gamma}{\kappa - \gamma} (1 + r_{\mathbf{d},t})^{1-\kappa/\gamma} \right].$$

Because $\mathcal{A}$ is constant, it cancels from all proportional changes and cross-economy ratios of B . I retain it in the level expressions below only for completeness. The proportional change in B is

$$\dot{B}_{\mathbf{d},t+1} \equiv \frac{B_{\mathbf{d},t+1}}{B_{\mathbf{d},t}} = \frac{(1 + r_{\mathbf{d},t+1}) + \chi^\kappa \frac{\gamma}{\kappa - \gamma} (1 + r_{\mathbf{d},t+1})^{1-\kappa/\gamma}}{(1 + r_{\mathbf{d},t}) + \chi^\kappa \frac{\gamma}{\kappa - \gamma} (1 + r_{\mathbf{d},t})^{1-\kappa/\gamma}}. \quad (\text{C.1})$$

Proposition C.1. *Given the temporary-equilibrium allocation at t , the temporary equilibrium at $t+1$ can be computed without knowing the levels of the fundamentals. It is characterized by the following system of equations:*

$$\begin{aligned} \dot{w}_{j,n,t+1}^M &= \dot{p}_{j,n,t+1} \dot{A}_{j,n,t+1}^{(\sigma-1)/\sigma} \dot{M}_{j,n,t+1}^{-1/\sigma} \dot{Y}_{j,n,t+1}^{1/\sigma}, \\ \dot{w}_{j,n,t+1}^F &= \dot{p}_{j,n,t+1} \dot{A}_{j,n,t+1}^{(\sigma-1)/\sigma} \dot{F}_{j,n,t+1}^{-1/\sigma} \dot{Y}_{j,n,t+1}^{1/\sigma}, \\ \dot{Y}_{j,n,t+1} &= \dot{A}_{j,n,t+1} \left(s_{j,n,m,t} \dot{M}_{j,n,t+1}^{(\sigma-1)/\sigma} + s_{j,n,f,t} \dot{F}_{j,n,t+1}^{(\sigma-1)/\sigma} \right)^{\sigma/(\sigma-1)}, \end{aligned}$$

$$\begin{aligned}
\dot{P}_{j,n,t+1} &= \left(\sum_{i=1}^L \sum_{k=1}^K \pi_{ik,jn,t} (\dot{p}_{i,k,t+1})^{1-\eta_n} \right)^{1/(1-\eta_n)}, \\
\pi_{ik,jn,t+1} &= \pi_{ik,jn,t} \left(\frac{\dot{p}_{i,k,t+1}}{\dot{P}_{j,n,t+1}} \right)^{1-\eta_n}, \\
\dot{P}_{j,t+1} &= \prod_{n=1}^K \dot{P}_{j,n,t+1}, \quad \dot{E}_{j,t+1} = \dot{P}_{j,t+1} \dot{Q}_{j,t+1}, \\
\dot{p}_{i,k,t+1} \dot{Y}_{i,k,t+1} &= \sum_{j=1}^L \sum_{n=1}^K \varphi_{ik,jn,t} \frac{\pi_{ik,jn,t+1}}{\pi_{ik,jn,t}} \dot{E}_{j,t+1},
\end{aligned}$$

for all region-sector pairs (j, n) , where $s_{j,n,m,t} \equiv \frac{w_{j,n,t}^M M_{j,n,t}}{p_{j,n,t} Y_{j,n,t}}$ and $s_{j,n,f,t} \equiv \frac{w_{j,n,t}^F F_{j,n,t}}{p_{j,n,t} Y_{j,n,t}}$ are the male and female shares of sectoral revenue, and $\varphi_{ik,jn,t} \equiv \frac{\pi_{ik,jn,t} \iota_n P_{j,t} Q_{j,t}}{p_{i,k,t} Y_{i,k,t}}$ is the share of origin-sector revenue from (i, k) sold to destination sector (j, n) . Moreover, for each destination bundle relevant to couples, $\dot{r}_{d,t+1} = \dot{w}_{j,n_f,t+1}^F / \dot{w}_{j,n_m,t+1}^M$, $r_{d,t+1} = r_{d,t} \dot{r}_{d,t+1}$, $\rho_{d,t+1} = 1 - \chi^\kappa (1 + r_{d,t+1})^{-\kappa/\gamma}$, and $\dot{B}_{d,t+1}$ follows from Equation C.1.

Building on Proposition C.1, I next characterize the sequential equilibrium in time differences. Equation 18 implies that the ratio of within-location shares between $t + 1$ and t is

$$\frac{\lambda_{od|j,t+1}^h}{\lambda_{od|j,t}^h} = \frac{\exp [\delta (V_{d,t+2}^h - V_{d,t+1}^h)]^{1/v}}{\Omega_{oj,t+1}^h / \Omega_{oj,t}^h}, \quad \frac{\Omega_{oj,t+1}^h}{\Omega_{oj,t}^h} = \sum_{z \in \mathcal{S}_h(j)} \lambda_{oz|j,t}^h (\dot{u}_{z,t+2}^h)^{\delta/v},$$

where the second equality follows from the time invariance of μ_{oz} . Define the proportional change in the inclusive value of location j as

$$D_{oj,t}^h \equiv \sum_{z \in \mathcal{S}_h(j)} \lambda_{oz|j,t}^h (\dot{u}_{z,t+2}^h)^{\delta/v}. \quad (\text{C.2})$$

The location shares in Equation 17 and the value function in Equation 16 depend on continuation values only through the inclusive values. Their time differences can therefore be expressed entirely in terms of $D_{oj,t}^h$.

Proposition C.2. *Given the economy's initial allocation, including the initial location shares $\lambda_{oj,0}^h$ and within-location shares $\lambda_{od|j,0}^h$, together with an anticipated convergent sequence of productivity changes, the sequential equilibrium in time differences can be solved without recovering the levels of the fundamentals. The solution satisfies the following system of*

nonlinear equations:

$$\begin{aligned}
\lambda_{\mathbf{o}\mathbf{d}|j,t+1}^h &= \frac{\lambda_{\mathbf{o}\mathbf{d}|j,t}^h (\dot{u}_{\mathbf{d},t+2}^h)^{\delta/v}}{\sum_{z \in \mathcal{S}_h(j)} \lambda_{\mathbf{o}z|j,t}^h (\dot{u}_{z,t+2}^h)^{\delta/v}}, \quad \forall h, \\
\lambda_{\mathbf{o}j,t+1}^h &= \frac{\lambda_{\mathbf{o}j,t}^h (D_{\mathbf{o}j,t}^h)^{v/\vartheta}}{\sum_{\ell=1}^L \lambda_{\mathbf{o}\ell,t}^h (D_{\mathbf{o}\ell,t}^h)^{v/\vartheta}}, \quad \forall h, \\
\dot{u}_{\mathbf{o},t+1}^h &= \dot{W}_{\mathbf{o},t+1}^h \left(\sum_{\ell=1}^L \lambda_{\mathbf{o}\ell,t}^h \left(\sum_{z \in \mathcal{S}_h(\ell)} \lambda_{\mathbf{o}z|\ell,t}^h (\dot{u}_{z,t+2}^h)^{\delta/v} \right)^{v/\vartheta} \right)^{\vartheta}, \quad \forall h, \\
N_{\mathbf{d},t+1}^h &= \sum_{\mathbf{o} \in \mathcal{S}_h} \lambda_{\mathbf{o}\mathbf{d}|j,t}^h \lambda_{\mathbf{o}j,t}^h N_{\mathbf{o},t}^h, \quad \mathbf{d} \in \mathcal{S}_h(j), \quad \forall h, \\
M_{j,n,t+1} &= N_{j,n,t+1}^{s_m} + \sum_{n_f=1}^K N_{j,n,n_f,t+1}^c, \\
F_{j,n,t+1} &= N_{j,n,t+1}^{s_f} + \sum_{n_m=0}^K \rho_{j,n_m,n,t+1} N_{j,n_m,n,t+1}^c,
\end{aligned}$$

for all feasible bundles and all sectors at each t , where $D_{\mathbf{o}\ell,t}^h$ is defined in Equation C.2 and the sectoral prices, trade shares, and participation objects solve the temporary-equilibrium system in Proposition C.1.

Because the migration costs $\mu_{\mathbf{o}\mathbf{d}}$ are time-invariant, they cancel from all ratios and therefore need not be recovered in levels. The flow term in the couple's continuation value is the only one that contains an additional factor. In particular, because the couple's indirect utility is $W_{\mathbf{o},t}^c = B_{\mathbf{o},t} w_{i,k_m,t}^M / P_{i,t}$, its proportional change is $\dot{B}_{\mathbf{o},t+1} \dot{w}_{i,k_m,t+1}^M / \dot{P}_{i,t+1}$. Thus, the relevant term is the *ratio* defined in Equation C.1, rather than the level $B_{\mathbf{o},t+1}$.

Algorithm for Solving the Sequential Competitive Equilibrium. Given an initial equilibrium at $t = 0$ and an anticipated convergent sequence of sectoral productivity changes $\left\{ \dot{A}_{j,n,t} \right\}_{j=1, n=1, t=0}^{L,K,T}$, the algorithm is as follows:

1. Fix a sufficiently large terminal horizon T and guess a path for the changes in continuation values $\left\{ \dot{u}_{\mathbf{o},t+1}^{h,(0)} \right\}_{t=0}^T$, subject to the terminal condition $\dot{u}_{\mathbf{o},T+1}^{h,(0)} = 1$ for every feasible bundle $\mathbf{o}$.
2. Decompose the initial migration matrix into location shares $\lambda_{\mathbf{o}j,0}^h = \sum_{\mathbf{d} \in \mathcal{S}_h(j)} \lambda_{\mathbf{o}\mathbf{d},0}^h$ and within-location shares $\lambda_{\mathbf{o}\mathbf{d}|j,0}^h = \lambda_{\mathbf{o}\mathbf{d},0}^h / \lambda_{\mathbf{o}j,0}^h$. Using Proposition C.2, update these objects forward in the following order: the within-location shares, $D_{\mathbf{o}j,t}^h$, and then the location shares.

3. Starting from the initial allocation, apply the law of motion to recover the entire path of household masses $\left\{N_{\mathbf{d},t}^{sm,(0)}, N_{\mathbf{d},t}^{sf,(0)}, N_{\mathbf{d},t}^{c,(0)}\right\}_{t=0}^T$, and aggregate these masses into sectoral male labor supplies $\left\{M_{j,n,t}^{(0)}\right\}_{t=0}^T$.
4. Solve the temporary equilibrium sequentially for each period $t \geq 0$:
 - (a) Guess the participation rates $\{\rho_{\mathbf{d},t+1}^{(0)}\}$ and use them to construct the sectoral female labor supplies $\{F_{j,n,t+1}^{(0)}\}$.
 - (b) Conditional on the labor aggregates and the path of fundamentals, solve the production block for $\{\dot{Y}_{j,n,t+1}^{(0)}, \dot{w}_{j,n,t+1}^{M,(0)}, \dot{w}_{j,n,t+1}^{F,(0)}\}$, the trade block for $\{\dot{p}_{j,n,t+1}^{(0)}, \dot{P}_{j,n,t+1}^{(0)}, \pi_{ik,jn,t+1}^{(0)}\}$, and the regional aggregation block for $\{\dot{P}_{j,t+1}^{(0)}, \dot{E}_{j,t+1}^{(0)}\}$.
 - (c) Recover $\{\dot{r}_{\mathbf{d},t+1}^{(0)}\}$, construct the levels $r_{\mathbf{d},t+1}^{(0)} = r_{\mathbf{d},t} \dot{r}_{\mathbf{d},t+1}^{(0)}$, and update $\{\rho_{\mathbf{d},t+1}^{(1)}, \dot{B}_{\mathbf{d},t+1}^{(1)}\}$.
 - (d) Iterate over steps (a) through (c) until the participation block converges.
5. Given the temporary-equilibrium path and the guessed continuation values, solve backward from T for the updated path $\{\dot{u}_{\mathbf{o},t+1}^{h,(1)}\}_{t=0}^T$ using the $\dot{u}^h$ equation in Proposition C.2. This backward solution requires both the location shares and the within-location shares.
6. Use the updated path as the new guess and repeat steps 2 through 5 until the sequential equilibrium converges.

C.2 Solving the No-Shock Equilibrium

The path of sectoral productivity fundamentals in the no-shock economy, $\{A'_{j,n,t}\}_{t \geq 1}$, defines the corresponding productivity hats $\{\hat{A}_{j,n,t}\}_{t \geq 1}$. Under CDP's timing, households do not anticipate the change at $t = 0$. Therefore, $\lambda'_{\mathbf{o}j,0} = \lambda_{\mathbf{o}j,0}^h$, $\lambda'_{\mathbf{od}|j,0} = \lambda_{\mathbf{od}|j,0}^h$, and $N'_{\mathbf{o},1} = N_{\mathbf{o},1}^h$.

Dividing each no-shock migration equation in Proposition C.2 by its with-shock counterpart, and using $\dot{\lambda}_{\mathbf{od}|j,t}^h = \lambda_{\mathbf{od}|j,t}^h / \lambda_{\mathbf{od}|j,t-1}^h$ and $\dot{\lambda}_{\mathbf{o}j,t}^h = \lambda_{\mathbf{o}j,t}^h / \lambda_{\mathbf{o}j,t-1}^h$, yields the following no-shock analogue of Equation C.2:

$$\hat{D}_{\mathbf{o}j,t}^h \equiv \sum_{\mathbf{z} \in \mathcal{S}_h(j)} \lambda'_{\mathbf{oz}|j,t-1} \dot{\lambda}_{\mathbf{oz}|j,t}^h (\hat{u}_{\mathbf{z},t+1}^h)^{\delta/\nu}. \quad (\text{C.3})$$

Proposition C.3. *Given the with-shock economy and a convergent sequence of productivity changes in the no-shock economy relative to the with-shock path, the no-shock sequential equilibrium can be solved without recovering the fundamental levels. For $t \geq 2$, it is characterized by the following system of nonlinear equations:*

$$\begin{aligned}
\lambda'_{od|j,t} &= \frac{\lambda'_{od|j,t-1} \dot{\lambda}_{od|j,t}^h (\hat{u}_{d,t+1}^h)^{\delta/v}}{\hat{D}_{oj,t}^h}, \quad \forall h, \\
\lambda'_{oj,t} &= \frac{\lambda'_{oj,t-1} \dot{\lambda}_{oj,t}^h (\hat{D}_{oj,t}^h)^{v/\vartheta}}{\sum_{\ell=1}^L \lambda'_{o\ell,t-1} \dot{\lambda}_{o\ell,t}^h (\hat{D}_{o\ell,t}^h)^{v/\vartheta}}, \quad \forall h, \\
\hat{u}_{o,t}^h &= \hat{W}_{o,t}^h \left(\sum_{\ell=1}^L \lambda'_{o\ell,t-1} \dot{\lambda}_{o\ell,t}^h \left(\sum_{z \in \mathcal{S}_h(\ell)} \lambda'_{oz|\ell,t-1} \dot{\lambda}_{oz|\ell,t}^h (\hat{u}_{z,t+1}^h)^{\delta/v} \right)^{v/\vartheta} \right)^{\vartheta}, \quad \forall h, \\
N'_{d,t+1} &= \sum_{o \in \mathcal{S}_h} \lambda'_{od|j,t} \lambda'_{oj,t} N'_{o,t}, \quad d \in \mathcal{S}_h(j), \quad \forall h, \\
M'_{j,n,t+1} &= N'_{j,n,t+1}^{s_m} + \sum_{n_f=1}^K N'_{j,n,n_f,t+1}^c, \\
F'_{j,n,t+1} &= N'_{j,n,t+1}^{s_f} + \sum_{n_m=0}^K \rho'_{j,n_m,n,t+1} N'_{j,n_m,n,t+1}^c.
\end{aligned}$$

In period $t = 1$, when households first learn the new path, the migration equations instead become

$$\begin{aligned}
\lambda'_{od|j,1} &= \frac{\theta_{od|j}^h (\hat{u}_{d,2}^h)^{\delta/v}}{\sum_{z \in \mathcal{S}_h(j)} \theta_{oz|j}^h (\hat{u}_{z,2}^h)^{\delta/v}}, \\
\lambda'_{oj,1} &= \frac{\lambda_{oj,1}^h \left(\sum_{z \in \mathcal{S}_h(j)} \theta_{oz|j}^h (\hat{u}_{z,2}^h)^{\delta/v} \right)^{v/\vartheta}}{\sum_{\ell=1}^L \lambda_{o\ell,1}^h \left(\sum_{z \in \mathcal{S}_h(\ell)} \theta_{oz|\ell}^h (\hat{u}_{z,2}^h)^{\delta/v} \right)^{v/\vartheta}}, \\
\hat{u}_{o,1}^h &= \hat{W}_{o,1}^h \left(\sum_{\ell=1}^L \lambda_{o\ell,1}^h \left(\sum_{z \in \mathcal{S}_h(\ell)} \theta_{oz|\ell}^h (\hat{u}_{z,2}^h)^{\delta/v} \right)^{v/\vartheta} \right)^{\vartheta}, \quad \theta_{od|j}^h \equiv \lambda_{od|j,1}^h (\hat{u}_{d,1}^h)^{\delta/v}.
\end{aligned}$$

The no-shock temporary equilibrium is characterized by

$$\begin{aligned}
\hat{w}_{j,n,t}^M &= \hat{p}_{j,n,t} \hat{A}_{j,n,t}^{(\sigma-1)/\sigma} \hat{M}_{j,n,t}^{-1/\sigma} \hat{Y}_{j,n,t}^{1/\sigma}, \\
\hat{w}_{j,n,t}^F &= \hat{p}_{j,n,t} \hat{A}_{j,n,t}^{(\sigma-1)/\sigma} \hat{F}_{j,n,t}^{-1/\sigma} \hat{Y}_{j,n,t}^{1/\sigma}, \\
\hat{Y}_{j,n,t} &= \hat{A}_{j,n,t} \left(s_{j,n,m,t} \hat{M}_{j,n,t}^{(\sigma-1)/\sigma} + s_{j,n,f,t} \hat{F}_{j,n,t}^{(\sigma-1)/\sigma} \right)^{\sigma/(\sigma-1)},
\end{aligned}$$

$$\begin{aligned}
\hat{P}_{j,n,t} &= \left(\sum_{i=1}^L \sum_{k=1}^K \pi_{ik,jn,t} (\hat{p}_{i,k,t})^{1-\eta_n} \right)^{1/(1-\eta_n)}, \\
\pi'_{ik,jn,t} &= \pi_{ik,jn,t} \left(\frac{\hat{p}_{i,k,t}}{\hat{P}_{j,n,t}} \right)^{1-\eta_n}, \\
\hat{P}_{j,t} &= \prod_{n=1}^K \hat{P}_{j,n,t}^{\iota_n}, \quad \hat{E}_{j,t} = \hat{P}_{j,t} \hat{Q}_{j,t}, \\
\hat{p}_{i,k,t} \hat{Y}_{i,k,t} &= \sum_{j=1}^L \sum_{n=1}^K \varphi_{ik,jn,t} \frac{\pi'_{ik,jn,t}}{\pi_{ik,jn,t}} \hat{E}_{j,t}, \\
\hat{r}_{\mathbf{d},t} &= \frac{\hat{w}_{j,n_f,t}^F}{\hat{w}_{j,n_m,t}^M},
\end{aligned}$$

and the corresponding no-shock level objects satisfy

$$r'_{\mathbf{d},t} = r'_{\mathbf{d},t-1} \dot{r}_{\mathbf{d},t} \hat{r}_{\mathbf{d},t}, \quad \rho'_{\mathbf{d},t} = 1 - \chi^\kappa (1 + r'_{\mathbf{d},t})^{-\kappa/\gamma}, \quad \hat{B}_{\mathbf{d},t} = \frac{B'_{\mathbf{d},t}}{B_{\mathbf{d},t}},$$

with $B'_{\mathbf{d},t}$ evaluated at $r'_{\mathbf{d},t}$. The with-shock revenue shares $s_{j,n,m,t}$, $s_{j,n,f,t}$ and revenue-allocation shares $\varphi_{ik,jn,t}$ are those of Proposition C.1, taken from the with-shock equilibrium.

Solution Algorithm for the No-Shock Equilibrium. Given the with-shock sequential equilibrium and a convergent sequence of changes in fundamentals, $\{\hat{A}_{j,n,t}\}_{t \geq 1}$, that defines the no-shock economy, the no-shock equilibrium is computed as follows:

1. Because the temporary equilibrium and migration shares at $t = 0$ are unchanged, initialize all no-shock objects at their with-shock values.
2. Fix a sufficiently large horizon T and guess a path $\{\hat{u}_{\mathbf{o},t}^{h,(0)}\}_{t=1}^{T+1}$, imposing $\hat{u}_{\mathbf{o},T+1}^{h,(0)} = 1$ for all feasible bundles $\mathbf{o}$ and household types h .
3. Use the guessed continuation values to compute the path of no-shock migration shares. At $t = 1$, construct $\theta_{\mathbf{o}d|j}^h$ using the with-shock within-location shares and $\hat{u}_{\mathbf{d},1}^{h,(0)}$, and apply the period-1 equations in Proposition C.3. For $t \geq 2$, update, in sequence, the within-location shares, $\hat{D}_{\mathbf{o}j,t}^h$ from Equation C.3, and the location shares.
4. Starting from the same initial allocation as in the with-shock economy, use the no-shock shares to compute the implied path of no-shock household populations $\left\{ N'_{\mathbf{d},t}{}^{sm,(0)}, N'_{\mathbf{d},t}{}^{sf,(0)}, N'_{\mathbf{d},t}{}^{c,(0)} \right\}_{t=1}^{T+1}$.

5. Given the no-shock household masses, recover the path of no-shock sectoral male labor supply $\left\{M_{j,n,t}'^{(0)}\right\}_{t=1}^{T+1}$.
6. Solve the no-shock temporary equilibrium period by period. For each $t \geq 0$, proceed as follows:
 - (a) Guess the no-shock participation rates $\left\{\rho_{\mathbf{d},t+1}'^{(0)}\right\}$.
 - (b) Use these participation rates to compute no-shock female labor supply $\left\{F_{j,n,t+1}'^{(0)}\right\}$.
 - (c) Construct $\left\{\hat{M}_{j,n,t+1}^{(0)}\right\}$ and $\left\{\hat{F}_{j,n,t+1}^{(0)}\right\}$. Given these objects and the changes in fundamentals, recover $\left\{\hat{Y}_{j,n,t+1}^{(0)}\right\}$ from the production block. Next, solve the goods-market-clearing block for $\left\{\hat{p}_{j,n,t+1}^{(0)}, \hat{P}_{j,n,t+1}^{(0)}, \hat{P}_{j,t+1}^{(0)}, \pi_{ik,jn,t+1}'^{(0)}\right\}$ and recover $\left\{\hat{w}_{j,n,t+1}^{M,(0)}, \hat{w}_{j,n,t+1}^{F,(0)}\right\}$ from the firms' first-order conditions. Finally, recover $\left\{\hat{r}_{\mathbf{d},t+1}^{(0)}\right\}$.
 - (d) Recover $\left\{r_{\mathbf{d},t+1}'^{(0)}\right\}$ and update $\left\{\rho_{\mathbf{d},t+1}'^{(1)}, B_{\mathbf{d},t+1}'^{(1)}\right\}$. If the updated $\rho_{\mathbf{d},t+1}'^{(1)}$ differs from its initial guess, return to step (b) and iterate until the participation block converges.
 - (e) Repeat steps (a) through (d) sequentially for each period t .
7. Given the no-shock temporary-equilibrium path and the guessed continuation values, solve backward from T for the updated path $\left\{\hat{u}_{\mathbf{o},t}^{h,(1)}\right\}_{t=1}^{T+1}$ using the $\hat{u}^h$ equations in Proposition C.3. At $t = 1$, $\hat{u}_{\mathbf{o},1}^h$ appears on both sides through ϑ , so solve this equation as a fixed point.
8. Use the updated path as the new guess and repeat steps 3 through 7 until the continuation-value path converges.
9. Compute welfare using Equation 29, with $\hat{\lambda}_{\mathbf{o}i,s}^h$ and $\hat{\lambda}_{\mathbf{o}\mathbf{o}|i,s}^h$ constructed from the no-shock and with-shock location and within-location staying shares.

D Quantification of the Model

This appendix details three elements of the model’s quantification. First, I explain how I assign each non-employed married woman to her potential sector. Then, I describe how I build the migration matrices by household type. Finally, I describe the calibration of the couple’s maximization problem.

D.1 Assignment of Non-Employed Wives to Their Potential Sector

The model assigns every wife to a productive sector, but the ACS does not indicate which sector a non-employed married woman would work in. In the ACS, which I use to construct the initial distribution of couples across states and sector pairs, $\{N_{o,0}^c\}$, a married woman who is out of the labor force is recorded as non-employed. Thus, I do not observe the counterfactual sector in which she would have worked if the couple had decided that she would participate in the market. In the model, couples choose the wife’s productive sector first and her participation second, so she is never assigned to the non-employment sector itself. I therefore need to distribute the non-employed wives across productive sectors.

If I were to distribute them across productive sectors in the same proportions as employed wives, I would ignore a selection problem. The sectors in which married women are more likely to be employed are also the ones in which they are less likely to be non-employed. Following the distribution of employed wives would therefore assign a larger share of non-employed wives to the sectors that are least likely to be their potential sector.

To recover the potential sector, I return to the model. I drop the time index and write the labor supply of married women in state j and sector n as

$$F_{j,n}^c = \rho_{j,n} N_{j,n}^c.$$

The mass of non-employed wives in the same cell, which I denote by $\text{Non-}F_{j,n}^c$, is

$$\text{Non-}F_{j,n}^c = (1 - \rho_{j,n}) N_{j,n}^c.$$

The ratio of non-employed to employed wives is therefore

$$\frac{\text{Non-}F_{j,n}^c}{F_{j,n}^c} = \frac{(1 - \rho_{j,n})}{\rho_{j,n}},$$

which implies that the mass of non-employed wives in that cell is

$$\text{Non-}F_{j,n}^c = \underbrace{F_{j,n}^c}_{\text{observed}} \times \underbrace{\frac{1 - \rho_{j,n}}{\rho_{j,n}}}_{\text{the model's odds of non-employment}}.$$

Dividing by $\rho_{j,n}$ therefore corrects the selection problem and assigns each non-employed wife to her potential productive sector.

These cell-level masses are in shares rather than headcounts, so nothing in the model's odds forces them to sum to the number of non-employed married women actually observed in state j . I therefore normalize them within each state, rescaling the $\{\text{Non-}F_{j,n}^c\}_n$ proportionally so that they add up to the observed state total.

D.2 Building the Migration Matrix

Since I solve the model with Dynamic Hat Algebra, the time-invariant mobility costs μ_{od} cancel from the recursion that governs the choice probabilities (Proposition C.2). The recursion requires only two initial objects for each household type: the location shares $\lambda_{oj,0}^h$ of Equation 17 and the within-location shares $\lambda_{od|j,0}^h$ of Equation 18.

Following CDP's Appendix E, I build the initial migration matrix as the product of an interstate probability from the ACS and a sectoral transition probability from the Current Population Survey (CPS), where an interstate mover's sectoral transition is imputed from the destination state's sector matrix:

$$\lambda_{(i,k)(j,n),0}^h = \underbrace{f_{ij}^h}_{\text{ACS}} \times \underbrace{P_j^h(n | k)}_{\text{CPS}}.$$

Summing over the destination sectors of location j gives $\lambda_{oj,0}^h = f_{ij}^h$, and dividing by that sum gives $\lambda_{od|j,0}^h = P_j^h(n | k)$.³⁴

I build the matrices in three steps: interstate flows from the ACS, sectoral transitions from the CPS, and a mapping of both onto the model's full grid. First, I derive interstate flows from the ACS migration question for 2001–2007, which records the state of residence one year earlier. The resulting state-to-state matrix is therefore annual, whereas the model period is quarterly. Following CDP's Appendix E, I assume that interstate moves are evenly distributed throughout the year and that a household makes at most one such move annually, so the quarterly matrix retains one-fourth of the moves and imputes the remaining three-

³⁴Where an origin puts no mass on a location, the within-location share is undefined. I set it to zero rather than to a uniform distribution, since the two factors only ever enter as a product, and a uniform distribution would place mass on destinations that the data indicate are unreachable once the location share moves away from zero.

fourths as non-moves:

$$f_{i \rightarrow j}^q = \frac{1}{4} f_{i \rightarrow j} \quad (j \neq i), \quad f_{i \rightarrow i}^q = \frac{1}{4} f_{i \rightarrow i} + \frac{3}{4}.$$

I apply this adjustment to every quarter from 2000 to 2007. The adjustment is quantitatively important: combining the *annual* interstate probability with the *quarterly* sector matrix without it would inflate mobility by roughly a factor of six, implying a quarterly own-state retention rate of 0.959 for single men against 0.994 in CDP. The error would fall entirely on the location margin, which is calibrated to the population’s response to the shock.

Next, I estimate sectoral transitions from the CPS on fewer sectors than the model’s 22 sectors plus non-employment, because disaggregating by household type would otherwise leave too few observations per cell. I therefore pool the 12 manufacturing sub-sectors into a single manufacturing sector and the 8 service sub-sectors into a single service sector, leaving an aggregated five-sector structure: non-employment, manufacturing, retail, construction, and services. I estimate transitions on this aggregation separately for single men, single women, and couples, with couples moving on the joint space of both spouses’ groups.

Combining the first two steps gives a transition matrix across five sectors and 50 states, by household type, which I then map onto the model’s grid of 22 productive sectors. Let $\tilde{\mathbf{o}}$ denote the origin bundle in the five-sector grid and $\tilde{\mathbf{d}}$ denote the corresponding destination bundle. The production side considers 22 productive sectors across 50 states. Applying the same sectoral aggregation to the production side would lose the dispersion across manufacturing sectors in the China Trade Shock. I map the matrices onto the full grid by factorizing each transition probability into a group-level move and a within-group destination. For an individual of gender $g \in \{m, f\}$ in a household of type h ,

$$\underbrace{P_{h,g}(\mathbf{d} \mid \mathbf{o})}_{\text{model's grid}} = \underbrace{P_h(\tilde{\mathbf{d}} \mid \tilde{\mathbf{o}})}_{\text{ACS + CPS, by household type}} \times \underbrace{P_g(\mathbf{d} \mid \tilde{\mathbf{d}}, \mathbf{o})}_{\text{within-group composition, by gender}},$$

so that household type governs *which* broad sector, and which state, a household moves to, and the within-group composition governs *where within* that sector it lands. Because this factorizes a single conditional distribution, the rows of the resulting matrix sum to one by construction. The within-group composition only redistributes mass among sectors of the same group *in the same state*, so it leaves the location shares unchanged and acts only on the within-location shares.

To build the within-group composition, I split the CPS sample by sex and compute, within manufacturing and within services, the share of men and of women going to each sub-sector. This step matters: men and women sort differently within both groups, with men concentrated in heavy and durable manufacturing and women in lighter, non-durable

goods. Without it, the model would miss one of the channels through which the shock affected men and women differently.

D.3 Calibration of the Couple’s Maximization Problem

This subsection documents the calibration of the couple’s maximization problem in Equation 20, with $\beta \sim \text{Pareto}(\chi, \kappa)$. Four parameters govern it: the wife’s weight in the household objective, α ; the public-good expenditure share, γ ; and the lower bound and shape of the home-production shock, χ and κ . Because I follow Chiappori and Mazzocco (2017)’s method to solve the couple’s problem, I identify the four parameters *sequentially*, each from its own source and each mapped onto a distinct piece of variation.

D.3.1 α and γ from Expenditure Data

The optimal allocations for private and public consumption in Equation 21 are Cobb-Douglas shares, so both parameters map exactly onto expenditure shares. The parameter α maps onto the share of female private consumption in total private consumption, and γ maps onto the share of public expenditure in total expenditure:

$$\alpha = \frac{c_f^*}{c_f^* + c_m^*}, \quad \gamma = \frac{q^*}{q^* + c_f^* + c_m^*}.$$

Data. I use the Consumer Expenditure Interview Survey (CEX), 2000–2007, to construct three inputs.³⁵ First, the *assignable private goods* c_m and c_f are adult apparel and footwear.³⁶ Second, the *public good* q is the market input to home production: because $Q = \beta^e q$, the object I need is what the household combines with the wife’s home time. It comprises shelter, fuels and utilities, food at home, cable and internet services, do-it-yourself repair materials, and household operations such as domestic services, lawn care, and appliance repair. Household operations are the *market substitutes* for home time, so omitting them would measure q selectively on precisely the margin the model is about. Third, I construct *total expenditure* Y as consumption only.³⁷

The sample consists of household-year pairs with complete twelve-month expenditure records. Couples are intact married couples with both spouses present, aged 25–65; singles are one-person consumer units in the same age band. I drop households containing a 16- or 17-year-old, because the CEX books apparel for ages 16 and over under the adult men’s and

³⁵The complete list of expenditure codes entering each of the three objects is available upon request.

³⁶Children’s, boys’, girls’, and infants’ apparel are excluded, since the model has no children.

³⁷The CEX expenditure file also carries asset stocks, debt balances, loans, principal repayments, and asset-sale receipts. All of these are excluded, as are transfers out of the household and gifts.

women’s codes, so a teenager’s clothing would otherwise be read as a spouse’s assignable private consumption. The couples and singles samples are disjoint by construction.

Estimating α . I estimate the wife’s weight in the household objective, α , from apparel, the only assignable good in the CEX, correcting for the fact that women spend more on apparel than men. Apparel is only a *subset* of private consumption, so the raw apparel ratio, 0.60 in this sample, is not α . Taken at face value, the raw ratio would imply that the wife has the larger Pareto weight. To recover α , I follow Browning et al. (2013): I estimate an Engel curve for apparel among *singles* of each gender and recover the resource shares ϱ_f and ϱ_m such that, evaluated at each spouse’s own resources $\varrho_g Y$, these Engel curves reproduce the aggregate apparel spending observed in couples. Using both ϱ_f and ϱ_m , I recover $\alpha = \varrho_f / (\varrho_f + \varrho_m)$. The correction is substantial: it moves α from 0.60 to 0.47, which *reverses* which spouse has the larger bargaining power.

Voena (2015) estimates a value of 0.3, which is one-third smaller than the one I estimate. The two numbers do not align because α captures different objects in the two papers. In Voena (2015)’s world, α prices the wife’s outside option given changes in divorce laws across US states, while in my model, α reflects her observed consumption. In addition, she estimates α using data from the PSID between 1980 and 1990, years when female labor force participation was lower than in the 2000s, and women’s bargaining power is expected to rise as they enter the labor force.

Estimating γ . I measure the public-good expenditure share, γ , directly from expenditure data rather than as the residual implied by the resource shares. One alternative would be to set γ equal to the residual $1 - \varrho_f - \varrho_m$. The residual is 0.13, implying that the couple spends roughly \$5,400 a year on public goods, while couples are *observed* spending \$22,300 a year on shelter, utilities, and food alone. The residual is this small because the apparel Engel curves are essentially flat ($R^2 = 0.004$). Browning et al. (2013)’s method allows me to recover the *ratio*, but it carries almost no information about the *level* of resources, and the residual absorbs the error. I therefore measure γ directly as the share of total spending devoted to market inputs for home production, including shelter, utilities, food at home, and market substitutes for home time, such as domestic services and appliance repair. This yields $\gamma = 0.4887$.

Del Boca and Flinn (1995) estimate γ to be 0.204. The two values differ because 0.204 is an estimate based on divorced parents, whose only joint consumption concerns their children; they cannot enjoy economies of scale in housing, utilities, or transportation, among others. For couples who live together, joint consumption is therefore expected to account for a

significant share of total consumption.

Standard errors. The CEX is a stratified, multistage, clustered sample, and the BLS does not publish the design variables needed for an analytic variance. Thus, following the BLS recommendation, I re-run the *entire* estimator on each of the 44 replicate weight vectors and take the spread across replicates as the sampling variance. The resulting standard errors are 0.021 for α and 0.006 for γ .

D.3.2 κ/γ from the China Trade Shock in the PSID

As shown in Section 4, the participation rule satisfies

$$\log(1 - \rho(r)) = \underbrace{\kappa \log \chi}_{\text{intercept}} - \underbrace{(\kappa/\gamma)}_{\text{slope}} \log(1 + r). \quad (\text{D.1})$$

This equation is a structural relation, not a regression. Cross-sectional variation in the relative wage r is not exogenous: r is an equilibrium object. In addition, I observe r only when the wife participates in the market. I therefore use the empirical design in Section 2 as the identification strategy for the participation block.

In particular, I identify κ/γ from the experiment in which the *husband's* industry is exposed. The wife is not directly displaced, so the shock affects her participation decision only through r , which is exactly the exclusion restriction required by Equation D.1. Because the equation is linear, the parameter is a Wald ratio:

$$\frac{\kappa}{\gamma} = - \frac{\Delta \log(1 - \rho)}{\Delta \log(1 + r)}. \quad (\text{D.2})$$

To construct the numerator of Equation D.2, I use the estimates in Panel B of Table 3. Wives of exposed husbands raise labor-force participation by 4.2 p.p. from a baseline of 0.78, so ρ increases from 0.78 to 0.822. Then, the numerator is:

$$\Delta \log(1 - \rho) = \log(1 - 0.822) - \log(1 - 0.78) = -0.212.$$

To construct the denominator, I estimate the static version of Equation 5 on $\log(\text{Hourly Wage})$ for each spouse. The sample consists of individuals who have at least one observation with positive wages before and during the shock. Table D.1 shows that when the husband is exposed to the shock, his hourly wage decreases by 16.4 log points while her wage decreases by 4.2 log points. Since the husband's wage falls by more than his wife's, the relative wage she faces rises. Using the estimates in Table D.1, r increases by 12.2 log points, which implies an increase of 13%. The r implied by the baseline mean is 0.88. However, this number includes only employed individuals, while the numerator considers all

individuals in the sample. Then, the correct value for r_0 is 0.607. Thus, the denominator of Equation D.2 is

$$\Delta \log(1 + r) = \log(1 + 0.686) - \log(1 + 0.607) = 0.0477.$$

Substituting the numerator and the denominator into Equation D.2,

$$\frac{\kappa}{\gamma} = -\frac{-0.212}{+0.048} = 4.44.$$

With $\gamma = 0.4887$, I obtain $\kappa = \gamma \cdot \frac{\kappa}{\gamma} = 2.17$, which satisfies the integrability restriction $\kappa > \gamma$.

Table D.1: Husband's Exposure to the China Trade Shock and log(Hourly Wage)

	(1) Husband	(2) Wife
Exposed Husband=1	-0.078* (0.042)	0.150*** (0.052)
(Exposed Husband=1)x(Post=1)	-0.164* (0.086)	-0.042 (0.070)
Observations	12,023	9,979
R^2	0.35	0.31
Baseline Mean	3.31	2.91

Notes. Robust standard errors are two-way clustered at the husband's industry and individual levels. Given that I restrict the sample to individuals who have at least one observation with positive wages before and during the shock, the number of observations in this table does not match the number of observations in Fact 3. * for $p < 0.10$, ** for $p < 0.05$, *** for $p < 0.01$.

D.3.3 χ as a Level Restriction

The participation rule in Equation D.1 is log-linear in $1 - \rho$: its *slope* in $\log(1 + r)$ identifies κ/γ , and its *intercept* identifies χ . With κ/γ estimated above and $\kappa = \gamma \cdot (\kappa/\gamma)$, χ is the single remaining free parameter, and I set it so that the model reproduces the observed *level* of married women's participation in 2000Q1, $\rho_0 = 0.621$. Married women's participation is therefore a *targeted* moment and is not reported as an untargeted success of the model.

I calibrate χ only on couples in which the husband is employed in a productive sector, because the model misses the participation of wives of non-employed husbands. In the data,

wives of non-employed husbands work *less*, whereas the model predicts that they work *more*. Since ρ is monotone in r , and a non-employed husband's income is the benefit b_j , the ratio $r = w_f/b_j$ is mechanically the largest anywhere in the model, inducing wives to work. The data instead reflect strong positive assortative matching in the marriage market (Browning et al., 2014), which the model abstracts from. I denote the bundles in which the husband is employed by $\bar{o}$.

Let $m_{\bar{o},0}$ be the mass of couples in bundle $\bar{o}$ at $t = 0$ and $r_{\bar{o},0} = w_{\bar{o},0}^F/w_{\bar{o},0}^M$ its wage ratio. I then set χ such that the mass-weighted average participation rate the model generates equals the one in the data:

$$\sum_{\bar{o}} m_{\bar{o},0} \rho(r_{\bar{o},0}; \chi) = \rho_0. \quad (\text{D.3})$$

Since ρ is strictly decreasing in χ , Equation D.3 has a unique root, and it yields $\chi = 1.92$. This satisfies the support restriction $\chi \geq 1$ and leaves the participation margin interior in every cell.

E Additional Results in the Quantitative Exercises

E.1 Robustness Exercises

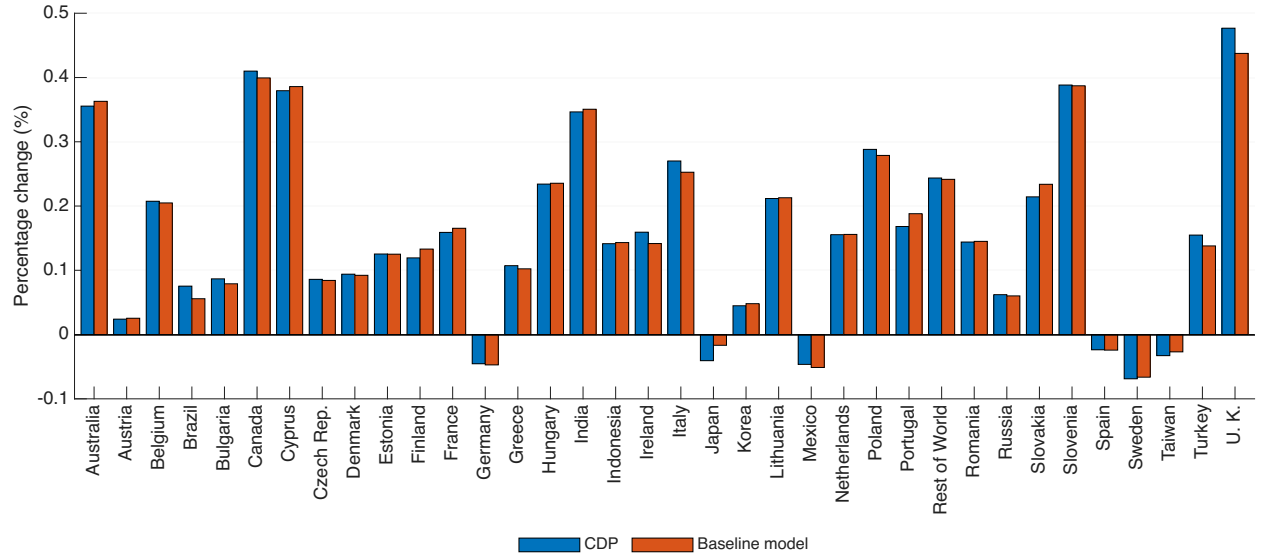
The household block changes who bears the China Trade Shock, but not the size of the shock where trade operates. In Table E.1, I compare CDP with the baseline economy, the single-agent economy, and the three counterfactual exercises. Manufacturing jobs per capita fall by between 0.200 and 0.210 p.p. in every economy I study, within 2.6% of the baseline. Welfare in the rest of the world does not move either: the mean gain of the other 36 regions is between 0.156% and 0.158% in every economy, and their gains correlate with those in CDP at 0.997. The trade block of my model is, therefore, still that of CDP.

Table E.1: Same Shock, Same Trade Block

	CDP	Baseline	Single-agent	C1	C2	C3
<i>Panel A. Manufacturing, $t = 199$</i>						
Manuf. jobs per capita (pp)	-0.228	-0.206	-0.202	-0.202	-0.200	-0.210
% change vs baseline	+11.0	—	-1.6	-1.7	-2.6	+2.0
<i>Panel B. Rest of the world (36 regions)</i>						
Mean welfare gain (%)	0.158	0.156	0.158	0.156	0.156	0.158
Corr. with CDP	—	0.997	0.997	0.997	0.997	0.997

Notes. Responses at $t = 199$. Manufacturing jobs per capita is manufacturing employment divided by population, expressed in p.p.; % change vs. baseline is $100 \times (X/\text{baseline} - 1)$. CDP is the economy of Caliendo et al. (2019). Rest of the world: consumption-equivalent welfare gains of the 36 regions (35 countries and the rest of the world; China excluded), computed as in Caliendo et al. (2019) and summed over $t = 2, \dots, 199$.

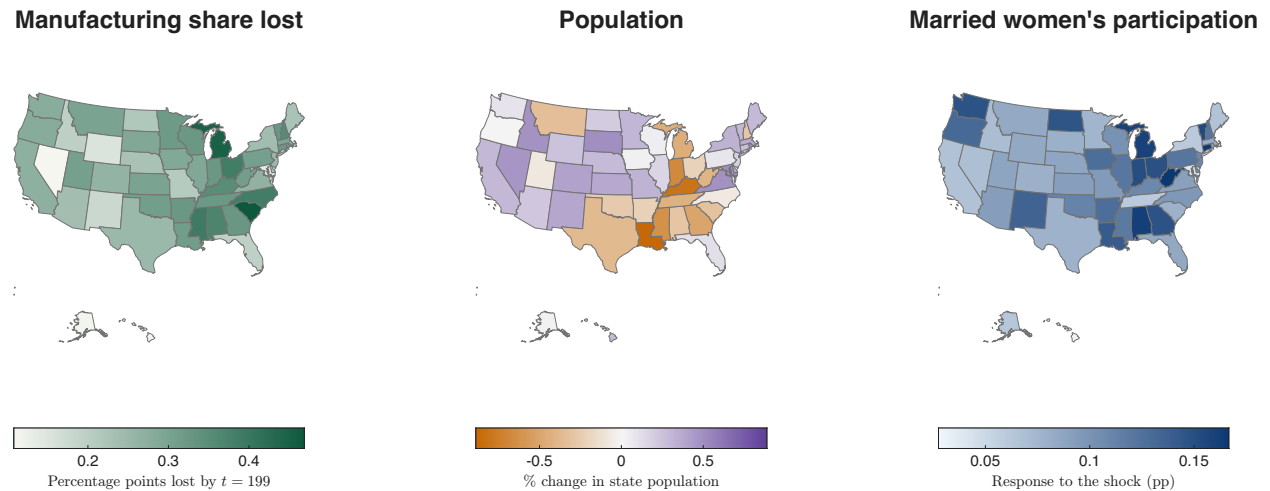
Figure E.1: Comparison of Welfare Effects in the Rest of the World



Notes. Consumption-equivalent welfare gains calculated in both models as in Caliendo et al. (2019), summed over $t = 2, \dots, 199$.

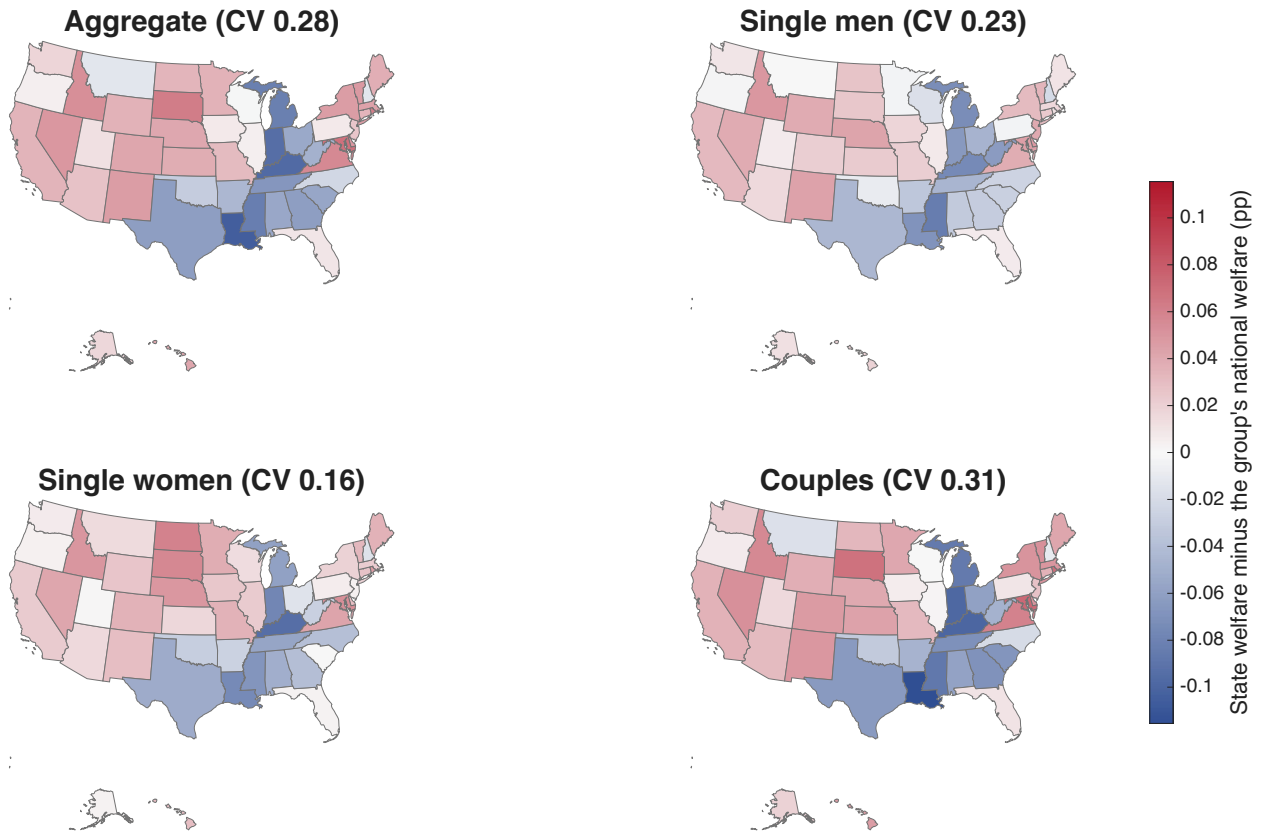
E.2 Additional Results

Figure E.2: Manufacturing Employment Decline and Migration and Participation Responses



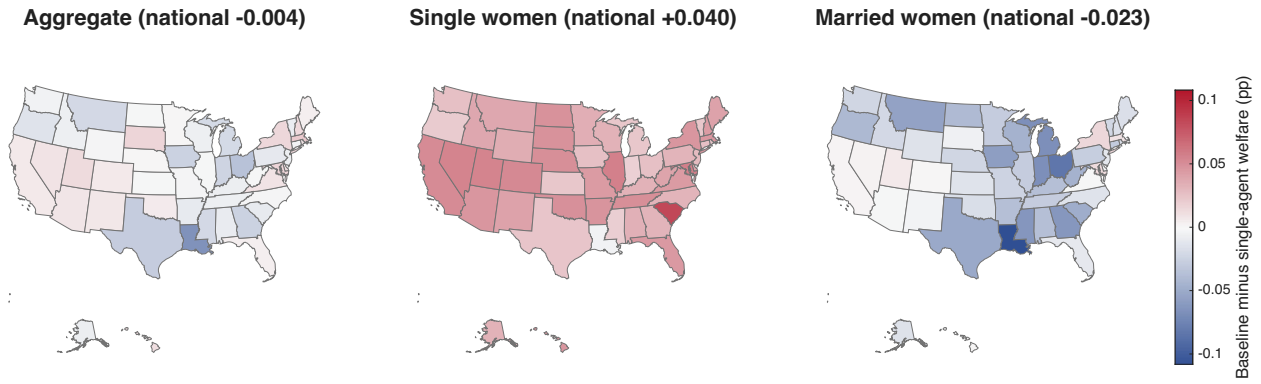
Notes. Effect of the China Trade Shock by $t = 199$. The left panel plots the decline in manufacturing's share of the state's employment, in p.p., shown as a positive number. The middle panel plots the change in the state's population as a percentage of its no-shock level. The national population is fixed, so this panel is pure reallocation across states: it sums to zero, and one state's gain is another's loss. The right panel plots the change in married women's participation rate, in p.p., averaged over the couples living in the state at $t = 199$.

Figure E.3: Distribution of Welfare Effects



Notes. State welfare minus the group's own national welfare, money-metric consumption-equivalent in p.p.. For each group, I report in the title the coefficient of variation of state welfare gains, weighted by the group's population.

Figure E.4: Distribution of Welfare Effects Gaps Across Economies



Notes. In each map, I plot the baseline model's state money-metric welfare minus the single-agent model's, in p.p., for the same people located at their $t = 1$ state, weighting both economies by the baseline money metrics. One shared scale; the national gap is in each title.

Table E.2: Parameter Values to Shut Down Margins

Counterfactual	Instrument	Value
C1. No migration response	$1/\vartheta \rightarrow 0$	$\vartheta : 10.67 \rightarrow 10^6$
C2. No participation response	$\kappa/\gamma \rightarrow 1^+$	$\kappa : 2.1695 \rightarrow 0.65$
C3. No gender-wage response	$1/\sigma \rightarrow 0$	$\sigma : 1.4 \rightarrow 100$

Notes. The first counterfactual is the only one that fully shuts down the margin. For counterfactual 2, since the model requires $\kappa > \gamma$, the most the response can be damped is $\kappa_0/\gamma = 4.44\times$; at $\kappa = 0.65$, the response is damped $3.39\times$. I also re-pin χ to keep the anchor participation rate unchanged. For the third counterfactual, I set $1/\sigma = 0.01$ against a calibrated 0.714, so 1.4% of the gender-wage response survives.

Table E.3: Effects of the China Trade Shock under Counterfactual 1

	Baseline	No migration resp.	Difference	% change
<i>Panel A. Employment, $t = 199$ (pp)</i>				
Manufacturing share of employment				
Aggregate	-0.293	-0.287	0.006	-2.2
Men	-0.383	-0.374	0.010	-2.5
Single men	-0.287	-0.278	0.008	-2.9
Married men	-0.412	-0.403	0.010	-2.4
Women	-0.191	-0.187	0.003	-1.7
Single women	-0.188	-0.185	0.003	-1.6
Married women	-0.192	-0.188	0.004	-1.8
Employment rate				
Aggregate	0.112	0.099	-0.013	-11.7
Men	0.099	0.079	-0.019	-19.6
Women	0.124	0.117	-0.007	-5.8
<i>Panel B. Married women's participation response, $t = 199$ (pp)</i>				
All couples	0.101	0.097	-0.003	-3.4
Both in manufacturing	0.103	0.100	-0.003	-3.0
Husband only	0.105	0.102	-0.003	-2.8
Wife only	0.101	0.097	-0.003	-3.4
Neither	0.100	0.096	-0.003	-3.5
<i>Panel C. Migration response, manufacturing – other, ever moved by $t = 199$ (pp)</i>				
Single men	+0.0206	—	—	—
Single women	+0.0115	—	—	—
Couples, by husband's sector	+0.0259	—	—	—
Couples, by wife's sector	+0.0053	—	—	—
<i>Panel D. Welfare (% CE)</i>				
Aggregate	0.175	0.176	0.002	+1.0
Single men	0.157	0.162	0.005	+2.9
Single women	0.234	0.236	0.002	+0.9
Couples	0.170	0.171	0.001	+0.7
Both in manufacturing	0.125	0.122	-0.003	-2.7
Husband only	0.140	0.135	-0.005	-3.4
Wife only	0.161	0.162	0.001	+0.7
Neither	0.180	0.182	0.003	+1.6

Notes. Responses at $t = 199$, except welfare. Difference is the exercise minus the baseline; % change is $100 \times (\text{exercise/baseline} - 1)$; a dash marks a margin that is shut down or a baseline that rounds to zero. Panel B: married women's quarterly participation rate, by the couple's exposure at $t = 1$. Panel C: difference between the shares of manufacturing and non-manufacturing workers who have changed state at least once by $t = 199$; couples are tagged by each spouse's sector at $t = 1$. Panel D: money-metric consumption-equivalent gains, weighting everyone by their baseline money metric at 2000Q1.

Table E.4: Effects of the China Trade Shock under Counterfactual 2

	Baseline	No participation resp.	Difference	% change
<i>Panel A. Employment, $t = 199$ (pp)</i>				
Manufacturing share of employment				
Aggregate	-0.293	-0.285	0.008	-2.9
Men	-0.383	-0.395	-0.011	+2.9
Single men	-0.287	-0.266	0.020	-7.1
Married men	-0.412	-0.433	-0.020	+5.0
Women	-0.191	-0.160	0.031	-16.3
Single women	-0.188	-0.227	-0.039	+20.9
Married women	-0.192	-0.127	0.065	-33.9
Employment rate				
Aggregate	0.112	0.085	-0.027	-24.3
Men	0.099	0.090	-0.009	-8.9
Women	0.124	0.080	-0.044	-35.7
<i>Panel B. Married women's participation response, $t = 199$ (pp)</i>				
All couples	0.101	0.041	-0.060	-59.2
Both in manufacturing	0.103	0.042	-0.061	-59.5
Husband only	0.105	0.042	-0.062	-59.6
Wife only	0.101	0.041	-0.060	-59.3
Neither	0.100	0.041	-0.059	-59.2
<i>Panel C. Migration response, manufacturing – other, ever moved by $t = 199$ (pp)</i>				
Single men	+0.0206	+0.0210	+0.0004	+1.8
Single women	+0.0115	+0.0096	-0.0018	-16.0
Couples, by husband's sector	+0.0259	+0.0269	+0.0010	+3.7
Couples, by wife's sector	+0.0053	+0.0057	+0.0004	+6.7
<i>Panel D. Welfare (% CE)</i>				
Aggregate	0.175	0.167	-0.008	-4.6
Single men	0.157	0.153	-0.004	-2.4
Single women	0.234	0.255	0.021	+8.8
Couples	0.170	0.157	-0.012	-7.2
Both in manufacturing	0.125	0.115	-0.010	-7.6
Husband only	0.140	0.126	-0.014	-10.2
Wife only	0.161	0.153	-0.008	-5.0
Neither	0.180	0.167	-0.012	-6.9

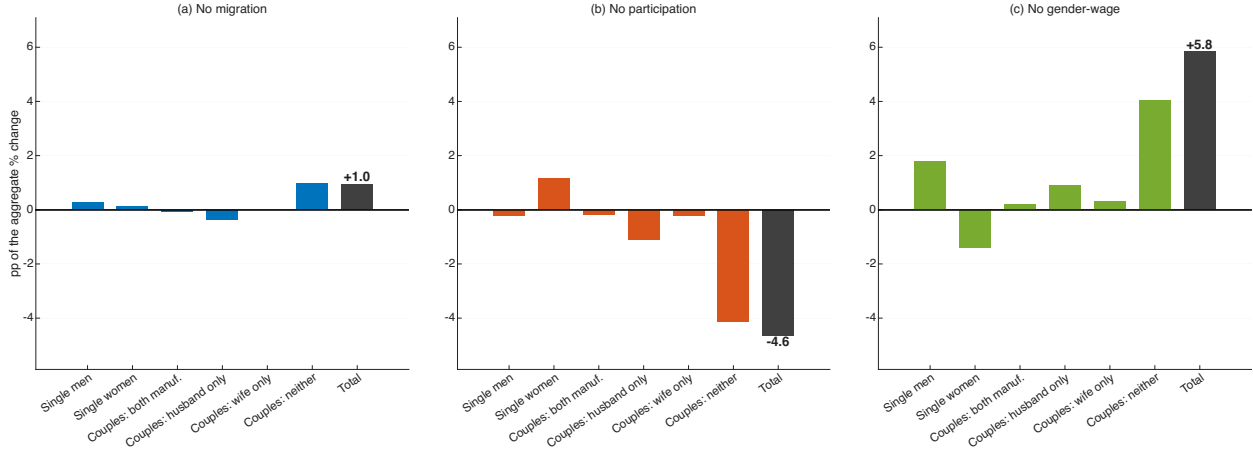
Notes. Responses at $t = 199$, except welfare. Difference is the exercise minus the baseline; % change is $100 \times (\text{exercise/baseline} - 1)$; a dash marks a margin that is shut down or a baseline that rounds to zero. Panel B: married women's quarterly participation rate, by the couple's exposure at $t = 1$. Panel C: difference between the shares of manufacturing and non-manufacturing workers who have changed state at least once by $t = 199$; couples are tagged by each spouse's sector at $t = 1$. Panel D: money-metric consumption-equivalent gains, weighting everyone by their baseline money metric at 2000Q1. On this economy's own money metrics, the aggregate gain falls by 6.9% rather than 4.6%.

Table E.5: Effects of the China Trade Shock under Counterfactual 3

	Baseline	No gender-wage resp.	Difference	% change
<i>Panel A. Employment, $t = 199$ (pp)</i>				
Manufacturing share of employment				
Aggregate	-0.293	-0.295	-0.002	+0.7
Men	-0.383	-0.401	-0.018	+4.7
Single men	-0.287	-0.304	-0.017	+6.0
Married men	-0.412	-0.430	-0.018	+4.4
Women	-0.191	-0.181	0.009	-5.0
Single women	-0.188	-0.172	0.016	-8.5
Married women	-0.192	-0.186	0.006	-3.3
Employment rate				
Aggregate	0.112	0.097	-0.015	-13.3
Men	0.099	0.127	0.028	+28.1
Women	0.124	0.070	-0.055	-44.1
<i>Panel B. Married women's participation response, $t = 199$ (pp)</i>				
All couples	0.101	0.037	-0.064	-63.3
Both in manufacturing	0.103	0.039	-0.064	-62.5
Husband only	0.105	0.040	-0.064	-61.4
Wife only	0.101	0.037	-0.064	-63.6
Neither	0.100	0.036	-0.064	-63.7
<i>Panel C. Migration response, manufacturing - other, ever moved by $t = 199$ (pp)</i>				
Single men	+0.0206	+0.0202	-0.0004	-2.1
Single women	+0.0115	+0.0111	-0.0004	-3.3
Couples, by husband's sector	+0.0259	+0.0251	-0.0008	-3.3
Couples, by wife's sector	+0.0053	+0.0051	-0.0003	-5.1
<i>Panel D. Welfare (% CE)</i>				
Aggregate	0.175	0.185	0.010	+5.8
Single men	0.157	0.187	0.030	+18.8
Single women	0.234	0.210	-0.024	-10.4
Couples	0.170	0.182	0.012	+7.1
Both in manufacturing	0.125	0.137	0.012	+9.4
Husband only	0.140	0.152	0.012	+8.4
Wife only	0.161	0.173	0.012	+7.5
Neither	0.180	0.192	0.012	+6.7

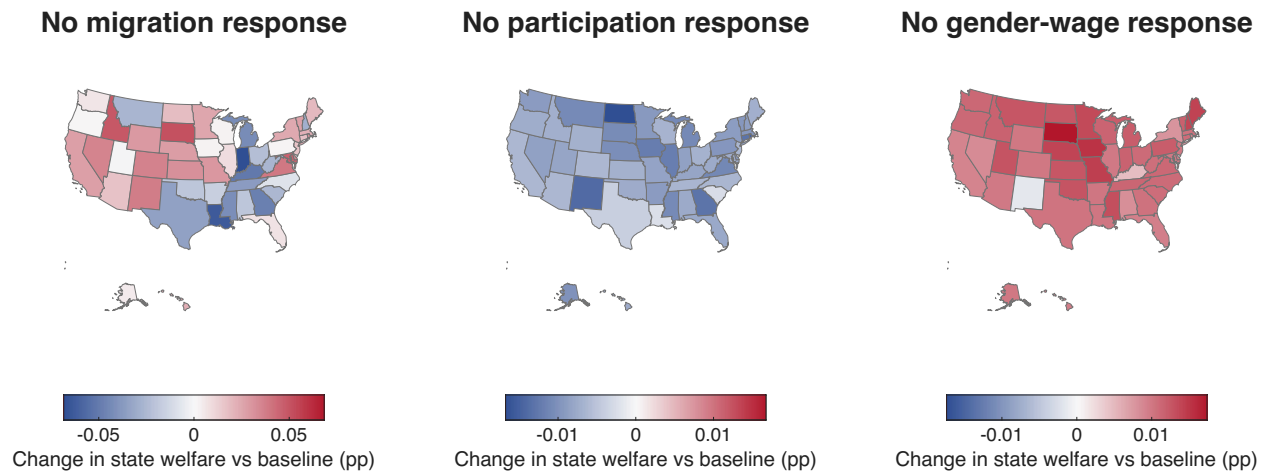
Notes. Responses at $t = 199$, except welfare. Difference is the exercise minus the baseline; % change is $100 \times (\text{exercise/baseline} - 1)$; a dash marks a margin that is shut down or a baseline that rounds to zero. Panel B: married women's quarterly participation rate, by the couple's exposure at $t = 1$. Panel C: difference between the shares of manufacturing and non-manufacturing workers who have changed state at least once by $t = 199$; couples are tagged by each spouse's sector at $t = 1$. Panel D: money-metric consumption-equivalent gains, weighting everyone by their baseline money metric at 2000Q1.

Figure E.5: Contributions to the Change in Aggregate Welfare



Notes. Each bar is the group's contribution to the percentage change in aggregate welfare relative to the baseline economy, $100 \times \omega_g(G_g^X - G_g^B) / \sum_h \omega_h G_h^B$, where ω_g is the group's baseline money metric at 2000Q1 and G is its welfare gain. The bars add up to the total.

Figure E.6: Distribution of Welfare Changes in each Counterfactual



Notes. Each map plots the change in state money-metric welfare relative to the baseline economy, in p.p.. Each panel carries its own scale, read off its own color bar, so magnitudes are not comparable by eye across panels.